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Neural Modeling of Musical Devices

Efficient and Low-Latency Data-Driven Emulation of
Nonlinear Time-Variant Instruments and Effects

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*Every day it gets a little easier... But you gotta do it every day—
that's the hard part. But it does get easier.*
– *Jogging Baboon (from BoJack Horseman)*

Abstract

Modeling musical devices, whether instruments or effects, acoustic or electronic, is a dynamic area of research and development characterized by significant scientific relevance, a rich history, and far-reaching artistic implications. This multidisciplinary field encompasses a diverse range of perspectives, from musical acoustics to digital signal processing. Musical devices present nonlinear and time-variant characteristics that provide unique soundings and behaviors that are challenging to replicate in digital emulation. Thanks to the increasing computational power of general-purpose computers, deep learning has emerged as an approach for modeling musical devices, particularly analog audio effects. In this thesis a step forward in this direction is taken, broadening the understanding of the use of neural networks to model musical devices that process or generate sounds, with a focus on complex devices that pose challenges for current neural modeling methods. In this research, novel neural architectures are investigated, placing great emphasis on essential aspects of modeling networks such as low latency, computational efficiency, and interactivity, all of which are crucial to achieving 'playable' emulations. Furthermore, the thesis develops architectures, subcomponents, and methods designed for wide applicability across different musical device categories with minimal need for customization. The primary focus lies on two case studies involving nonlinear, time-variant musical devices representative of the acoustic and analog realms: the compressor, as a processor, and the acoustic piano, as a generator. Besides proposing novel neural architecture for modeling musical devices, in this thesis, the understanding of learning strategies in neural models is advanced, and the application of neural techniques for parameter-sensitive musical behavior is refined. Additionally, it compiles a variety of datasets of the musical devices being studied.

Sammendrag

Modellering av musikkinstrumenter eller musikkeffekter, akustiske eller elektroniske, er et dynamisk område innen forskning og utvikling karakterisert ved betydelig vitenskapelig relevans, en rik historie og vidtrekkende kunstneriske implikasjoner. Dette tverrfaglige feltet omfatter et mangfold av perspektiver fra musikkakustikk til digital signalbehandling. Det handler om ikke-lineære og tidsvarierende egenskaper som gir unike lydopplevelser og oppførsel som er utfordrende å gjenskape i digital emulering. Takket være økt datakraft i generelle datamaskiner, har dyp læring blitt aktuell som en tilnærming for modellering, og særlig av analoge lydeffekter. I denne avhandlingen tas det et steg videre i denne retningen, ved å utvide forståelsen av bruken av nevrale nettverk for å modellere enheter som prosesserer lyd eller som generer lyd, med fokus på komplekse enheter som er utfordrende for dagens nevrale modelleringsmetoder. I forskningen undersøkes nye nevrale arkitekturer, med stor vekt på viktige aspekter ved modellering av nettverk som lav forsinkelse, beregningseffektivitet og interaktivitet. Dette er egenskaper som er avgjørende for å oppnå emuleringer som kan brukes i praksis ved spilling. Videre utvikles det i avhandlingen arkitekturer, subkomponenter og metoder designet for bred anvendelse på tvers av ulike kategorier av enheter med minimalt behov for tilpasning. Hovedfokuset ligger på to kasusstudier som involverer ikke-lineære, tidsvarierende systemer og som er representative for det akustiske og analoge området: kompressoren, som en prosessor, og det akustiske pianoet, som en generator. Foruten å foreslå nye nevrale arkitekturer for modellering av slike enheter, fremmes det i denne avhandlingen en forståelse av læringsstrategier i nevrale modeller og det gis bidrag til anvendelsen av nevrale teknikker for parameter-sensitiv musikalsk oppførsel. I tillegg er det samlet inn en rekke datasett fra de studerte enhetene.

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List of Papers

Paper I

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Paper VIII

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Chapter 1

Introduction

The digital emulation of physical musical devices, such as acoustic instruments, as well as analog synthesizers and effects, has been an active field of research and development since the early days of sound and music applications of digital signal processing. Despite the advent of numerical algorithms for digital sound synthesis, acoustic musical instruments and analog electronic musical devices, such as sound synthesizers and audio effects, still enjoy significant popularity due to the uniqueness, complexity, and richness of their timbre and sonic response. Their digital emulation has been introduced to primarily address accessibility issues. Indeed, for a significant percentage of users, there are non-negligible barriers in terms of cost, size, and availability. Most acoustic musical instruments and electronic hardware are usually expensive due to the costs associated with materials and the production process. They can also be large in size, posing significant space challenges for storage and transportation, especially for performances. In addition, many devices are no longer in production and are hard to obtain from the pre-owned market. Moreover, the digital emulation of acoustic instruments has also opened novel control and interaction opportunities because energy transfer from the player to the instrument's body is no longer necessary to generate the sound.

The digital emulation of acoustic and analog devices finds application beyond the context of music production and music performances. For example, musical education, training, preservation of music-related cultural and technological heritage, gaming, and interactive media are rapidly changing and gaining new relevance due to the recent advances in immersive technologies. In music performance and production, as well as in metaverse and virtual reality, it is essential to accurately reproduce the auditory, haptic, and visual aspects of the physical world. From a sound and music perspective, this reflects the need to emulate the device's sound generation and interactive functioning. If placed inside a virtual world, it also requires simulating wave propagation in space and interactions between objects to instill as much realism as possible. Achieving high-fidelity modeling can be particularly challenging, especially for replicating the broad spectrum of sonic nuances generated by the complex phenomena governing acoustic and analog musical devices.

Acoustic instruments are mechanical devices transferring energy and vibrations across multiple components and materials. Analog synthesizers and effects are circuits generating or processing a variable voltage signal in the audible frequency

range by orchestrating the flow of electrical signals in a network of active and passive components. Therefore, modeling techniques are not necessarily specific to the musical domain and can be borrowed or adapted from other acoustic and electronics application domains. However, emulating musical devices presents unique and demanding challenges in terms of interactivity and computation. The audio generation needs to be fast, timely, and uninterrupted. The computation must satisfy real-time audio constraints without introducing audible artifacts while minimizing response latency, which is a crucial factor in interactive music systems and digital musical instruments. Thus, this relatively small field of research has also contributed to pioneering advances that were later adopted more broadly in non-audio or non-musical contexts.

Methods analyzing the target's device functioning have been extensively proposed, proving to be highly realistic and accurate. However, limitations and challenges remain. Designing an accurate model for a specific device is costly in terms of time and resources. Conversely, machine learning techniques enable automatic processes that minimize the modeling costs and effort. Yet, these techniques require sufficient data from the target device, and, in many cases, their accuracy still falls short compared to analytical techniques.

1.1 Modeling Musical Devices

A model is a representation or idealization of a real-world system's structure, behavior, operation, or other characteristics. Physical or abstract models are at the core of science and engineering. They convey design information, simulate real-world behavior, or specify a process. Models have to be informative and accurate only concerning context-relevant characteristics, whereas, for simplicity, the remaining are often disregarded.

Modeling musical devices, whether electronic, electromechanical, or acoustic, is a well-established interdisciplinary field of research and development with a significant industry, market, and user base. The aim is to emulate the sound processing or generation of these devices. Musical devices are physical world systems that need to be analyzed and approximated with equations, which are then solved using computing systems to emulate them. The produced sound is heard and musically interpreted by the human auditory system. For this reason, psychoacoustics—which studies how our bodies receive sound and how the brain interprets it—plays a fundamental role in the modeling process. It can offer modeling simplifications and provide more accurate evaluation metrics. Humans use these devices and interact with them during musical performances. Therefore, human factors must also be considered in this highly specialized context of human-computer interaction.

Complex systems of equations usually describe musical devices, and their derivation can be particularly challenging. Nonlinear characteristics lie at the core of almost all acoustic and electronic analog instruments, generating a wide range of behaviors. Conversely to the linear case, in nonlinear systems, the output is not proportional to the input. This poses significant challenges to both the modeling process and the subsequent computational process needed to solve the

equations. Musical devices can also exhibit complex timing behaviors due to time-variant, input-dependent parameters and long transients, meaning that the instantaneous relationship between input and output can significantly change over time in response to both present and past inputs. In other words, the output can be different when the same input is applied at different times. Nonlinearities and time-variance are particularly challenging to incorporate into digital models and can considerably increase their computational complexity.

Musical devices must provide users with simultaneous multi-parametric control with sub-millisecond action-to-sound latency to allow expressive and intimate control. This interaction requirement further increases the challenges of meeting the strict constraints for live audio applications. Generating samples at an audio rate is a required but insufficient condition to use the models in most applications. The latency between the input, an audio signal, or control parameters, and the output must be minimized. Moreover, the response must be continuous, morphing between different model states without generating audible artifacts due to abrupt changes in the sonic response. On one side, these needs exclude certain modeling techniques because they are unsuitable for such specifications; on the other, they significantly increase the computational complexity of real-time interactive implementations of the models.

To date, numerous modeling methods and solutions have been proposed for musical devices. Several techniques based on internal physics knowledge or internal device measurements have also been successfully employed in commercial products. The choice of the modeling method depends on the device characteristics, the availability of the system's inner information, the available computational power for the implementation, and the target application. For example, when models do not need to be interactively played by a performer, sound can be synthesized in large blocks or even offline. In this context, action-to-sound latency is not important, and this significantly simplifies many aspects of the models and the associated computation.

Although physical modeling synthesis of acoustic instruments was introduced more than three decades ago, the number of commercial products using this highly accurate synthesis technique remains very limited. Indeed, the vast majority of acoustic instrument emulations still rely on pre-recorded samples, which offer more computationally efficient solutions, though at the expense of increased memory use, yielding reasonably realistic sounding results. On the other hand, analog sound synthesizers and effects are emulated using a broader spectrum of techniques. Common modeling methods are based on the analysis and simulation of circuits or on particular signal processing techniques where the parameters of a generic model are tuned using the device's measurements.

Musical device modeling has generally achieved high fidelity and realism. However, the current methodology presents limitations. Emulations based on the device's physics, whether acoustic or electronic, can be accurate but require significant effort to find a stable and computable numerical scheme to model each single and specific device. Formulating a suitable mathematical scheme to simulate the physical behavior is a complex task, often leading to models requiring high computational power to be executed in real-time. Approximations are generally

necessary to obtain systems of mathematical equations that can be solved in a reasonable amount of time. This becomes particularly challenging, especially when dealing with nonlinearity persisting over an extended period of time.

Comprehensive knowledge of the internal behavior of a device can also be missing. For example, the internal circuitry information of devices such as analog synthesizers and their effects is generally proprietary and not in the public domain. Moreover, it is not yet possible for several devices to describe internal behavior using mathematical formalism due to phenomena not yet fully understood at the physics level.

On the other hand, methodologies based on device measurements can be adapted to different modeling problems with minor changes. Moreover, they are relatively simpler because they primarily require a collection of measurements recorded from the device. These data-driven methods, however, have their drawbacks. Recording all possible responses of the device to input stimuli is unfeasible, so the model must accurately predict accurate outputs for unavailable input stimuli data. Therefore, these methods must approximate a continuous function based on examples typically recorded at discrete steps of real-valued inputs. This is particularly challenging for models that generate sound, such as those emulating acoustic instruments or analog synthesizers. Indeed, there is a lack of understanding regarding the integration of user control inputs in these models, especially when aiming for solutions that work across different classes of devices.

In the scope of this thesis, musical devices will refer to any electronic, electro-mechanical, and acoustic devices that process or create musical sounds. We refer to these as *processors* and *generators*. The first group, *processors*, refers to audio effects and devices that alter a sound signal. The sound alteration process is generally parametric, allowing users to change effect settings in real-time. The second group, *generators*, includes devices that produce musical sounds. Here, too, the sound generation is a parametric process, allowing users to interactively control how the sound is generated. However, these devices generally feature inputs that explicitly control the musical characteristics, such as pitch and duration, of the generated monophonic or polyphonic sound. In both cases, input and control parameters can be continuous or discrete, and low-latency response is necessary to enable use in music performances or live audio applications.

1.2 Motivation

Deriving a model of a musical device directly from data eliminates the need for specific expertise and the effort required to develop a mathematical model and find a stable, efficient computational solution. In theory, this procedure must be repeated for every new musical device that needs to be modeled. The advantages of a data-driven approach and its general-purpose qualities justify further investigation into this method. Artificial Neural Networks (ANN), particularly Deep Learning (DL), have shown promise in mitigating data-driven methodology limitations, such as noise sensitivity and generalization capability. These techniques offer end-to-end learning, enabling the modeling of all the components of a device simultaneously and employing backpropagation as an

optimization algorithm to compute the optimal values for the network parameter automatically. This approach provides a robust framework for tuning model parameters and is resilient to noisy data. In addition, neural networks can extract informative representations from data and approximate the underlying input-output relationship of the sampled space, which helps to generalize effectively and produce accurate predictions for new inputs. The strengths of these modeling techniques motivate further exploration of novel neural architectures in data-driven workflows for modeling musical devices.

Concerning modeling audio *processors*, several guitar amplifiers and nonlinear distortion circuits have been successfully modeled with DL approaches, at times also satisfying real-time and low-latency constraints. DL modeling of analog amplifiers and distortion pedals has been intensively investigated in recent years, with successful results determining the appearance of commercial products. However, these DL approaches still present limitations when modeling devices presenting time-variant characteristics and a large set of user-controllable parameters. Time-variant characteristics complicate the model estimation using data-driven approaches due to the need to track a variable amount of the device’s history without being detrimental concerning computational complexity, latency, and interactivity. Multiple parameters lead to the need to learn multiple representations of the devices, considering different sound processing based on different device settings.

The extent to which DL algorithms can provide an effective and efficient approach for modeling musical sound *generators* and enabling interactivity in live performances remains unknown. The use of DL in modeling analog sound synthesizers is still in its infancy, with some preliminary investigations conducted on acoustic musical instruments. Modeling acoustic drums has been explored, resulting in DL models capable of real-time sound synthesis. These devices generate inharmonic sounds with a relatively short duration. Instruments with more complex spectral richness, harmonic structure, and complex stimuli, such as guitar and piano, have also been explored. However, their accuracy and live playability are still significantly limited and require further refinement or alternative modeling solutions.

These aspects justify the research focus on two specific modeling tasks that tackle particularly complex musical devices, indicative of open challenges not yet resolved by state-of-the-art data-driven modeling techniques. The first task concerns dynamic range compressors, as a representative case of *processors*, and the second addresses pianos, as a representative case of *generators*. Dynamic range compression is a nonlinear time-variant audio effect. It reduces the amplitude of an input signal by an arbitrary amount when the input signal exceeds a threshold. It is a nonlinear system since the gain reduction follows a nonlinear function that depends on the input signal level. In turn, the compression of the sum of two signals is generally different from the addition of the two signals compressed individually. Dynamic range compression is a time-variant effect because it is not applied instantaneously upon exceeding the threshold level. Instead, it is applied and removed gradually over time, according to the attack and release times, which can be relatively long. Additionally, the compression characteristics are determined

by parameters that can be adjusted within wide ranges, leading to significantly different sound alterations.

On the other hand, the piano is another system that presents considerable complexity. The piano is a nonlinear system with nonlinearities present in its excitation mechanism—where the hammers interact with the strings—as well as within the strings themselves and in the interactions among various vibrating sub-components of the piano. The sound production also depends on the excitation status of the strings, specifically if they are at rest or already in motion due to past stimuli. Strings in close proximity also give subtle influences that, if vibrating, dissipate the energy to the bridge, and in turn, the bridge transfers energy to other strings. Furthermore, the piano hammer felt exhibits hysteresis in its deformation due to the properties of the wool felt and time-varying strings’ tension. Finally, both the compressor and the piano possess challenging characteristics in their mechanisms, making them ideal candidates for investigating novel neural techniques for modeling musical devices.

1.3 Research Aim and Objectives

This dissertation aims to expand knowledge and practices in the machine learning modeling of musical devices. The objective is to devise neural architectures and methods for data-driven frameworks suitable for both *processors* and *generators*. The research specifically targets parametric musical devices that present evident time-variant and nonlinear responses, narrowing the scope of the investigation to modeling techniques that offer computationally efficient and low-latency solutions. These methods aim to be sufficiently versatile to be applied within or across different classes of musical devices where the system responses demonstrate unique specificities. In the context of this thesis, the term ‘neural’ specifically refers to data-driven techniques based on artificial neural networks for the sake of brevity.

This dissertation also identifies the strengths and weaknesses of neural techniques in musical device modeling tasks and assesses their accuracy when models are subject to the same constraints applied in designing playable digital musical devices.

The overarching research question is

How can neural modeling techniques be further extended to emulate complex musical devices while retaining the characteristics necessary for a playable emulation?

Neural network-based models have been utilized to emulate audio effects and acoustic instruments, yet the state-of-the-art still faces significant limitations; only a few types of devices can be accurately modeled and used for real musical performances. Emulating complex systems, such as nonlinear and time-variant devices, remains a challenge, particularly when aiming for emulations suitable for real-time musical interactivity. Such models must respond quickly to various inputs, including audio signals, control parameters, or external stimuli. Furthermore, the model should enable smooth variations in these inputs,

generating audio signals that are perceived as continuous and coherent, thus emulating the behavior of their physical counterparts.

This led to three research questions that this dissertation aims to answer.

RQ1: How can we design neural architecture that can be trained to accurately emulate time-variant and nonlinear musical devices while satisfying the latency and computational requirements of digital musical devices?

Neural networks have proven to be effective for achieving perceptually accurate emulation of analog distortion circuits, offering an efficient alternative to white box modeling approaches that require the explicit solution of nonlinear equations. In this context, 'efficiency' means low computational complexity, meeting the demands of real-time audio processing. This is particularly challenging for complex tasks, such as modeling nonlinear and time-varying behavior intended to run on consumer-grade platforms, where achieving real-time performance is not always straightforward.

Distortion effects present relatively short-time dependencies between input and output signals, to the extent that allows memoryless sample-to-sample modeling using neural networks, as they present strong time-invariant nonlinearity, and there is no need to track past any long-term dependency between the input determining the output. This does not apply to all musical devices, whether they are analog or acoustic devices, that process or generate sound. When devices exhibit complex nonlinearities alongside intricate time behaviors and long temporal dependencies between inputs and outputs or exhibit time-variant behaviors, most neural network configurations struggle to achieve the necessary accuracy for online and causal computation. This issue has generally been addressed by designing networks that process large blocks of audio at once, providing sufficient 'temporal context' for the network to predict the output accurately. However, a major drawback of this approach is the introduction of excessive latency between inputs and output and, at times, a noticeable degradation of temporal resolution in response to parametric changes, both of which are detrimental to the use of these models in live performances. For these complex devices, discovering networks that are computationally efficient and with minimal latency that can accurately model such systems remains an open challenge. Although offline solutions can still be useful in specific contexts, a digital emulation that is playable in the same way as the actual physical device could have a significant impact, potentially enabling novel and compelling applications. This challenge is not meant to be solved solely by investigating network topologies and machine-learning algorithms. The development of associated datasets, data pre-processing, and training strategies are equally fundamental aspects in achieving high modeling fidelity. These are a complementary part of the proposals outlined in this thesis.

As discussed at the start of this chapter, musical devices typically feature a range of inputs, some of which significantly influence the device's behavior or response. Consequently, the second research question specifically investigates this aspect.

RQ2: How can neural models of musical devices be conditioned to adjust their responses according to inputs such as control parameters and stimuli, thereby enabling interactive user control?

In musical devices, both *processors* and *generators*, parameters and external stimuli can strongly influence the audio alteration and generation processes. Capturing all these diverse behaviors in a single model is crucial for complete emulation. This requires the neural network to learn multiple representations of sound processing or generation while also being capable of continuously morphing these processes in response to external inputs. The network must do so with sufficiently high temporal and numerical resolutions. This layer of network conditioning must maintain low latency and add minimal computational overhead to allow for musical interactivity, and it should not introduce audible artifacts. The neural modeling technique needs to accommodate various types and quantities of inputs, stimuli, and control parameters, enabling all the musical practices that are feasible with their physical counterparts. The development of appropriate datasets and training strategies is also crucial here. In particular, constructing datasets that adequately represent the parametric behaviors of the device without becoming excessively large presents a challenge, especially when dealing with high-dimensional input spaces.

The third question focuses on the generalization aspects of neural modeling techniques and any novel opportunities they may present.

RQ3: To what extent are the conceived neural architectures, or their internal components, sufficiently generic to model behaviors within or across different types of musical devices, and what openings do they offer for designing musical devices with novel creative features?

The two categories of musical devices considered in this thesis, *processors* and *generators*, exhibit significant differences, necessitating different neural modeling techniques. It is highly unlikely that the exact same methodology can be used for modeling both parametric audio-to-audio and parametric symbolic music-to-audio devices while still complying with the earlier explained requirements for efficiency, latency, and interactivity.

On the one hand, it is essential to investigate the extent to which the data-driven approaches proposed in this thesis can generalize to model other devices, such as those of the same type or within the same category. Moreover, incompatibility at the model level does not necessarily preclude the compatibility of sub-modules within the proposed architectures for various modeling tasks or for capturing common aspects shared by both *processors* and *generators*. Indeed, the overarching long-term objective, which extends beyond the scope of this thesis, is to establish a versatile, general-purpose architecture capable of adapting to various musical devices with minimal known modifications. Finally, given that neural networks are essentially universal interpolators, we explore how the developed neural-based solutions can enhance model capabilities with creative features not present in the physical counterparts. This innovation is pursued by experimenting with architectural designs and training frameworks, potentially boosting their sonic palette.

1.4 Contributions

The research questions are explored along two main avenues: dynamic range compressors as a significant open modeling challenge for *processors*, and pianos as a comparable challenge for *generators*. The investigation leading to the development of novel architectures and training approaches involved creating datasets through recording relevant devices, training and fine-tuning the proposed architectures, and assessing the models' outputs using both quantitative and qualitative methods. The findings of this dissertation contribute to advancing knowledge and practices in data-driven modeling of musical devices, particularly those with parametric, nonlinear, and time-variant characteristics. The contributions are summarized briefly here.

Deep learning techniques, including specialized architectures and training approaches, to model nonlinear and time-variant musical devices that present low computational complexity and minimal latency.

Within the scope of this research work, novel deep-learning architectures have been designed, developed, and evaluated to emulate Dynamic Range Compressors (DRCs), piano instruments, and other types of musical devices. Different architectures are conceived for real-time and interactive implementations. During this exploration, we gained knowledge on data-driven approaches on neural networks for modeling nonlinear and time-variant behavior in analog *processors* and using neural networks with physics-based equations for modeling nonlinear and time-variant behavior in acoustic *generators*. This knowledge can be applied, along with necessary extensions, to other modeling tasks involving parametric audio-to-audio and parametric symbolic music-to-audio contexts.

Approaches to condition deep learning models of musical devices to learn and replicate the different behaviors a musical device exhibits when changing input parameters and/or stimuli.

Conditioning techniques are designed and evaluated to infuse controllable input information into the network inference process. This conditioning enables a continuous change in the device's behavior in a manner identical to the inputs found on the physical device without generating audible artifacts. Models are conditioned by effect parameters for *processors* and by various input stimuli for acoustic *generators*, respectively.

Knowledge on the use of hyper-parameter and loss function for training parametric audio-to-audio and parametric symbolic music-to-audio neural networks.

The experiments aimed at designing novel neural architectures have been supported by comprehensive studies on loss functions and hyperparameters, which are critical for efficiently training the models. These are crucial and nontrivial,

given the myriad options available. Decisions are based on a robust understanding of the modeling task and empirical evidence, which can be applied to similar or equivalent neural modeling tasks.

Open source code for all proposed neural architectures and training approaches, public datasets, and associated trained models for a variety of musical devices, collected from hardware and software musical devices.

To conduct the extensive series of experiments and investigations required for this thesis, a multitude of datasets has been developed, along with implementations of both the proposed models and the baselines for comparison. Sharing these resources will facilitate further developments related to the contributions of this thesis, as well as their application in other contexts. Following open science principles, we made our research replicable and reusable by openly and publicly sharing the source code, trained models, and datasets. These resources are available at <https://github.com/RiccardoVib>.

1.5 Thesis Outline

The theoretical background supporting the work in this thesis, along with a review of relevant literature, is presented in Chapter 2. This chapter details traditional and state-of-the-art approaches for modeling various types of musical *generators* and *processors*. The review focuses on the key characteristics of these musical devices and the associated modeling challenges, discussing recent advancements and their limitations. Chapter 3 presents the research methodology employed in this work. The chapter begins by reviewing different types of artificial neural networks used as fundamental building blocks for the neural architectures proposed in this thesis. It then divides into two parts, one addressing *processors* and the other *generators*. Each part starts with a detailed description of the specific device selected as a representative target for this study, followed by descriptions of the developed datasets, proposed architectures, strategies for network training, experimental setups, and evaluation metrics. Chapter 4 summarizes the papers included in this article-based thesis, addressing the relevance of each to the thesis’s objectives and their key findings, followed by a collective discussion of the research contributions, outcomes, and limitations. The thesis concludes with Chapter 5, which discusses the findings in relation to the research questions outlined earlier and suggests directions for future research. Figure 1.1 illustrates the thesis organization.

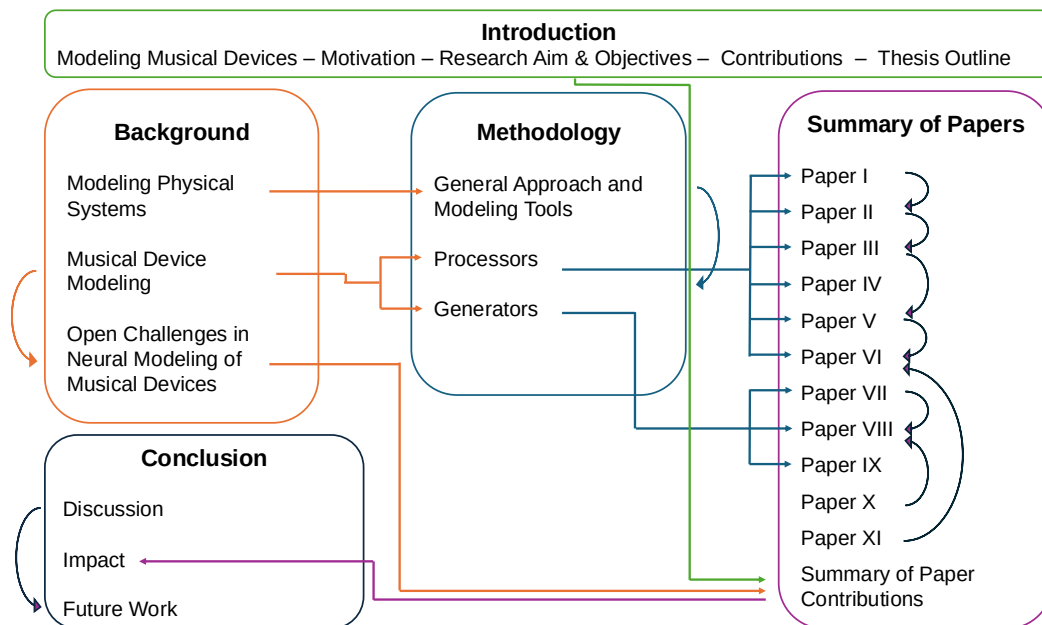


Figure 1.1: Chapter plan for this dissertation illustrating the pathways between sections.

Chapter 2

Background

This chapter provides the theoretical foundation for this thesis by systematically expanding upon the concepts introduced in the previous chapter with further mathematical formalism. It also includes a comprehensive review of existing approaches and state-of-the-art techniques in the data-driven modeling of musical devices. Section 2.1 introduces the musical devices as physical systems, discussing their nonlinear and time-variant characteristics. It subsequently provides an overview of the main techniques used to model generic physical systems, with a particular focus on machine learning. Section 2.2 reviews the state-of-the-art in modeling of audio effects, acoustic musical instruments, and analog sound synthesizers. The final part of the chapter, presented in Section 2.3, provides an analysis of the key functional requirements for emulating musical devices. It compares these requirements with the features offered by state-of-the-art machine learning emulation techniques, highlighting current limitations and open challenges.

2.1 Modeling Physical Systems

Musical devices, whether acoustic or analog electronic devices that generate or process sound, are dynamic physical systems (Fletcher and Rossing, 2012) producing time-dependent signals in the form of mechanical waves or electrical voltages. As such, their behaviors can be described by a system of equations that govern the underlying physical phenomena, often characterized by differential equations involving time derivatives. Real-world physical systems typically present *nonlinear* and *time-variant* behavior. Nonlinear dynamical systems are characterized by a system of nonlinear equations in which the unknown functions are involved in operations that lead to polynomials of a degree higher than one. In other words, the equations to be solved in a nonlinear system of equations cannot be written as a linear combination of the unknown functions that appear in them.

We can classify two types of nonlinear relationships. If the nonlinear relationship is related through a nonlinear differential equation, it is called a *dynamic* nonlinearity, while if it is between input and output that does not involve a differential equation but equations where the unknown appears as variables of a polynomial of a degree higher than 1 it is called a *static* nonlinearity.

Nonlinear, time-variant systems are commonly encountered in the real world. Modeling them with linear, time-invariant approximations is widespread in many application domains due to advantages such as predictability, computational efficiency, and the availability of better analytical tools. However, in the context of sound and music, where sonic aesthetics are paramount, this approach can be detrimental. Adhering too strictly to linear models may result in the loss of particular timbral characteristics and sonic nuances that are unique signatures of acoustic or analog musical devices. Modeling nonlinearities in the physical system, especially in this context, is crucial and of great concern for a proper emulation of the systems' functioning. Nonlinear model design is an active area of research and features prominently in many engineering problems (Isidori, 1985; Hauser, 2009). For example, nonlinear friction, stiffness, damping, saturation, hysteresis, and nonlinear interconnections can be found in mechanical or electronic systems.

As mentioned above, linearization is a popular initial approach to modeling physical systems, simplifying analysis and design, and enabling computationally efficient models. A linear system satisfies the following:

$$ay(t) + by(t) = H\{ax(t) + bx(t)\}. \quad (2.1)$$

where $x(t)$ and $y(t)$ are, respectively, the input and output, while a and b are constant factors. $H(\cdot)$ is the function describing the system. Linear systems benefit from two crucial properties: *homogeneity* and *additivity*. *Homogeneity* means that a change in the input signal's amplitude results in a corresponding change in the output signal's amplitude. While *additivity* or *superposition principle* means that the response caused by the sum of two or more signals simultaneously is the sum of the responses that would have been caused by each stimulus individually. Two significant consequences of these two properties are that a linear system cannot transfer energy from one frequency to another and that the output in the linear case is confined to a hyperplane in the N -dimensional space since a linear relationship between the lagged inputs and outputs can describe a linear dynamic system (Johan and Ljung, 2019).

Nonlinear systems, which do not satisfy the principles of linearity mentioned above, can considerably increase the difficulty associated with managing their descriptive equations. They can transfer energy from one frequency to another, and their output can be represented as a complex manifold in a high-dimensional space, making their characterization difficult. A linear hyperplane can be defined by only a few points, but for a complex manifold, it is challenging to predict the behavior outside the range where the function is sampled. Therefore, although linearization can aid in modeling nonlinear systems, the resulting model may show significant deviations from the actual physical system. Generally, linear simplifications involve neglecting aspects that are not considered fundamental to the specific context or application of the model so that the emulation is not excessively inaccurate. A trade-off between accuracy and complexity is invariably necessary, balancing the precision of the model with the growing number of terms required for a comprehensive mathematical description of nonlinear behavior.

Physical systems, in addition to exhibiting nonlinear characteristics, may also display complex temporal behaviors or responses that change over time. For time-

variant systems, the output response depends not only on the moment of input application but also on the moment of observation. This means that delaying the input does not simply shift the output in time, but it also alters other parameters and behaviors. Conversely, if a system's input and output characteristics remain unchanged over time, the system is considered time-invariant. Time-invariance implies that if $y(t) = H[x(t)]$, where $x(t)$ represents the input, $y(t)$ the output, and $H(\cdot)$ the mapping between them, then it must also be true that for any time shift τ :

$$y(t - \tau) = H[x(t - \tau)] \quad (2.2)$$

However, in the real world, systems are typically time-variant, which can be expressed as $y(t) = H[x(t), t]$. This time-dependence is often due to factors such as aging and decay of materials, as well as absolute or cyclic changes in environmental conditions. In many instances, these temporal variations are sufficiently minor or can be reasonably disregarded, which simplifies the system analysis by avoiding the introduction of unnecessary complexity.

Nevertheless, time-invariant systems can still manifest timing behaviors that are complex to model, often due to deliberate design choices or specific operational requirements. For example, a system's internal parameters, which determine its output responses, might change in response to the current or past inputs. These parameters might also undergo internal modulation over time or be linked to variable time constants. Consequently, the temporal dependencies between input and output can become complex and lengthy. This research primarily focuses on systems that exhibit such time-variant responses.

Modeling a physical system, particularly when it exhibits nonlinearities and time-varying characteristics, can be a complex endeavor. This dissertation addresses the modeling of musical devices that, despite being time-invariant systems, exhibit responses that significantly vary over time. Within the scope of this thesis, musical devices that display these properties are considered as nonlinear and time-variant.

There are two possible approaches for approximating a physical continuous-time system with discrete-time models, known as *white box* and *black box* approaches. As the names suggest, they exhibit stark differences in their approach to modeling. The main associated techniques are reviewed in the next two subsections. In a nutshell, the *white box* approach involves using physical knowledge to explicitly model the system, deriving a continuous-time description before converting it into discrete-time form. Conversely, the *black box* approach relies on measurements from the physical device to identify a function that approximates the observed behavior; it directly produces a discrete model from data by analyzing input/output observations and establishing a suitable mapping between them. These two diametric approaches are also known as physics-based and data-driven approaches, respectively. The latter is sometimes called model identification. A third approach, the *gray box* model, represents a hybrid that combines aspects of both *white* and *black box* approaches—for example, incorporating data in areas where physical knowledge is lacking or overly complex, and vice versa.

2.1.1 White Box Modeling

The *white box* approaches use the system’s knowledge to construct the discrete-time model. Traditional approaches involve deriving the equation governing the system’s functioning. In the case of a real-world system, it means to derive its Partial Differential Equations (PDEs) and subsequently discretize them to obtain the numerical solution of the implicit or explicit discrete nonlinear equations. Discretizing the continuous partial differential equations in time and space may lead to problems such as stability and aliasing.

The discretization method, as well as the sampling frequency, are directly related to the stability of the system. In the continuous world, we can assume arbitrarily short (infinitesimal) delays between spatial points of interest, and the order of events follows the causality principle of physics. In the discrete world, however, a single sample period is the shortest possible non-zero time interval in which explicit physical interaction can happen. This can lead to the problem of *delay-free* loops, i.e., implicit equations where the output of an operation needs an input value that may be dependent on the output value not yet known. Common discretization methods are the central-difference approximations, bilinear transform, the *trapezoidal method* and *implicit midpoint method* (Germain, 2017).

On the other hand, the *Aliasing* problem emerges where the bandwidth of the continuous system is wider than half of the sampling rate value, and it introduces additional frequency components in the discrete-time response. Since no physical system is truly bandlimited, aliasing will always occur unless bandlimited signals driving the system can be guaranteed. In the case of simulated nonlinearities, *aliasing* is a commonly encountered problem in audio signal processing due to harmonic distortion, which is not bandlimited. *Aliasing* reduction is an active research topic, and several methods have been proposed to overcome this problem. A common method is oversampling: using a digital lowpass filter, the input signal is upsampled to suppress the spectral components generated by harmonic distortion. Then, the output signal is downsampled back to the original rate. The amount of *aliasing* reduction will then depend on the oversampling factor and order of the antialiasing lowpass filter. Together, these two components determine the computational cost. A newer technique, applied for clipper systems and more abstract applications of nonlinear waveshaping, approximates the underlying continuous-time input signal with a piecewise linear function, applies a nonlinear function to it, and convolves the resulting signal with the continuous-time impulse response of a low-pass filter. This approach is called *Bandlimited Correction Functions* (Esqueda, Välimäki and Bilbao, 2015; Esqueda, Bilbao and Välimäki, 2016; Parker, Zavalishin and Le Bivic, 2016). A second and later approach is called *Antiderivative Antialiasing* (Bilbao, Esqueda et al., 2017). The approach is based on discrete differentiation of higher-order antiderivatives. First considered for a memoryless system, i.e., a system whose output depends only on the input at the current time-step, the antiderivative-based reduction has been explored for systems depending on previous information as well (Holters, 2020; Albertini, Bernardini and Sarti, 2020; Werner, Burlington and Azelborn, 2023).

Different paradigms can be considered in the context of white box modeling, including digital waveguides, wave digital filters, state-space, finite time domain

difference, modal, and port-Hamiltonian formulations. All reviewed methods are suitable for modelling both *generators* and *processors*.

Digital Waveguides & Wave Digital Filters

Digital waveguide (DWG) modeling is closely related to the lossless d'Alembert solution of the wave equation (Smith III, 2010). The idea is to simulate wave propagation in a medium by two delay lines, one for each directional wave component. A digital waveguide is thus defined as a bidirectional delay line with a specific wave impedance, denoted as R , where each delay line carries a sampled acoustic traveling wave. In this way, a delay block can, for example, represent pressure in an acoustic tube or transverse displacement in a vibrating string (Smith III, 1992). Lowpass and all-pass filters are usually cascaded with delay lines to simulate damping and dispersion.

The time evolution of a system's state is modeled as waves propagating energy. Waves are reflected or transmitted and propagated throughout a grid of scattering junctions. These junctions serve to simulate the behavior at an impedance discontinuity. This method is efficient and numerically robust because the stability and nonlinear interactions can be modeled using scattering junctions. The scattering junctions can be realized by a look-up table or using an analytical expression. On the other hand, this method does not easily allow the modeling of intrinsic nonlinearities.

While DWGs could be better suitable for distributed models (the system is defined over some region in space), wave digital filters (WDF) are for lumped elements. The WDF formalism represents the system using wave-variables (incident and reflected waves), and has been mostly used for the simulation of analog circuits (Dias de Paiva, 2013; D'Angelo, 2014). WDF is based on the following notation:

$$\begin{cases} a = V + RI \\ b = V - RI \end{cases} \quad (2.3)$$

where a is in-coming and b out-going wave in a port, V is voltage, I is current and R is port resistance. Using this paradigm, delay-free loops are avoided by the structure of the components and by the use of impedance-matched reflection-free ports (Fettweis, 1986). The WDF method supports combinations of parallel and series connections, and starting with including linear elements, WD was expanded to handle nonlinear elements, multiple/multiport nonlinearities, and arbitrary network topologies (Werner, Nangia et al., 2015; Rest, Dunkel et al., 2016; Olsen, Werner and Smith III, 2016; Werner, Olsen et al., 2017). Recent advances have facilitated the development of multidimensional simulations that incorporate multiple nonlinearities while addressing convergence speed limitations (Kolonko, Velten and Kummert, 2021). In (Bernardini, Bozzo et al., 2021), the authors demonstrated that quadratic convergence is heavily dependent on the choice of free parameters and proposed a novel formulation using WD formalism.

Finite Difference Time Domain

Finite Difference Time Domain (FDTD) approximation is another popular method of numerical integration of partial differential equations (Strikwerda, 2004). The time-dependent equations are discretized to the space and time partial derivatives. This method could be very flexible, but it suffers from ‘sporadic’ or ‘spurious’ oscillations that can appear in the FDTD structure (Sengupta et al., 2012). When using these schemes, the discretization error must be considered. The aliasing effect dominates the error due to the sampling of the input, output, and internal signals. The error is proportional to the step sizes, resulting in growing rapidly with f/f_s . Lastly, FDTD schemes are very flexible when considering nonlinear systems but can lead to delay-free loop problems, where variables can depend implicitly on themselves, and to high computation.

State-space models

A dynamical system can be described as a collection of mathematical expressions that relate input and output signals and variables characterizing the system’s behavior (Hendricks, Jannerup and Sørensen, 2008). Therefore, a collection of equations involving y , u , and x , the output signal(s), the input signal(s), and auxiliary signals, respectively, can be formulated. This formulation is the foundation of an Ordinary Differential Equation (ODE)-solving method called the State-Space (SS), where the systems’ equations can be written as:

$$\dot{x}(t) = f(x(t), u(t)) \quad (2.4)$$

$$y(t) = h(x(t), u(t)) \quad (2.5)$$

These expressions represent the nonlinear state-space description for the relationship from u to y . In the discrete domain, the system is formulated in a matrix form, creating a compact way of representing all the differential equations required to model the system behavior:

$$x(n+1) = Ax(n) + Bu(n) + F(x(n), u(n)) \quad (2.6)$$

$$y(n) = Cx(n) + Du(n) + G(x(n), u(n)) \quad (2.7)$$

where $y(n)$ is a function of past values of $u(n)$, and the nonlinear term $F(x(n), u(n))$ and $G(x(n), u(n))$ are multivariate nonlinear functions, that can be rewritten as a linear combination of nonlinear basis functions. State-space models use state variables to describe a system by a set of first-order differential or difference equations rather than by one or more n th-order differential or difference equations. Similar to an FDTD scheme, the nonlinear system described in this form cannot always be explicitly solved when discretized, suffering from the delay-free loop problem as well. In order to solve this, the *implicit function theorem* permits the solution of delay-free paths in the general nonlinear case. This approach is also named K-Method (Borin, De Poli and Rocchesso, 2000).

Modal Decomposition Methods

Contrary to the aforementioned methods, the modal decomposition method relies on frequency-domain formulations of systems (Van Den Doel and Pai, 2007; Cerioli, Ducceschi and Serafin, 2021). This method is based on the principle that any sound can be decomposed into a set of acoustic modes. Each mode represents a specific way in which the sound source vibrates and resonates at a particular frequency. Therefore, the behavior of a system is decomposed into decaying exponentials, whereby oscillatory components represent the eigenmodes of the system. The resulting model is a bank of damped harmonic oscillators excited by an external stimulus. The behavior of each mode is described by a scalar second-order ordinary differential equation that discretized gives the numerical solution:

$$M_d \frac{d^2 x_d(t)}{dt^2} + S_d \frac{dx_d(t)}{dt} + G_d x_d(t) = f(t) \quad (2.8)$$

where M_d , S_d and G_d are respectively the mass, elasticity, and damping of the d^{th} mode, and $f(t)$ is the forcing function that excites the system. The solution of the equation is a linear combination of the modes:

$$x_d(t) = a_d(t) \sin(2\pi\omega_d t + \phi_d), \quad (2.9)$$

with time-varying amplitude $a_d(t)$, frequency ω_d and initial phase ϕ_d . These modes are typically represented by sets of filters tuned to each mode's resonant frequency excited by an input source. The frequencies and dampings of the oscillators are determined by the geometry and material properties of the object, and the coupling gains are determined by the location of the force applied to the object. As FDTD, the method is highly flexible when dealing with nonlinearity, but a system is typically described by a large number of modes that are required to compute, leading to high computational costs.

Port-Hamiltonian

In port-Hamiltonian systems, the geometric structure derives from the interconnection of its sub-systems (Van Der Schaft and Jeltsema, 2014). The method uses the state-space description of power exchange between energy-storing parts and offers a systematic method for lumped and distributed models. It considers the physical system as an interconnection of ideal basic elements linked by energy flow. Therefore, the port-Hamiltonian modeling framework allows for models that preserve essential physical properties such as energy conservation or dissipative inequalities (Falaize and Hélie, 2016). Doing so ensures the stability of the obtained scheme, a critical aspect, especially for nonlinear cases where numerical stability is more challenging to obtain. If all subsystems are modeled as port-Hamiltonian systems and the inputs are related to the output linearly, the overall system can be written as a port-Hamiltonian system, which preserves the properties of the

underlying subsystems. The standard Hamiltonian equations are the following:

$$\dot{q} = \frac{\partial H}{\partial p}(q, p) \quad (2.10)$$

$$\dot{p} = -\frac{\partial H}{\partial q}(q, p) + F \quad (2.11)$$

where the Hamiltonian $H(q, p)$ is the total energy of the system, $q = (q_1, \dots, q_k)^T$ are generalized configuration coordinates for the mechanical system with k degrees of freedom, $p = (p_1, \dots, p_k)^T$ is the vector of generalized momenta, and the input F is the vector of external generalized forces. The state space of Eq. (2.10) with local coordinates (q, p) is referred to as the phase space.

2.1.2 Black Box Modeling

As discussed earlier in this chapter, explicitly deriving a model necessitates making certain assumptions and simplifications. Consequently, of the physics that we can observe and understand, only a fraction can be captured by models, and an even smaller portion can be accurately simulated numerically. In contrast, *black box* modeling approaches allow us to derive models from large sets of measurement data collected from the physical system, encompassing the entire input/output space. This enables black box models to deliver accurate predictions, even without a detailed understanding of the system's underlying principles. Black box models simplify the projection of inputs onto latent variables, often resulting in reduced computational complexity to produce the system's output. These models treat the system as a mapping between input and output data, striving to identify a function that closely approximates this relationship based on measured data. Consequently, the black box approach focuses on observing the correlation between input and output data to construct the mathematical model. It is characterized by four main components: the data, the set of candidate models, the estimation method, and the validation procedure.

Black box methods are employed when knowledge of the system's internal mechanisms is unavailable or when there is insufficient understanding of the underlying physics. In such cases, deriving a general mapping that approximates system behavior is desirable. The most common methods for black box modeling include the Hammerstein and Wiener Models, Volterra series, Gaussian Processes, and Artificial Neural Networks, which are introduced in the following subsections and can be considered suitable for modeling both *generators* and *processors*.

Hammerstein and Wiener Models

Popular examples of data-driven approaches are the block-oriented Hammerstein and Wiener models and their combinations (Schoukens and Tiels, 2017). In the Wiener model, a linear model is followed by a static nonlinearity, while the Hammerstein model is the opposite. These models provide a convenient representation, and they are derived by estimating the model coefficients of linear and nonlinear functions from real input-output measurements. Common

parametrizations for linear blocks are Rational Transfer Functions, Finite Impulse Response models, and Linear State Space models, while nonlinear ones are the Chebyshev Polynomials, B-splines functions, and Wavelets basis functions. The estimation of the coefficient is pursued with some error minimization techniques. An iterative search algorithm minimizes the chosen loss function. Recent work proposes to classify whether the nonlinear system should rather be modeled by a Wiener or a Hammerstein model in (Primavera et al., 2018).

Volterra Models

A Volterra Model (Roze and Hélie, 2014) is a representation of the system using multidimensional impulse responses called Volterra Kernels. The model has the following form:

$$\hat{y}(t) = \sum_{n=1}^{+\infty} y_n(t) \quad (2.12)$$

$$= \int_{-\infty}^{+\infty} h_1(t, \tau_1) x(\tau_1) d\tau_1 \quad (2.13)$$

$$+ \int \int_{-\infty}^{+\infty} h_2(t, \tau_1, \tau_2) x(\tau_1) x(\tau_2) d\tau_1 d\tau_2 \quad (2.14)$$

$$+ \dots \quad (2.15)$$

where the $h_n(t, \tau_1, \tau_2, \dots, \tau_n)$ terms are the kernels. $y_1(t)$ represents the linear part of the system, the impulse response. Volterra kernels can be identified from a relatively short data set, and the choice of the maximum degree of the kernels is critical and involves a trade-off between model complexity and model error.

Volterra models are non-parametric and require computing the system's impulse responses. The number of Volterra Kernels depends on the strength of the nonlinear characteristics of the systems. The method's main disadvantage is that kernels must be stored in memory, and to compute the output, the input must be convolved with all the kernels. Therefore, this approach is device-specific because it relies on the measurement of the kernels. The number of terms $O(m^M)$ increases the number of considered past times ('memory'). For this reason, it can be applied to a problem with short memory m and moderate polynomial of order M . The most common method to determine the nonlinear response of audio systems is the swept-sine technique (Farina, 2000; Farina, Bellini and Armelloni, 2001). The swept-sine method starts by injecting a test signal having an instantaneous frequency given by a logarithmically increasing sweep:

$$f_i(t) = \sin \left[\frac{f_1 T}{\ln \left(\frac{f_2}{f_1} \right)} e^{\frac{t}{T} \ln \left(\frac{f_2}{f_1} \right)} - 1 \right] \quad (2.16)$$

where f_1, f_2 are the initial and end frequency, and T the signal length. Since the instantaneous frequency of a harmonic frequency of $f_i(t)$ is given by a constant time shift, if the output signal of the system is filtered by a signal that inverts the group delay response of the excitation signal, the impulse response of each harmonic will

be separated in time at time instants. In this way, the impulse responses of each individual harmonic can be obtained. The inverse filter is either obtained by time-inverting the instantaneous frequency response with energy compensation for the high frequencies or by inverting the excitation signal in the frequency domain.

Gaussian Processes

Gaussian Processes are a generalization of the Gaussian distribution to infinite dimensions. They can be interpreted as a way to specify a probability distribution over functions and offer a non-parametric approach to model the map $y = H(x)$, where y is the output, x is the input and $H(\cdot)$ the mapping function. The fundamental concept of this technique is that the function to be estimated is embedded within a stochastic process. Specifically, this involves a distribution with zero mean and large variance, accompanied by a covariance function that characterizes the smoothness of H . Then, when $y_k = H(x_k), k = 1, \dots, N$ have been measured, the posterior distribution $H^p(x|y)$ can be formed for any x . The mean of the function estimates the function H and is formed by interpolation and extrapolation $[y_k, x_k]$ using the probabilistic relationships.

Artificial Neural Networks

An Artificial Neural Network (ANN) (Hopfield, 1988), conceptualized in 1943 with the perceptron (McCulloch and Pitts, 1943), is a supervised Machine Learning (ML) algorithm inspired by the human brain, mimicking the way biological neurons signal to one another. It is an intricate network of interconnected nodes or neurons consisting of a certain number of layers. The model's internal parameters are tuned during the learning process through an optimization process. When ANNs employ multiple layers to model complex relationships between inputs and outputs, these are known as deep networks, and the specific type of machine learning is referred to as Deep Learning (DL). ANNs play a central role in modern Artificial Intelligence (AI), which encompasses the broader concept of machines performing tasks that typically require human intelligence, such as speech recognition, image classification, decision-making, and natural language processing. ML algorithms, more specifically ANNs, have enabled the development of AI systems that learn directly from data without needing explicit programming.

ANN consists of basic units called neurons, which are organized into layers. Figure 2.1 illustrates an example of a neuron. An artificial neuron has one or more inputs x_i , each associated with a corresponding weight w_i , a bias b , an activation function $\phi(\cdot)$, and output y . The output is the weighted sum of input and bias, which is then processed by the generally nonlinear activation function:

$$y = \phi\left(\sum_{i=1}^N w_i x_i + b\right) \quad (2.17)$$

where N denotes the number of inputs, and i indexes the i -th input of the neuron. In the most basic network configuration, the output of one layer is fed forward to the next from the input to the output layer. Layers in between are called

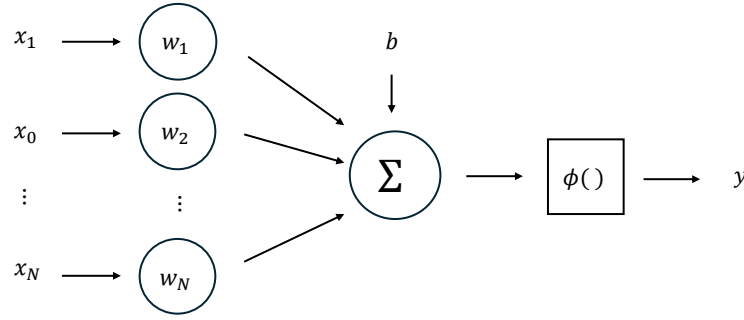


Figure 2.1: Diagram of a neuron, which is described by the Equation (2.1.2).

hidden layers. If the output of any individual node is above the specified threshold value, determined by the bias of each neuron, that node is activated, enabling the connection to the next layer of the network. In fully connected networks, layers are highly interconnected: each neuron in layer l is connected to all neurons in layers $l - 1$ and $l + 1$. Artificial neural networks are adaptive; training algorithms allow them to modify their internal coefficients as they see input-output examples. Similarly to block-oriented models, during training, an error is calculated by measuring the difference between the output of the network and the expected value. The error is computed using a loss function, determining the aspect the networks optimize during training. The error term is back-propagated through the connecting layers to update each neuron's associated weights, searching for the values that minimize the error. Therefore, the loss influences the direction and the magnitude of the weights' modification inside a high-dimensional space. The training process involves an optimization task that searches for the global minimum of the loss function. However, reaching the global minimum is only guaranteed if the function being modeled is convex—a condition that is not commonly met. Otherwise, the optimization process will likely converge to a local minimum. The back-propagation algorithm leverages gradient information to move towards a minimum of the loss function; this specific approach is known as gradient descent. Nevertheless, alternative methods exist, such as fuzzy logic, genetic algorithms, or Bayesian optimization.

With this learning process, ANNs are particularly well-suited for modeling tasks, as all steps of the learning process occur simultaneously and automatically within the end-to-end input/output processing chain. On the other hand, tuning the model's parameters to achieve negligible residual prediction errors can be challenging. Furthermore, identifying computationally efficient models is not always straightforward.

Generative Frameworks

Within the field of AI, so-called Generative models enable the creation of new content based on a variety of inputs. The most popular and effective among these are Generative Adversarial Networks (GANs), Diffusion models, Variational Autoencoders (VAEs), and Flow-based models.

GANs (Goodfellow et al., 2014) are a type of machine learning algorithm in

which two neural networks engage in a zero-sum game framework - one network's gain is the other's loss. The GAN framework is designed to generate new data that mimic the statistical properties of the training dataset. It consists of a discriminator, a neural network that determines whether the samples are from the original dataset, and a generator, which is trained to produce samples that are indistinguishable from authentic data, thus 'fooling' the discriminator. The two models compete against each other, forming an adversarial relationship that underpins GANs (Donahue, McAuley and Puckette, 2019; Engel, Agrawal et al., 2019; Bitton et al., 2019).

VAEs (Kingma and Welling, 2014) employ an encoder-decoder network architecture. The encoder transforms each data point (e.g., an image or a sound) from a complex dataset into a set of parameters defining a probabilistic latent space, often modeled as a multivariate Gaussian distribution. This process yields a variational distribution within the latent space instead of a single fixed point. Conversely, the decoder reconstructs the input data from the encoded latent representation, sampled from the probabilistic distribution. In generative contexts, the trained decoder is utilized independently to generate new, high-dimensional data by sampling from the coordinates in the lower-dimensional latent space.

Finally, Flow-based models (Jakub, 2021) explicitly model a probability distribution by leveraging normalizing flow, which uses the change-of-variable law of probability to transform a simple distribution into a more complex one. These models employ a sequence of invertible transformations to generate the samples and learn the probability distribution of the data. A normalizing flow can be framed as an autoregressive model, in which each sample's generation is conditioned on the ones previously generated, similar to the approach used in one of the earliest examples of such an approach, WaveNet (Oord et al., 2016).

Diffusion Models (Ho, Jain and Abbeel, 2020) operate by gradually corrupting training data through the successive addition of Gaussian noise and then learning to reconstruct the data by reversing this noising process. After training, the model generates data by subjecting randomly sampled noise through the learned denoising process.

Models based on the aforementioned frameworks, with the exception of autoregressive flow-based one, produce an entire example in a single step (e.g., an image or a sound). Moreover, there is no link or relationship of any kind between consecutively generated examples. This aspect restricts their use to scenarios where the independent generation of fixed-dimension examples is necessary, in stark contrast to musical devices' requirements.

2.1.3 Gray Box Modeling

Hybrid approaches, also known as *gray box* models, blend white and black box modeling techniques to combine the advantages provided by both. Typically, they involve dividing the problem into sub-components: the understood and mathematically solvable aspects are modeled using *white box* techniques, while the unknown or difficult-to-model efficient phenomena are addressed with *black box* methods.

In theory, *gray box* models can be created through various combinations of the previously detailed *white box* and *black box* methods. This results in a wide array of possibilities, most of which have been applied only in a few specific cases and do not conform to any widely recognized technique that has been formally named.

Differentiable Digital Signal Processing

A popular gray box technique involves using automatic gradient backpropagation to estimate audio parameters in traditional digital signal processing algorithms. This approach, named Differentiable Digital Signal Processing (DDSP) from (Engel, Hantrakul et al., 2020), integrates machine-learning techniques with digital signal processing to create models that consist of a chain of differentiable signal processing blocks. DDSP models demonstrated remarkable advancements, contributing to achieving computational efficiency and low-latency models.

Nevertheless, it is crucial to have and utilize specific knowledge about the modeling problem when designing a signal processing block capable, at least theoretically, of synthesizing the desired sound. The DDSP framework is typically employed to construct differentiable models from traditional synthesis methods, such as additive, subtractive, and FM synthesis, as well as algorithms for effects such as filters, reverb, and compression. DDSP can accommodate nonlinearities and time-variant behavior if these characteristics are initially encapsulated in the DSP algorithm definition. It is important to note that while this method cannot learn behavior beyond what the model's equations explicitly describe, it can optimize the parameters of these equations to resemble the desired behavior closely.

2.1.4 Comparison of Modeling Approaches

White box models are built on a deep insight into the system's internal behavior. Detailed physical descriptions are made at the level of the microscopic subsystems, and then they are linked together in a macroscopic model. Therefore, white modeling is utilized when comprehensive knowledge about the target system is available. This entails knowing the physics equation governing the system's functioning, how the sub-components are interconnected, and their specific values. In the context of analog electronic audio effects and sound synthesizers, it is essential to have access to the schematics of the internal circuits and the values of their electrical and electronic components. For acoustic musical instruments, detailed knowledge of the instrument's geometry and the physical properties of the materials used is required. This information is often not in the public domain, and a reverse engineering process, if possible, can be costly. Afterwards, a numerical scheme must be formulated, and the resulting discrete equations must be solved iteratively to emulate the device's sound. These models have the advantage of flexibility since their parameters can be varied with little additional effort, allowing them to represent a family of similar musical devices.

White box models achieve high accuracy when based on detailed physical descriptions. However, these approaches are specific to the target system, as different systems are governed by different equations. Furthermore, substantial

effort is required for the mathematical formulation of the problem, especially in determining the precise values of the necessary parameters for their equations. Accurate models include nonlinear and complex characteristics, typically leading to computationally demanding models. Lastly, having a solvable mathematical scheme is not trivial; in the case of an implicit ordinary differential equations system, the Newton-Raphson iterative method is commonly used to find the solutions that typically converge quadratically, although such convergence is not guaranteed in all cases.

Black box models offer an advantage when developing a physical model is prohibitively expensive or when the inner workings of the target system are unmeasurable or incomprehensible. Data-driven approaches rely solely on input-output measurement data. However, these methods require access to substantial and accurate data, which may not always be publicly available in the necessary form or volume. Collecting the required data can be time-consuming and may demand precise calibration to ensure the quality of the resulting model; erroneous data can compromise its validity. Data-driven approaches aim to replicate only the input-output relationships of a system and are calibrated directly from measurement data without regard to the internal mechanisms. Although each model is specific to the target system, the tools used in the modeling process can often be universally applied to other musical devices, regardless of their characteristics.

On the other hand, measurements provide a static snapshot of the target's behavior, which can make accurately emulating dynamic aspects more challenging. Data-driven methods also suffer from a lack of interpretability and flexibility, as the internal parameters remain obscure and difficult to adjust.

Gray box methods combine the previous approaches to bridge their respective gaps. They take advantage of data-driven techniques to model aspects that are not understood or not efficiently implementable while using numerical-derived schemes for easier sub-modules. Typically, a component that is modeled using linear schema is combined with data-driven model nonlinearity. Gray box models add flexibility and interpretability, benefiting from incorporating external knowledge or manual adjustment, given their comprehensible internal structure, unlike black box models. Since external knowledge is infused, the optimization problem is bound, requiring less data to achieve good accuracy.

Gray box methods consist of various techniques and may potentially inherit and combine the disadvantages of these techniques. Specifically, the model's ability to emulate components treated as black boxes can be hindered by the limited accuracy of the parts modeled as white boxes.

2.2 Musical Devices Modeling

This section examines the most recent advancements in modeling musical devices, particularly emphasizing data-driven techniques, which are central to this thesis. The review is organized into two subsections: the first addresses *processors*, while the second focuses on *generators*. Each subsection delves into various modeling tasks and techniques, with a special consideration of nonlinear and time-variant

characteristics. Additionally, the section includes comprehensive reviews of recent works on modeling analog dynamic range compressors and pianos, which are discussed in detail since they are the primary targets for modeling in this thesis.

2.2.1 Processors

White Box Models

Analog audio effects are the most common type of *processors* for which the development of a digital model has been an active field of research and development for more than two decades. Analog audio effects are electronic circuits designed to operate at audio frequencies, and among white box methods, Wave digital filters (WDF) techniques have been particularly popular for their modeling. This approach has been recently exploited for vacuum tube-based circuits as the preamplifier circuit of an amplifier (Dunkel et al., 2016) and cascaded amplifiers (J. Zhang and Smith III, 2018), where the simulation of the triodes are based on the Dempwolf triode model equations (Dempwolf and Zölzer, 2011). WDFs have also been used for an envelope filter guitar effect circuit consisting of operational transconductance amplifiers (OTAs) (Bogason and Werner, 2017) and a booster guitar pedal based on field-effect transistors (FETs) (Bogason, 2018). Bernardini, Werner et al. (2018) presented a ring modulator constituted of four diodes and two multi-winding transformers model based on WDF, and a state-space model for a Booster guitar pedal was introduced by Holmes and Van Walstijn (2016). Chowdhury (2019) proposed a Finite Difference Time Domain (FDTD) scheme to model an analog tape machine, where the Runge-Kutta 4th-order method (Yeh, 2009) is used to find an explicit solution for the equation governing the hysteresis loop of the magnetization recorded to tape. Werner and Abel (2016) simulate the sonic effect of the Hammond tonewheel organ and apply it to adapt saturating nonlinearities to mimic the memory-less pickup nonlinearities distortion of tone wheels. State-space representations constructed from the circuit diagram using circuit analysis and representing the circuit using Kirchoff-variables (voltage and current) are in (Macak, Schimmel and Holters, 2012; Holters and Zölzer, 2015; Holters and Zölzer, 2017). A port-Hamiltonian formulation is proposed by Najnudel et al. (2021), in which a nonlinear passive peaking EQ mode is proposed. The nonlinear RLC circuit is modeled using a combination of the port Hamiltonian formulation with kernel-based methods.

Black Box Models

Data-driven methods for distortion and amplification circuits have recently gained great popularity and achieved a high degree of fidelity. Parametric block-oriented models, using an iterative error minimization procedure (as the Levenberg-Marquardt minimization procedure), have been considered for distortion pedals for guitarists and amplifiers, specifically a Wiener–Hammerstein model for guitar distortion pedals (Eichas, Möller and Zölzer, 2015), a Hammerstein model for a guitar amplifier (Schuck Jr and Bodmann, 2016) and a Wiener model for a distortion circuit (Eichas and Zölzer, 2016a). Eichas, Möller and Zölzer (2017),

modeled guitar amplifiers with a cascade of block-oriented models using mapping functions based on polynomials and hyperbolic tangents. In these cases, the nonlinear block for amplification (pre- and post-gains) consists of a polynomial mapping function, while the distortion units consist of sigmoid/hyperbolic tangent functions. Ebers–Moll equation (Ebers and Moll, 1954) is used by Schuck Jr, Ferreira et al. (2019) to design a parametric nonlinear transfer function for Wiener–Hammerstein models. Köper and Holters (2020) modeled the guitar overdrive effect pedal based on transistors with the same approach but, in this case, using a derived model for Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFETs). The MOSFETs are modeled by analyzing and comparing the device’s measurement data. Darabundit, Wedelich et al. (2019) propose a heuristic search to determine the bias and scaling factor of the $\tanh()$ function, modeling the nonlinearity that manifests as clipping during the pre-amplification stage of an analog effects foot pedal.

Other popular data-driven models are based on neural networks. Indeed, machine and deep learning have been intensively explored for distortion and amplifier units. The earlier attempt involves a multilayer feedforward network to learn a high-pass filter and the Ibanez Tube-Screamer distortion effect (Mendoza, 2005). The author experimented with a network having one input layer, two hidden layers with sigmoid neurons, and one output layer with a linear activation function. The hidden and output layers present the same size as the input layer, and two different input sizes were considered in the experiments: 512 and 60 samples recorded at 44.1-kHz sampling rate. The training was performed using the MSE as a loss function and feeding the network with overlapped segments; in particular, the hop size was half of the input size. The high-pass filter training was carried out on the frequency domain, which led to unsatisfactory results, while the distortion effect was trained in the time domain, which demonstrated the ability of the network to learn distortion effects. A spurious frequency was found to occur due to the overlapping process since the transition among frames was not sufficiently smooth.

A vacuum-tube amplifier model is presented in (John and Livingston, 2013), where authors investigated different networks and input sizes in two scenarios: slightly distorted and overdriven signals. The networks consisted of 5 or 10 units, while the networks’ input included a variable number of audio samples. The network is autoregressive, which means it generates the output sample at the current time step using current and past input samples but also output samples generated at previous time steps. The number of past samples can differ from the input and output signal, and in this case, the experiments considered 5, 10, or 50 input samples and 5 or 10 output samples. The work presents preliminary experiments that resulted in poor performance, but it demonstrates the feasibility of modeling vacuum-tube amplifiers with Recurrent Neural Networks (RNNs.) Larger networks generated lower prediction errors, and the overdrive case resulted in more challenges in modeling accurately.

Damskägg, Juvela, Thuillier et al. (2019) used convolutional-based networks to model the preamplifier circuit from the Fender Bassman 56F-A vacuum-tube amplifier. The dataset was built using a circuit simulator, and the network

consisted of a stack of dilated causal convolution and fully connected layers. The first fully connected layer uses the gated activation function. The second layer has a hyperbolic tangent activation. Finally, a linear layer outputs the predicted sample. The error-to-signal ratio is the measure used as loss. The selected number of convolutional layers is 10 with a filter width of three and the dilation pattern $d_k = 1, 2, 4, \dots, 512$. This configuration leads to a receptive field of 2046 samples, corresponding to 46 ms of latency between the input and corresponding output sample, as it uses a sampling rate of 44.1-kHz. The model presents the fully connected layers with 8 and the convolutional layers with 16 units each. The number of samples used as input is set equal to the receptive field. The dataset was simulated with different values of the gain, which is a variable resistor, and used to condition the model. The conditioning is realized in the convolutional layers and using the gated activation (Oord et al., 2016). The work shows that Convolutional Neural Networks (CNNs) accurately emulate the reference device better than a fully connected feedforward network and can respond to user control changes. The Ibanez Tube Screamer, the Boss DS-1, and the Electro-Harmonix Big Muff Pi distortion pedals have been modeled as well by using the same architecture (Damskägg, Juvela and Välimäki, 2019). The datasets, in this case, are collected directly from the devices. Here, the authors investigated more configurations: considering different dilation patterns, activation functions, such as rectified linear unit (ReLU), tanh, and gated activation, numbers of convolutional layers (10, 18, 24) and number of convolution channels (2, 4, 8, 16, and 32). In this work, the conditioning is not considered. A model with gated activation, 18 layers, and 16 convolution channels resulted in lower ESR and real-time capability. (A. Wright, Damskägg and Välimäki, 2019) compared the previous models with RNNs, in particular Long short-term memory (LSTM). The architecture consists of one RNN layer and one linear fully connected layer as an output layer with one unit. LSTM and Gated recurrent unit (GRU) are compared as RNN layers varying the number of units (32, 64, and 96). The input signal is extended to include an additional value representing the user control's parameter. The results show that RNNs achieve similar accuracy to the CNN model while requiring significantly less computational cost and latency since requiring only the current input sample to predict the current output. LSTM performed better than GRU networks, and the error decreased when the number of units was increased. Chowdhury (2020) introduced another RNN-based model utilized for the Klon Centaur guitar pedal circuit, which is also compared with a WDF model of the same circuit. Machine learning implementation resulted in better computational performance than the WDF model but sounded very similar, although the ML implementation slightly damped high frequencies. The architecture consists of a GRU layer of 8 units and one linear fully connected layer with one unit as the output layer. Also, here, a single variable control parameter of the audio effect is added as an extra input to condition the inference of the networks.

Ramírez and Reiss (2019) take a different approach for modeling audio effects, utilizing a combination of different deep learning architectures. They divide the model into three parts: an adaptive front-end, latent space, and synthesis back-end. The front-end performs time-domain convolutions with the raw audio to map

it into a latent space. It also generates a residual connection, which facilitates the reconstruction of the waveform by the back-end. The adaptive front-end consists of a convolutional encoder. It contains two CNN layers, one pooling layer, and one residual connection. The first layer has 128 filters and 64 channels and is followed by the absolute value as a nonlinear activation function. The second layer has 128 filters of size 128. The latent-space DNN contains two layers. The first layer is based on locally connected dense layers of 64 units, and the second layer consists of a fully connected layer of 64 units. Both of these layers are followed by the softplus function. The DNN modifies the latent representation of the input audio into a new latent representation, which is fed into the synthesis back-end. The synthesis back-end consists of an unpooling layer, a deep neural network with smooth adaptive activation functions (DNN-SAAF), and a single CNN layer. DNN-SAAF consists of four fully connected dense layers of 128, 64, 64, and 128 hidden units, respectively. All dense layers are followed by the softplus function except the last layer. The SAAF (Hou et al., 2017) is used as the nonlinearity for the last layer. The model is first trained without the latent-space DNN and DNN-SAAF, and then these modules are incorporated for a second training. The loss function to be minimized is the mean absolute error (MAE). In both training procedures, the input and target audio are sliced into frames of 1024 samples with a hop size of 64 samples. Distortion, overdrive, and EQ effects have been modeled using this method, with a sampling rate equal to 16 kHz.

Martínez, Benetos and Reiss (2020) further explores the aforementioned architecture and compares it with various modifications as well as with the CNN architecture designed by Damskägg, Juvela, Thuillier et al. (2019). The variations mainly consist of incorporating Bi-LSTMs or CNNs into the latent space. In this work, the target devices were Universal Audio Vacuum-Tube Preamplifier 610-B, Universal Audio Transistor-Based Limiter Amplifier 1176LN, and 145 Leslie Speaker Cabinet. The highest score in the listening test was obtained by the variation incorporating Bi-LSTM. The input frame size was 4096 sampled with a hop size of 2048 samples. Another variation based on Bi-LSTM and SAAF as latent-space modules is used for plate and spring reverberators (Ramírez, Benetos and Reiss, 2020). Here, the loss function to be minimized is the sum of MAE in the time domain and MSE of the log power magnitude spectra of the target and output. These architectures represent a general-purpose deep learning framework for modeling audio effects, which is capable of accurately modeling different types of effects. On the other hand, the models are large networks with high computational complexity, leading to difficulty when considering efficient implementation. In addition, since the Bi-LSTM layer, the architecture is non-causal, does not allow real-time applications, and also lacks conditioning possibility and focuses on modeling a static representation of the devices (i.e., fixed parameters).

Gray Box Models

A delay-based effect is investigated in (Otto et al., 2023), where a tape-based delay is modeled using GRU networks using a 44.1kHz sample rate. Here, the delay trajectory is analyzed and extracted using impulse train signals. This trajectory is used to demodulate the signal before training the models or used to directly

guide a delay-line built in a differentiable way and based on (Engel, Hantrakul et al., 2020) work. The second strategy produced smaller prediction errors. The authors include the modeling of the tape’s noise and the trajectory using diffusion techniques to approximate the probabilistic distribution. The hidden size of the GRU layer was set to 64, and attempts with larger sizes led to no improvements.

In (A. Wright and Välimäki, 2021), authors found it beneficial to use as extra input the Low-Frequency Oscillator (LFO) signal representing the time-varying frequency response to model LFO delay-based effects, such as flanger and phaser. The LFO signal is first extracted by analyzing the recording at 44.1kHz and then used to help the LSTM during the learning process. The extra input has shown to be beneficial to learning the LFO-based effect and can be freely controlled after model training, although in the study is considered only static parameter values and not LFO signal changing depth and width over time. The error-to-signal ratio (ESR) is used as a loss function, and the model is an LSTM layer followed by a linear fully connected layer with one unit. The size of the LSTM layer is 16 or 32 or 48.

Parker, Esqueda and Bergner (2019) uses fully connected neural networks together with measurements within the circuits. In this work, the structure includes 2 to 3 layers embedded within a discrete-time state-space system and explored for diode-based guitar distortion circuits and the lowpass filter of the Korg MS-20 synthesizer. The method required measurements from the devices’ circuits, which are not always accessible, and is moving away from black-box models. Continuing towards gray box approaches, (Kuznetsov, Parker and Esqueda, 2020) is based on the Differentiable Digital Signal Processing (DDSP) framework and applies this concept to RC lowpass filter. Subsequently, Esqueda, Kuznetsov and Parker (2021) applied a similar modeling concept to the parametric tone section of a guitar amplifier and a nonlinear guitar overdrive circuit. The technique learns the parameters of a circuit model from measurements of real devices using a Wiener-Hammerstein model consisting of two Infinite impulse response (IIR) filters and a feedforward artificial neural network. The IIR filters are, in this way, trainable and derived from the analysis of the circuit. This approach is also used by Nercessian, Sarroff and Werner (2021), where a cascade of biquad filters is trained, employing a Fourier transform, to model the BOSS MT-2 pedal distortion pedal. The model includes four tunable parameters. Differentiable Digital Filters have been applied for the case of a diode clipper circuit (Chowdhury and Clarke, 2022), showing higher accuracy and similar computation time when compared to traditional physics-based models. Wilczek et al. (2022) propose learning the ordinary differential equations that govern the first-order and second-order diode clipper using fully connected layers and RNNs. Their formulation utilizes discrete-time state-space modeling in a manner akin to that described by (Parker, Esqueda and Bergner, 2019). Neural networks have also been investigated in the case of wave digital filter formulations (Darabundit, Roosenburg and Smith III, 2022). A tube amplifier circuit is simulated using multiport nonlinearities inside a wave digital filter and fully connected layers to learn features from a Kirchhoff-domain dataset. Similarly, Miklanek et al. (2023) combine recurrent neural networks (RNNs) with the Kirchhoff nodal analysis to create a gray-box guitar amplifier

model. Lastly, a digital model of a Moog ladder filter has been developed using a simplified digital-analog model and deriving the differentiation expressions of the filter, with respect to the angular cutoff frequency and resonance controllable parameters (Gerat, Bhattacharya and Zölzer, 2023).

Dynamic Range Compressors

To date, only a few examples of analog dynamic range compressor modeling can be found in the literature. The circuit of the compressor pedal MXR DynaComp, based on operational transconductance amplifiers (OTAs), has been modeled using a nonlinear state-space system (Kröning, Dempwolf and Zölzer, 2011). The resulting model is inefficient since it needs to solve a nonlinear system to compute the next sample. Also, the emulation of tube amplifiers placed in dynamic range controllers has been modeled utilizing a polynomial approximation of the transfer characteristics (Schimmel, 2003). The optocoupler, the optical isolator used in optical compressors, is a two-port circuit element that electrically isolates different audio system parts while maintaining one-directional coupling via the optical channel. The analog optical isolator consists of a light-dependent resistor and a light-emitting diode in an opaque enclosure. A model of the optocoupler has been introduced by Eichas and Zölzer (2016b), where they use input/output measurements to extract the static and dynamic characteristics of the device. The circuit elements are measured and analyzed, and an iterative error minimization procedure is utilized to refine the parameters of the digital model. This data-driven approach does not need extensive circuit analysis, but from listening tests, an audible difference emerged between the proposed emulation and the actual analog reference device.

The first attempt to model a dynamic range compressor using ANN (Steinmetz and Reiss, 2022) utilizes convolutional-based networks, particularly the Temporal Convolutional Networks (TCNs). The device used to evaluate this approach is the Teletronix LA-2A Leveling Amplifier; a database was recorded from it at 44.1 kHz, the same sampling rate for the model. TCNs present dilated convolutional layers with rapidly growing dilation factors between 2^2 and 2^{10} . Increasing the dilation factors increases the receptive field and the networks' ability to deal with long dependencies without increasing the number of layers. This method allowed the design of a model with lower computational requirements but still with significant input signal samples. The proposed models indeed present between 100 and 1000 ms of input-output latency on the audio signal. This intrinsic latency is equal to the length of the receptive field, and it is detrimental to live audio applications. On the other hand, TCN results have a similar accuracy to recurrent neural networks but are more computationally efficient. TCN block, in this work, consists of a convolutional layer, batch normalization, conditioning using FiLM, and PRelu activation function. The authors compared networks with 4, 5, 10 TCN blocks, 32 convolutional channels for each convolutional layer, and 5, 13, and 15 as the kernel size. The resulting models present receptive fields of 100, 300, and 1000 ms, respectively. The model with a receptive field of 300 ms provided the best accuracy when evaluated with MAE. On the other hand, LSTM networks with 32

units present better STFT and LUFS errors. Finally, listening tests showed that LSTM performed similarly to TCN, with 300 ms of latency.

A gray box model of the Teletronix LA-2A Leveling Amplifier has also been proposed (A. Wright and Välimäki, 2022). The compressor is modeled starting from the structure of a traditional digital compressor while predicting the compression curve parameters, threshold, ratio, and knee utilizing a combination of Fully Connected (FC) and Recurrent Neural Networks (RNNs). Similarly, the model predicts the timing parameters for the equation describing the attack and release envelopes. This approach incorporates domain knowledge and helps generate audio with fewer artifacts and parameters than (Steinmetz and Reiss, 2022), but the accuracy is limited by the utilized approximations of the compression equations. The LA-2A does not have controllable attack and release time parameters. Due to its internal design, the actual attack and release times, especially the latter, depend on the current and past input signal levels. In this work, three different approaches are used to model the timing of the compression. The first considers attack and release time to be equal and not a function of the input. This means the model features tunable values that are used in the equation governing the compression timing. The second considers different attack and release times, but they are still not a function of the input. The third considers the attack equal to release time but a function of the input, and this time is predicted by a neural network. These assumptions do not accurately represent the actual functioning mechanism of the device, as attack and release times significantly differ from each other and depend on the input signal energy. As a result, the model is wrongly biased, which impacts its accuracy.

2.2.2 Generators

Musical devices in the *generators* category include both electronic sound synthesizers and acoustic musical instruments. For sound synthesizers, the primary modeling interest lies in analog synthesizers, as digital ones are already based on numerical models. Most *generators* exhibit modularity in their construction, with individual components often displaying a high degree of independence. As such, not only can these devices be fairly complex, but there is often a significant dimensional imbalance between the input and output data—with the latter typically being greater (a disparity not seen in processors). This factor, along with the modularity and independence of components, means that research and investigations into modeling often focus on individual sub-modules.

Analog Generators

Electronic *generators*, such as analog synthesizers, typically employ a subtractive sound generation process, in which frequency content is removed through the filtering of one or more tunable, harmonic-rich signals. The three fundamental components of analog sound synthesizers are Voltage Controlled Oscillators (VCOs), Voltage Controlled Filters (VCFs), and Voltage Controlled Amplifiers (VCAs). The digital generation of the periodic waveforms matching the classic analog waveforms (sawtooths, square waves, and triangle waves) and the problem

of emulating VCOs have been studied together with the aliasing reduction problem. This is because the original analog waveforms had an infinite bandwidth, while the digital emulations must be bandlimited to half the sampling frequency; otherwise, the output will exhibit aliasing distortion. For this reason, modeling approaches have primarily focused on finding efficient algorithms for generating periodic waveforms that match the analog ones rather than on accurately emulating the circuits. Pekonen (2014) provide a review of commonly used techniques, including the two recent modeling approaches for the Moog sawtooth oscillators (Pekonen et al., 2011). The first approach works in the time domain and uses phase distortion synthesis, and the second matches the spectrum of the existing antialiased sawtooth oscillator to the corresponding analog spectrum using a first-order IIR filter.

Wavefolding is a type of nonlinear waveshaping in which portions of a waveform are inverted or "folded back" on itself. A model of the wavefolding-based of the Buchla 259 has been designed with a memoryless mappings of the input-output voltage relationships inside the circuit after deriving the formulation of the explicit equations Esqueda, Pöntynen, Välimäki et al. (2017). Furthermore, the Lockhart Wavefolder and Serge Triple Waveshaper have been modeled and derived an explicit formulation for the output voltage using the Lambert-W function¹ from the nonlinear implicit relationship between the input and output voltages (Esqueda, Pöntynen, Parker et al., 2017a; Gormond et al., 2018; Esqueda, Pöntynen, Parker et al., 2017b).

VCF modeling works have often focused on the filters used in synthesizers manufactured by Moog due to their popularity. A Moog analog filter, based on transistors and which presents saturating distortion, has been modeled with the Finite Difference Time Domain (FDTD) technique (Paschou et al., 2017). Later, another model using the cascade of 2nd-order state variable filters (SVFs) and a global feedback path has been proposed (Werner and McClellan, 2020). The port-Hamiltonian approach has also been applied to the OTA-based Korg35 filter circuit and the Moog 4-pole ladder filter circuit (Danish, Bilbao and Ducceschi, 2021). Wave digital filters (WDF) implementation of the voltage-controlled low-pass filter of the Korg MS-50 synthesizer, built around a nonlinear diode bridge is presented in (Rest, Parker and Werner, 2017). Lastly, the low-pass gate filter of the Buchla synthesizer is modeled by solving the system through instantaneous linearization around the point of operation of the nonlinearity and calculating its tangent (Parker and D'Angelo, 2013).

The Roland TR-808 Rhythm Composer has also been the focus of several analog modeling works. The TR-808 is a drum machine producing synthetic percussive sounds such as bass drum, snare, toms, conga, rimshot, claves, handclap, maraca, cowbell, cymbal, and hi-hat. The circuit has been analyzed and decomposed into sub-circuits, which are taken as a case study to apply Wave Digital Filter in scenarios with different numbers of nonlinearities (Werner, 2016).

¹<https://mathworld.wolfram.com/LambertW-Function.html>

Acoustic Generators

White Box Models

Nonlinear behavior lies at the core of acoustic instruments' sound generation, particularly for the excitation part, giving them unique and discernible timbre. The nonlinearities of vibrating strings, membranes, and tubes, as well as the nonlinear aspects of collision interactions between strings, frets, fingers, and fingerboards, are essential components for understanding the behavior of musical instruments, particularly concerning timbral aspects. Most models rely on a subdivision of the instrument into excitation mechanism and resonator (Fletcher, 1999). In sustained-tone instruments, such as violins, flutes, and trumpets, the resonators are only approximately linear, and their operation relies upon the extreme nonlinearity of their driving mechanisms. Instead, in an impulsively excited instrument, the energy is transferred to the vibrating element in a short time compared with the decay time of the vibration. Therefore, although the impulse could be nonlinear, the nonlinearity is more concerned with the vibrating element in this case.

Traditionally, the excitation is considered a nonlinear lumped block, while the linear resonator is distributed and characterized by a number of natural frequencies or modes. Such a characterization has been exploited extensively for efficient synthesis purposes but still becomes imprecise in some contexts as it neglects some aspects dominating essential spectral phenomena. Considering the string as a resonator, recent studies have demonstrated that the linear model of the string fails to account for the nonlinear mixing of transverse and longitudinal vibration (Etchenique, Collin and Moore, 2015; Giordano and Korty, 1996) observed in strings at high amplitudes, which has been identified as a source of so-called phantom partials. For woodwind instruments, nonlinear steepening effects occur along cylindrical segments of the bore when subjected to high blowing pressures (Bilbao, 2014). Such steepening effects are intimately related to 'brassy' timbres.

While not all nonlinear aspects of the sound generation have been modeled yet, incomplete models can be found in the literature. For bowed string instruments, such as violin, viola, cello, and double bass, a digital waveguide is utilized to model the bow-string nonlinear interaction using a scattering junction (Sterling and Bocko, 2010). This is implemented as a memoryless lookup table and relates the incoming and outgoing velocity waves at the bowing point and the velocity of the bow. Alternatively, a finite-difference approximation simulates bow dynamics and solves the bow-string interaction (Maestre, Spa and Smith III, 2014). In particular, the nonlinear friction problem is graphically solved by the so-called 'Friedlander construction' method (Friedlander, 1953; Woodhouse, 2003).

An excited string presents two polarizations of transverse waves parallel to the bowing direction (horizontal) and perpendicular to the bowing direction (vertical). In the vertical polarization, considered by Desvages and Bilbao (2014) and Desvages and Bilbao (2015), a nonlinear collision models the action on the string in the vertical polarisation while a nonlinear friction curve models the horizontal polarization. Consequently, a finite difference scheme is established

based on the initial set of partial differential equations. The two nonlinear systems describing the contact force and tangential friction are solved using the Newton-Raphson algorithm. To achieve real-time capability, non-iterative methods to solve the nonlinear collision interactions (Ducceschi and Bilbao, 2019b) have been exploited for a singular triangular bowed string instrument called Tromba Marina (Willemsen, Serafin et al., 2020). This model uses a Finite Difference Time Domain (FDTD) scheme, which is obtained by discretizing the partial differential equations governing the system in time and space. Finally, a more efficient model for nonlinear interaction with the bow combining modal approach and semi-discretization is proposed in (Russo, Ducceschi and Bilbao, 2022). This work investigates the cases of an ideal and stiff string coupled with a bow. Finally, a recursive scheme based on the Impulse Pattern Formulation (IPF) approach (Linke, Mores and Bader, 2022) was designed to model the bifurcation phenomenon. Bifurcation occurs when a small smooth change made to some system's parameter values (called bifurcation parameters) causes a sudden qualitative or topological change in its behavior. This method describes musical instruments as self-organized systems that are driven by impulses, as described in an abstract top-down modeling approach (Bader, 2013).

Wind instruments include reed woodwinds, brass instruments, and flutes. The nonlinear collision of the reed with the lay-in reed woodwind instruments, causing the generation of reed-beating effects, has been modeled using a finite difference time domain schema, and the numerical stability ensured using the Hamiltonian formulation (Bilbao, Torin and Chatziioannou, 2015). Nonlinear losses are included using the Hunt and Crossley model (Hunt and Crossley, 1975). Nonlinear equations are solved using the Newton-Raphson method, in which convergence was not proved. Similarly, Chatziioannou, Schmutzhard and Bilbao (2017) simulated the clarinet reed. An energy-based approach ensures stability. The coupled system, where the air column has been discretized using the finite difference method and an iterative solver, is used to solve the nonlinear system of equations. In the latter case, the authors focus on the numerical collision model that incorporates impact damping. They also calculate the maximum number of required iterations for the convergence. The nonlinear relationship between mouthpiece pressure and mouthpiece flow is modeled by Guillemain (2004) by using the convolution with the reflection function of the tube. Its input impedance is calculated, including viscous and thermal losses at the walls and radiation losses at the open end.

The reed-valve end termination of the alto saxophone's pipe had been modeled via a nonlinear scattering element that is interfaced with an air column model (Maestre, Scavone and Smith III, 2018). The joint modeling of input impedance and sound pressure radiation has been built using parallel digital filters, with the filter coefficients derived from experimental data and computed iteratively, minimizing the error between the frequency response of an input impedance measurement and that of a digital impedance model of the parallel filter structure. The digital model includes the transition between fingering positions by using linear interpolation of input impedance and sound pressure radiation models. The impedance-radiation model is based on a digital filter realized as a digital waveguide termination that is interfaced with a reed model based on nonlinear

scattering.

Models of brass instruments have mainly exploited the FDTD approach, and only a few take into account the nonlinearities of the instruments. A power-balanced model of the exciting mechanism has been developed using the Port-Hamiltonian formalism (Lopes and Hélie, 2014). This work focuses on the design of nonlinearity due to the nonlinear coupling involved by the jet between the lips. A nonlinear collision modeling lip reed using a finite difference scheme has also been developed for the trombone (Willemsen, Bilbao et al., 2021). Instead, a numerical scheme for nonlinear wave propagation inside the bore of the instrument has been proposed for the nonlinear acoustical resonator (Berjamine et al., 2017). Viscothermal losses in the bore and the continuous variation of the instrument's cross-section are also considered. Lastly, nonlinear models of flutes have yet to be proposed in scientific literature.

Impulsively excited-string instruments have plucked and hammered excitation. Hammered excited instruments such as pianos are described in the next section. Nonlinear interaction between the string and other parts of the structure, such as the frets (Woodhouse, 2017), is described by a nonlinear state-space system discretized employing the bilinear transformation (Evangelista, 2011). The pitch glide effect in plucked instruments is emulated with a numerical purely kinematic string-barrier interaction model (Kartofelev, Stulov and Välimäki, 2015). A modal representation of the displacement with an energy-conserving scheme has also been developed (Issanchou et al., 2016). This approach allows the simulation of a stiff, damped string vibrating against an obstacle with arbitrary geometry, and it can produce high-frequency components caused by the nonlinear contact. Similarly, a modal approach for the nonlinear contact has been designed for tanpura string vibrations (Van Walstijn and Chatziioannou, 2014). In this case, a real-time model for tanpura that considers this nonlinear interaction with the bridge is proposed. A modal model has also been used for the tension modulation effect (Bridges and Van Walstijn, 2017). Another modal-based numerical scheme of the interaction between a string and a plate coupled via a nonlinear bridge element is proposed by Van Walstijn and Mehes (2017), focusing on the implementation of the numerical algorithm such that any of its parameters can be adjusted in real-time.

Woodhouse (2022) experimentally examined and numerically modeled a plucked harp string. This work presents a modal linear model incorporating coupling between transverse and longitudinal displacements and an investigation via measurements of the nonlinear excitation of longitudinal vibration.

Bensoam and Roze (2013) proposed a nonlinear string model using the Volterra series. Since the Volterra kernels are only time-dependent and the Volterra series cannot be used to solve interaction with a space-varying force distribution, a different nonlinear physical model which extends the Green's function and the Volterra series has been later developed (Roze and Bensoam, 2014; Roze and Hélie, 2014).

Modeling of percussive instruments has also been extensively investigated. An FDTD model for the interaction between the membranes and the snares is introduced in (Bilbao, 2012), where the finite difference techniques are applied to a full 3D system, examining the coupling between membranes. A nonlinear FDTD

model of a tom-tom based on von Kármán equations (Torin and Newton, 2014) includes geometrical drum membranes nonlinearities and full air coupling between them. An Inverse Fast Fourier Transform synthesis-based model is proposed for tom-tom drum synthesis, encoding all sample characteristics in a single, real-valued, frequency domain signal (Kirby and Sandler, 2020). The tom-tom sound features are modeled with chirps, where the width corresponds to the magnitude of the pitch glide effect, and the sinusoidal peaks correspond to the high-frequency attack sound. The interaction mallet/membrane has been studied and modeled via Hamiltonian framework (Bilbao, Torin and Chatziioannou, 2015). A friction drum, a percussive instrument consisting of a single membrane stretched over a sound box, has been modeled as two primary components and an excitation mechanism: a membrane connected to an acoustic tube, where the membrane can be bowed via a nonlinear elastoplastic friction model (Onofrei, Willemsen and Serafin, 2021).

In plates, modeled nonlinear phenomena are crashes and pitch glides. In the latter, the amplitude-dependent frequency of vibration is perceived as a pitch-bend. They are characterized by a cascade of energy from large to small wavelengths, perceived as a crash or a shimmering sound. This phenomenon is referred to as Wave Turbulence. This has been investigated by Ducceschi, Cadot et al. (2014) with a finite difference, energy-conserving scheme. The schema simulates the effect together with realistic boundary conditions.

A sound synthesis environment composed of several plates under nonlinear conditions coupled with the surrounding acoustic field has been proposed by Torin and Bilbao (2013) using the Von Kármán equations (Timoshenko, Woinowsky-Krieger et al., 1959). These equations govern the behavior of elastic plates undergoing displacements, which are implemented numerically using FDTD methods. A similar work, but based on a modal approach, includes the nonlinear coupling between in-plane and transverse motions (Ducceschi and Touzé, 2015a), with later improvement on the scheme (Ducceschi and Touzé, 2015b). Port-Hamiltonian formalism has also been used for Chinese opera gongs (Jossic et al., 2017). This work focuses on the simulation of the control of the pitch glide effect: for this purpose, the study focuses on a single-degree-of-freedom nonlinear mode described by a Duffing equation.

Black Box Models

Due to the complex nature of the modeling task, the number of black-box emulations of sound *generators*—particularly those utilizing Artificial Neural Networks (ANNs)—is still limited, and their performance has room for improvement. An exhaustive literature review on this topic, with a broader scope, is included in (Simionato, Fasciani and Holm, 2024: Paper VII), which is attached to this thesis. Below is a summary derived from the paper, focusing on a narrower scope.

As already discussed, in recent years, ANNs and Deep learning (DL) architectures have been successfully employed to learn directly from raw audio (Dieleman and Schrauwen, 2014) and adopted for nonlinear time series prediction as well as for complex nonlinear audio effects. Among the few attempts, deep

learning techniques have been applied primarily for modeling acoustic instruments instead of analog synthesizers. One challenge in this context is collecting suitable datasets that relate the stimuli a player imparts to the instrument’s body to the generated sound. Indeed, the input represents the excitation mechanism, which is generally human-driven and lies in a domain different from the generated sound. Capturing such stimuli accurately with non-intrusive sensing technologies can be difficult, as can selecting a finite, representative set of stimuli-to-sound examples. Another challenge is designing an ANN that is able to learn the mapping between input and output, significantly different from each other, with the latter often presenting a much higher dimensionality.

Among the deep learning approaches utilized for audio generation, WaveNet represents the earliest attempt (Oord et al., 2016). It is a convolutional-based autoregressive model. Autoregressive models are trained to predict subsequent audio samples using the prior predictions as inputs. A key strength of WaveNet is the use of dilated causal convolution. Dilation increases the model’s receptive field by skipping input values with a certain stride, while causality ensures that future context is not considered when making predictions. A sound synthesis application using a WaveNet-based architecture is proposed in (Engel, Resnick et al., 2017), where the model was trained with the NSynth dataset (Engel, Cinjon et al., 2017). Subsequently, Kalchbrenner et al. (2018) integrated a Recurrent Neural Network (RNN) into the autoregressive architecture, introducing WaveRNN and utilizing it for text-to-speech synthesis. Later application of the WaveRNN model involved the NSynth dataset, generating string, brass, woodwind, and mallet sounds (Hantrakul et al., 2019). In this case, the synthesis is conditioned by audio features represented as a time series with a much lower sampling frequency (250 Hz) compared to the audio rate.

SampleRNN (Mehri et al., 2017) is an RNN-based model that uses three modules organized in a hierarchical structure, each operating at a different temporal resolution. The lowest module processes individual samples, while each subsequent higher module operates on an increasingly longer timescale and lower temporal resolution. Each module conditions the module below, with the lowest module generating sample-level predictions. The audio generation is not conditioned by any parameters, hence offering no means for control or interaction with this sound generator. The architecture is evaluated on two speech-related datasets and on one containing a collection of Beethoven’s piano sonatas.

Autoregressive methods enable low-latency implementations as they predict one or a few samples at a time, leveraging short input audio frames from previous predictions. On the other hand, they suffer from significant computational inefficiency and data flow dependencies, preventing the use of parallel computing. Teacher forcing is a possible strategy to overcome data dependency, which uses ground truth as input instead of the model’s prior output. With this approach, errors can accumulate quickly along the prediction sequence. To overcome this issue, a curriculum learning strategy can be used to change the training process from a fully guided scheme based on previous true tokens to a less guided scheme (S. Bengio et al., 2015), to the disadvantage of significantly extending the duration of the training phase.

Generative frameworks have been applied to audio as well. Unlike autoregressive models, Generative adversarial networks (GANs) typically generate one complete audio example in a single pass rather than in a sequence of inferences, producing a few audio samples at a time. Indeed, an entire sound example is produced in a single or in a few inference steps. This represents a significant drawback in terms of latency for interactive sound synthesis applications. However, existing implementations have demonstrated that real-time audio computation is feasible. Donahue, McAuley and Puckette (2019) generated raw audio examples with a duration of one second, synthesizing a variety of sounds, such as drums, bird vocalizations, speech, and piano. Improvements have been proposed in (Engel, Agrawal et al., 2019), demonstrating that working in the spectral domain can provide higher fidelity and better locally coherent sound generation. In (Bitton et al., 2019), the adversarial framework uses Wasserstein autoencoders (WAEs) (Tolstikhin et al., 2018) and extends the generation using expressive modulations of target musical attributes. The Wasserstein generative adversarial network is also used in (Drysdale, Tomczak and Hockman, 2020), where the network is trained with a small dataset of drum sounds and conditioned by high-level parameters to generate different drum sounds, such as kick, snare, and cymbals. Nistal, Lattner and Richard (2020) present DrumGAN, which improves drum sound synthesis by conditioning the model on perceptual features using generative adversarial networks. Finally, StyleWaveGAN (Lavault, Roebel and Voiry, 2022) generates audio examples conditioned on the drum type and a set of audio descriptors.

WAE and a Multi-Head Convolutional Neural Network (MCNN) are utilized in (Aouameur, Esling and Hadjeres, 2019) to generate percussive sounds. The WAE learns to generate Mel-scaled magnitude spectrograms of the sound signal, while an MCNN estimates the corresponding audio samples from the magnitude spectrogram. The dataset used to train this system is a collection of sound examples across 11 drum categories, including kicks, claps, snares, open and closed hi-hats, tambourines, congas, bongos, shakers, snaps, and toms.

A feedforward CNN, based on U-Net (Ronneberger, Fischer and Brox, 2015), is used in (Ramires et al., 2020) to synthesize percussive sounds by mapping high-level timbre characteristics of the sound to the corresponding audio signals. In particular, the probability distribution of the audio signal is modeled as a function of the timbral features and time-domain amplitude envelope. The training dataset used in this case contains percussive audio examples such as kicks, snares, cymbals, and bells.

Gray box models

Hybrid models are often trained on synthetic datasets derived from physics simulations, making them particularly useful for developing faster and more efficient solvers for existing digital models. However, this approach cannot synthesize aspects of known acoustic phenomena that lack available mathematical models or are unsolvable with current numerical techniques.

Residual networks have been employed as a black-box differential equation solver for ordinary differential equations (R. Chen et al., 2018). They have been used in structures like RNN to solve acoustic problems, such as the 2D wave

equation for linear and nonlinear strings (Parker, Schlecht et al., 2022). Moreover, ANNs have been integrated into a popular Port-Hamiltonian framework to learn dynamical systems (Desai et al., 2021).

In an early application of this method, Radial Basis Function Networks are utilized to model the nonlinear components in an exciter-resonator scheme, emulating the sounds of violins and oboes (Drioli and Rocchesso, 1998). This led to the development of a system that, instead of defining a signal processing structure explicitly, could learn the shape of the nonlinearity to reproduce target sounds.

In more recent examples integrating machine learning with signal processing techniques, Wilkinson, 2019 have used the Gaussian process to learn physical modal synthesis parameters from sound recordings. This approach models nonlinear audio spectral mappings using examples of natural sounds. Specifically, the authors estimate the exponential decay rates for the vibrational models of various instruments, including the clarinet, oboe, and piano, as well as metal and wooden impact sounds. Neural networks have also been leveraged to estimate physical model parameters (Gabrielli, Tomassetti, Squartini et al., 2017; Gabrielli, Tomassetti, Zinato et al., 2018). In particular, a CNN is linked with the physical model of a pipe organ, which includes an exciter simulating wind jet oscillation, a resonator that uses a digital waveguide structure to replicate the bore, and a noise model that simulates air noise modulated by the wind jet.

Another approach for sound synthesis is based on a source-filter model with parameters tuned using end-to-end training in the audio domain. The earliest neural source-filter approach has been successfully applied to speech synthesis (X. Wang, Takaki and Yamagishi, 2019). In this case, the neural source filter is trained using spectrum-based criteria that return acoustic features as conditioning. Then, it uses a trainable filter module to transform a sine-based excitation signal into the model’s output.

Recently, the use of the Differentiable Digital Signal Processing (DDSP) framework in combination with Spectral Modelling Synthesis (Serra and Smith III, 1990) (SMS) has been particularly common. This type of synthesis considers the sound signal as the sum of harmonic and noise parts. The harmonic content can be synthesized with a sum of sine waves whose parameters are directly extrapolated from the target.

A differentiable subtractive-based synthesizer has been proposed for singing voice synthesis (Wu et al., 2022). The harmonic part of voices is synthesized by filtering a sawtooth source signal with a linear time-variant finite impulse response filter. The neural networks estimate the filter’s coefficients using the mel-spectrogram as input. Similarly, digital infinite impulse response (IIR) filters have been integrated into deep-learning networks, where (Kuznetsov, Parker and Esqueda, 2020) linked IIR filters and RNNs, presenting as an application example a Wiener-Hammerstein model using differentiable IIRs.

The DDSP method has been used for percussive sound synthesis Shier et al., 2023, integrating a Temporal convolutional neural network. to emulate the signal’s transients. Two models for the guitar are (Wiggins and Kim, 2023; Jonason et al., 2024). Wiggins and Kim (2023) presented a polyphonic model

composed of a monophonic synthesis module for each guitar string. Parameters are predicted from MIDI information and drive inharmonic oscillator and filtered noise synthesizer modules. They also showed how to vary the model parameters to simulate the use of a capo and alternate fretboard positioning. Jonason et al. (2024) compared four different system configurations, starting from a base model having a control model which, from MIDI information, predicts four features for each string: the fundamental frequency, the loudness, the periodicity, and the spectral centroid. These features are used to predict the synthesis parameters for the harmonic, noise, and reverb modules. The authors concluded that skipping the prediction of features but directly predicting the synthesis parameters yields better results.

Acoustic Pianos

The size, weight, and cost of acoustic pianos have led to the development of digital counterparts that mimic the sound of acoustic pianos as closely as possible. Most commercial digital pianos employ sampling-based sound synthesis, where the sounds of the acoustic piano are recorded and reproduced, and then the corresponding key is pressed. With this approach, phenomena such as the free vibration of the strings when the sustain pedal is pressed, the coupling between the strings of the sounding notes, or the restriking of an already sounding string cannot be easily and concisely reproduced by recorded samples.

Nonlinear string vibration poses a complex problem for real-time computations without the use of simplifying assumptions. Among the non-sample-based techniques for piano synthesis, the digital waveguide method is among the most common. Digital waveguide synthesis models the motion of the string based on known traveling wave solutions where each traveling wave has its own delay line to describe its spatial and temporal coordinates. Daudet (1999) use a digital waveguide to synthesize the sound generated by string vibrations, considering both polarizations. In a related approach, Aramaki et al. (2001) synthesizes the sound produced by the vibrations of two piano strings that are tuned to very close pitches and coupled at the bridge level. This method has also been adopted in several other works (Bank, 2000; Bensa, Bilbao, Kronland-Martinet et al., 2003; Bensa, Bilbao, Kronland-Martinet et al., 2005). A commercial synthesizer featuring this synthesis method is the Yamaha VL1 (Bilbao, Torin and Chatziioannou, 2015).

Digital waveguides are capable of the highly efficient modeling of linear string behavior, but they are not very well suited to model nonlinear string vibration. Nonlinear longitudinal vibrations of piano strings' low and middle range are very important for realistic piano sounds (Bank and Lehtonen, 2010). Another popular method is the finite difference method, which transforms a partial differential equation into an ordinary differential equation that can be solved by approximating the derivatives in the differential equations (Bilbao, 2009). This method has been applied in (Chaigne and Askenfelt, 1994; Giordano and Jiang, 2004). Modal-based models have been frequently employed (Bank and Sujbert, 2005; Bank, Zambon and Fontana, 2010), with applications also in the commercial software

piano emulation Pianoteq, produced by Modartt². One of the main benefits of the modal string model is its complete control over the behavior of partials, which allows for the accurate matching of the sonic properties of a specific piano. However, this method assumes linearity and time invariance, rendering it unable to precisely model certain important effects, such as the resonance within the same string or nonlinear string vibrations.

A complete piano model, consisting of a hammer, string, soundboard, and the room/surrounding air, has been proposed by Giordano and Jiang (2004). The string is based on a linear stiff string equation struck by a hammer model proposed in (Stulov, 1995). The string is coupled to a simple soundboard model that includes the ribs and bridges such that the stiffness of the soundboard is position-dependent. The room model is the three-dimensional linearised Euler equation, which is driven by the vibration of the soundboard. All of the components are coupled and solved by finite difference methods. The main challenge is that there are a large number of room elements to be solved, and the soundboard model is highly dispersive and requires small time steps for stability. Multiple time steps are used: the soundboard and strings are iterated 6 and 4 times before an iteration is carried out for the room.

The string model is significantly enriched by Chabassier, Chaigne and Joly (2013), which also introduced a complete piano model. In this case, the string considers the stiffness thanks to shear waves, longitudinal waves arising from geometric nonlinearities, and the string cross-sectional area's rotation. The string model is based on Timoshenko beam theory and is modeled by a one-dimensional damped system of equations. The interacting force between the hammer and string is a nonlinear function of hammer compression. The hammer model is still based on (Stulov, 1995). By solving for the longitudinal displacement, the model can thus replicate the precursor signal and phantom partial phenomena. The soundboard is described using Reissner-Mindlin equations, which consider shear deformations through the thickness of a plate (Cohen and Grob, 2007). Similar to (Giordano and Jiang, 2004), ribs and bridges are modeled by modifying the thicknesses and material parameters at different parts of the soundboard to account for its heterogeneous material properties. Energy-preserving schemes guarantee stability, ensuring the conservation (or dissipation) of a discrete equivalent of the continuous energy across time steps (Chabassier and Joly, 2010). The soundboard is solved semi-analytically. The modes are first determined before it is coupled to the nonlinear string. To ensure stability for the coupling, the vertical string forces are solved at interleaved time steps before it is input to the soundboard. Finally, the soundboard is coupled to the air, which is explicitly solved by the finite element method.

Other works proposed modeling approaches only for specific components of the piano. Longitudinal displacements are modeled by Bank and Sujbert (2005) using the digital waveguide method and by Chabassier and Joly (2010) using the finite difference method and a Hamiltonian framework. Similarly, Bilbao, Torin and Chatziioannou (2015) introduced a hammer-string contact emulation using Finite Difference Time Domain (FDTD) schema and the Hamiltonian formulation for

²www.pianoteq.com

stability. In this case, the string is considered linear, and the Hunt and Crossley model is used for nonlinear losses.

Ducceschi and Bilbao (2019a) presented a prepared piano model including nonlinear geometric effects in the strings and the string-hammer interaction, using an energy-preserving scheme. The string and hammer interaction are modeled using finite difference schema, while the soundboard is in a modal form. A clavichord model features a modal representation that has been proposed by Jiolat et al. (2021). The geometrical deformation of the string is considered and modeled in the instrument’s dynamics using the Kirchhoff–Carrier nonlinear string model (Carrier, 1945), which includes the nonlinear geometric terms for computing the dynamic string tension. In this work, all couplings between transverse and longitudinal motions have been neglected.

Tan (2017) proposed a finite difference schema based on the Kirchhoff-Carrier string model to include the double transverse polarisation after investigating the phenomena experimentally. The second polarization gains energy and begins to vibrate if the first polarisation has a lower fundamental frequency and is excited by a sufficiently large amplitude. In addition, the double decay phenomenon is largely influenced by the vertical oscillations of the bridge, which have a resonant frequency between the first two frequencies of the string. Horizontal and rotational oscillations are instead insignificant.

Chabassier (2014) examined the influence of hammer shank vibration in the hammer-string contact. Here, an energy-based numerical scheme is provided and coupled to a global energy-preserving numerical solution for the whole piano. The numerical simulations show that hammer shank vibration clearly influences the spectrum of the piano sound of equally loud isolated notes. The finite element model for the soundboard of Mamou-Mani, Frelat and Besnainou (2008) emulates the effect of the tension of the strings downbearing on its vibration, considering the ribs, the bridges, and the crown of the soundboard. Downbearing is modeled using the prestress theory. The effects of downbearing in modal frequencies and mobility are investigated, as the importance of crown on the downbearing effects. The initial crown modifies largely the eigenfrequencies movements, introducing an initial decrease of them before they increase.

Finally, although several white box models of acoustic pianos have been proposed, many physical features are still missing, preventing comprehensive piano emulations. For example, key restriking, sympathetic strings, duplex stringing (Bank and Chabassier, 2018), are yet to be included in models, and they could lead to a significant increase in their computation complexity. However, most reviewed white box approaches for acoustic piano modeling still present an audible lack of fidelity, which is often evident upon listening due to the aforementioned missing features. In addition, the human ear is very sensitive to decay rates of sounds (Järveläinen and Tolonen, 2001), which are linked to dissipation phenomena that are yet not fully understood and modeled.

Neural network-based approaches for piano modeling have also been explored, though to a lesser extent. Autoregressive Discrete Autoencoders (ADAs) have been proposed to capture long temporal correlations in audio signals (Dieleman, Aäron and Simonyan, 2018), where the model unconditionally generates 25 seconds long

examples of piano music at 16 kHz sampling rate using three separately trained autoregressive models, each working at a different levels of abstraction. An encoder computes a high-level representation of the input, and then a decoder, consisting of a local autoregressive model and a modulator, generates the audio signal. The modulator guides the local model in reproducing the audio signal. Inspired by text-to-speech models such as Tacotron (Shen et al., 2018), Hawthorne et al. (2019) propose a WaveNet model for piano synthesis that is directly conditioned on MIDI data. To train the model, they developed the Maestro dataset (Curtis et al., 2018), which includes paired audio and MIDI recordings from human performances on an acoustic piano outfitted with electronic sensors. This approach requires two networks: the acoustic model to produce the conditional features and the neural vocoder to produce the final raw audio output.

DDSP concept has been adapted for acoustic piano modeling (Renault, Mignot and Roebel, 2022), featuring a system architecture based on high-level acoustic knowledge of the instrument, which allows the generation of polyphonic sound. This model generates audio at 16 kHz sampling rate, and it was trained using the Maestro dataset and conditioned by MIDI data, including inputs from pedals and a piano ID to emulate different types of pianos. The model generates the quasi-harmonic and noisy components for all individual voices and applies the estimated impulse response of the recording environment to their sum. The components of the model emulate the mechanism of sound generation in a piano. In this context, each sub-model contributes to the overall timbre of the sound. The inharmonicity sub-model, for instance, incorporates the *partials* of a piano key, which are not pure harmonics of the key’s fundamental frequency, primarily because of the stiffness of piano strings. Meanwhile, the detuner model accounts for the *partials* beating caused by the presence of multiple strings per key. Finally, a monophonic network models the spectral envelopes of the notes and tracks their temporal evolution.

2.3 Open Challenges in Neural Modeling of Musical Devices

In recent years, significant advancements have been made in neural modeling of distortion pedals and amplifiers, achieving perceptually accurate results and real-time computation. The proposed modeling solutions also present low latency due to the sample-by-sample processing capabilities of the employed neural models based on recurrent neural networks. However, with existing neural modeling approaches, other types of audio effects can not be accurately and efficiently modeled, particularly time-variant effects such as dynamic range compressors. These devices exhibit time dependencies that complicate neural network learning, necessitating either large networks or a substantial amount of past input signals. A few works have proposed alternative neural architectures capable of achieving sufficiently good accuracy with some of these effects. However, they still present computational complexity or causality limitations that prevent the further development of these ideas as usable neural audio effects. For example, neural methods for dynamic range compressors, as proposed by (Steinmetz and Reiss,

2022), cope with challenging time-behavior modeling by utilizing networks that use large input-output segments of audio samples. This is detrimental to the latency of the compressor emulation, as well as to the time resolution at which control parameters can be changed. Alternatively, low latency and efficient networks have been obtained at the expense of significant biases in the models, which tend to exhibit unsatisfactory accuracy (A. Wright and Välimäki, 2022). Dynamic range compression, despite appearing to be a simple audio effect that only applies a gain reduction factor, is, in fact, a complex and challenging device to model with neural techniques. It presents significant nonlinearities and time-variant behavior, which is why it has been selected as the main *processor* investigated in this thesis.

Tracking temporal dependencies over numerous past network inference steps is crucial for neural modeling of dynamic range compression, particularly with optical analog compressors, due to their slow attack and release times. However, improving this aspect while working with relatively short input sample segments for the neural model presents a significant challenge. The ability of CNN-based networks to remember information from past steps is confined to the kernel's receptive field, which means that the extracted features are local. Consequently, capturing long-term dependencies using short input segments is not possible. Attention layers, successfully employed in many sequence-to-sequence modeling tasks, leverage sequence similarities across all temporal steps, thereby enhancing the modeling of long-term dependencies. However, this approach is not suitable for audio effects that require real-time processing, and it can be utilized only for offline applications. Attention-based models typically learn to predict the entire output sequence in one inference, given the complete input sequence. If the prediction is broken down into smaller sequences, as needed for audio effects that operate on small buffers of input/output samples, the attention mechanism fails to maintain dependencies across these sequences because it is memoryless. Recurrent Neural Networks (RNNs), thanks to their internal state mechanism, enable the modeling of audio effects even when processing sequences of short segments of audio samples. However, RNNs are limited by their poor ability to maintain information in internal states over long periods, which diminishes their practical effectiveness in handling long-term dependencies. Although variants of vanilla RNNs have been proposed to improve their memory capabilities, these models still struggle to retain information over extended periods. Further developments in their memory mechanisms or the creation of alternative, more sophisticated neural architectures are necessary to model a broader range of audio effects. Traditionally, RNNs feature nonlinearities in their state computations, which have been shown to be less effective for long-term memory tasks. Recent advancements indicate that removing nonlinearity from the recurrent path can enhance the networks' memory capabilities, leading to the development of newer architectures focusing on this improvement. In particular, State Space Models (SSMs) have shown particularly promising results.

Another aspect that needs further improvement in neural modeling approaches for *processors* is the conditioning of the networks, which is required for a full emulation of parametric audio effects. Although several proposed models have attempted to incorporate the effect's parameters into the neural network, these parameters are often treated as generic inputs, similar to the audio samples,

which at times yield limited results in terms of model completeness, temporal resolution, accuracy, and the ability to predict the correct audio alteration for parameter combinations not included in the training data. A dedicated conditioning technique might offer a more robust and flexible solution, particularly if it can be applied consistently across a range of musical devices. Indeed, recent generic modeling techniques demonstrated that neural networks can be effectively influenced by conditioning information using ad-hoc layers. However, there is still a need for a more efficient and structured strategy for the case of audio effects. The Dynamic Range Compressor (DRC) features four controllable parameters: two predominantly influence its nonlinear behavior, while the other two affect the time-variant characteristics. The combination and interdependence of these parameters create numerous challenging configurations, making the compressor one of the most complex and representative parametric audio effects to model with a single network.

When examining neural models of *generators* in the literature, there are fewer works compared to those on *processors*. The focus has mostly been on drum and percussion modeling. Synthesizing harmonic sounds with longer and more complex timbral structures remains a challenge for neural techniques. Harmonic sounds have primarily been modeled using the Differentiable Digital Signal Processing (DDSP) framework, which has shown good accuracy and flexibility. In particular, the guitar and piano were among the first instruments to be modeled (Renault, Mignot and Roebel, 2022; Jonason et al., 2024; Wiggins and Kim, 2023). However, these models are limited regarding the physical features of the instruments they include; many features are still missing, presenting significant potential for expansion and refinement. The proposed models learn from datasets comprising recorded songs and melodies. This methodology introduces considerable complexity to the task and may impede the network’s ability to generalize to scenarios not represented in the datasets. For example, the reproduction of isolated notes or chords can be particularly challenging if essential harmonic content is masked by other simultaneous sounds in the dataset. Furthermore, capturing perceptually critical attributes, such as the decay rate of a note, can prove difficult and demands precise modeling. Moreover, learning complex patterns from recorded songs necessitates a substantial number of internal trainable parameters and an extensive amount of training data. Therefore, these architectures result in a high computational cost to function. Lastly, the methodology to train such models to achieve an interactive and playable synthesizer is still to be determined. The piano, with its nonlinear and time-variant characteristics, is a well-known challenge modeling task. Consequently, it has garnered significant interest and is often considered a benchmark for musical instrument modeling research. In this thesis, the piano has been chosen as the primary subject for the investigation of *generators*. Additionally, the extent to which these techniques can be applied across different musical instruments with minimal changes would benefit from further experimentation.

Generative adversarial network (GAN) approaches have been applied significantly to drum sound generation, delivering realistic results in real-time. However, similar to other generative frameworks, they typically produce extended audio seg-

ments without dependency between consecutive segments, with durations ranging from hundreds of milliseconds to several seconds. If these approaches were used for synthesizing playable emulations of musical instruments, the resulting delay in the action-to-sound response would be unacceptable, thus limiting their use to offline sound generation. The combination of an autoregressive model with an acoustic model that maps MIDI information to audio features, guiding the sound generation process as proposed by Hawthorne et al. (2019), appears promising and may open new opportunities in this context. However, the ability to condition such a model on user input — of the kind typically provided to musical instruments — seems to be limited. Moreover, the options to reduce latency are also restricted because the additional mapping stage between MIDI data and audio features can introduce significant delays. DDSP-based approaches, as shown by recent proposals, appear to offer a superior method for the data-driven modeling of acoustic musical instruments. This technique enables the granular modeling of specific sub-components, which is useful for complex systems and is capable of achieving low latency and generating complex waveforms from MIDI-like inputs that describe control signals. The piano has been extensively investigated, as discussed in the works reviewed previously, with several white box models and analyses available that describe its functioning. Integrating these insights into gray box models can be beneficial. Similarly to dynamic range compressors, acoustic pianos present a high number of different scenarios due to the variety of input stimuli and their combinations. An efficient and structured strategy is required to integrate these variables into a single model—a strategy that may differ from the one designed for *processors*.

2.3.1 Modeling Requirements

Musical devices are physical objects that, when played by performers, generate and/or process sound. While some models of musical devices may not be intended for live human interaction, designing a playable emulation presents significant challenges due to stringent functional requirements. A model designed for live playability can generally be adapted for other uses; however, the reverse is not always true. For instance, models created solely for offline emulation may not be suitable for live performance. This thesis focuses solely on the numerical model required to generate synthetic sound, excluding the physical interface, which is beyond its scope. Moreover, it exclusively focuses on modeling techniques that enable the playable emulation of musical instruments. Three key requirements must be considered: *real-time* computation, *interactive* capability, and *low latency*. These requirements should be addressed during the early stages of conceptualizing modeling techniques, as certain approaches may inherently conflict with live playability and cannot be easily modified once chosen. Taking these factors into account undoubtedly increases the complexity and challenges involved in searching for and designing suitable neural modeling techniques.

An emulation of a musical device is no different from other *real-time* audio applications in that the correctness of the output depends not only on its value but also on the timing of its production; it must not exceed a given deadline. Real-

2.3. Open Challenges in Neural Modeling of Musical Devices

time audio systems operate with a constant sampling rate, which, for musical devices, is at least 44,100 Hz or 48,000 Hz. The sampling rate dictates the temporal interval within which the system must produce one or more output samples, a requirement contingent on the input/output buffering configuration. The deadline must consistently be met to avoid audible glitches, classifying musical devices as hard real-time systems. Whether a model can compute audio samples within real-time constraints depends on the computational complexity of the synthesis algorithm (i.e., the model) and the selected execution platform. Both can typically be gauged in terms of the number of Floating-Point Operations per Second (FLOPS). However, this metric may not be fully informative, as hardware operators may offer a level of parallelism that certain algorithms cannot exploit due to strictly sequential data dependencies. Moreover, complex models that fulfill real-time constraints by utilizing nearly all computational resources of a powerful workstation may be of interest from a research perspective but have limited short-to medium-term applicability for general users. After all, dozens of software emulations of musical devices can generally run concurrently on consumer-grade personal computers, commonly hosted on Digital Audio Workstations (DAWs). In contrast, consumer digital emulations running on dedicated hardware, which often benefit from real-time operating systems, operate on digital signal processors with significantly less computational power than personal computers. Low- to mid-end digital signal processors typically used for audio can handle up to 100 MFLOPS. This figure serves as the upper-bound reference for the neural modeling approaches explored in this research work.

To be playable, emulations of musical devices must be *interactive*, offering users the same inputs found on their physical counterparts. These inputs are generally time-continuous and value-continuous; therefore, modeling algorithms must allow the updating of these inputs with every generated audio sample. However, many current implementations of musical device emulations employ much lower time and value resolutions, as these still provide a perceptually continuous sonic variation in response to continuous multi-parametric input variations (Rovan et al., 1997). Nonetheless, for the models proposed in this thesis, our goal is to achieve inputs and parameters that can be updated at the single-sample level. Value resolution is generally not a challenge since modern computers and signal processors can implement entire algorithms using floating-point numerical representations. With respect to the type of inputs, *generators* and *processors* differ significantly: *generators* have inputs—whether discrete or continuous—that specify the characteristics of the sound to be generated, including both timbral and musical aspects such as pitch. *Processors*, on the other hand, have discrete or continuous parameters that determine aspects of sound processing, in addition to the input audio signal that must be altered. For acoustic musical instruments, the input stimuli provided by the player may vary widely, depending on the principle of sound generation and how the performer transfers energy to the instrument’s body. Capturing these stimuli accurately is challenging without invasive sensing technologies. Moreover, the possibility of altering the timbre of acoustic instruments is somewhat limited, especially when compared to analog sound synthesizers. In such cases, it is common to use standardized digital formats

for the stimuli, such as MIDI. MIDI provides a symbolic representation of music for aspects like pitch and duration and can also emulate continuous-like parameters for other musical inputs that musicians apply to the instrument, such as the use of piano pedals.

While real-time constraints define the data throughput that the implementation of musical device emulation must handle continuously, *latency* refers to the time delay between the input and corresponding output. Acoustic and purely analog musical devices typically exhibit negligible latency. Similarly, digital musical devices present a minimum latency of at least two sampling periods, which is also negligible. However, samples are often processed in batches for computational and interfacing efficiency reasons (involving analog-to-digital and digital-to-analog converters). The input and output buffering associated with batch processing introduces latency. Generally, equal input and output buffer sizes are used, and the introduced latency amounts to twice the buffer size multiplied by the sampling period. Excessive latency hinders the musical usability of most devices, particularly for performances and live audio applications. Studies suggest that latency for digital musical instruments (i.e., *generators*) should be no more than 10 ms (Wessel and M. Wright, 2002). However, smaller latencies can be noticeable, though often manageable, especially for struck instruments with sharp attack sounds like percussion, as well as in audio effects. Therefore, sub-millisecond latencies are generally preferred. Besides input/output buffering, latency can also arise from the white or black box models of musical devices that are causal but not memoryless. They use past input information to compute the current output, necessitating the implicit or explicit introduction of an internal buffering stage that contributes to latency. The research work described in this thesis has been focused on designing and experimenting with neural modeling approaches for musical devices that can accommodate a maximum latency of 64 audio samples, equivalent to 1.33 ms at 48 kHz sampling rate.

2.4 Summary

This chapter presented the primary techniques for modeling physical systems, emphasizing the differences between the main approaches: *white* box, *black* box, and *gray* box. The chapter focused particularly on the motivations for employing the *black* and *gray* box method, especially within the context of machine learning. It then provided an extensive overview of the literature, distinguishing between the categories of *processors* and *generators*. In the final part, the chapter discussed the open challenges and opportunities identified through the review and highlighted the essential requirements to consider for modeling musical devices: *latency*, *real-time* computation, and *interactivity*.

Chapter 3

Methodology

This chapter provides an overarching summary of the research methodology used in the papers included in this article-based thesis. The chapter begins with a description of the general approach taken, including the different neural network layers used as basic building blocks in the proposed architectures, as well as the computational tools used for implementation and experimentation. The three research questions detailed in Section 1.3 have been primarily addressed separately for the two categories of musical devices under scrutiny: *processors* and *generators*. The specific methodologies for each category are outlined in their respective subsections. Each subsection is divided into five parts, describing the device selected for the modeling tasks, the dataset collection, the neural architectures, the training strategy, and the approach to the evaluation method.

3.1 General Approach and Modeling Tools

In the previous chapter, current challenges and limitations of neural techniques for modeling musical devices were highlighted, particularly when aiming to obtain emulations that fulfill requirements for real-time, interactivity, and low latency. The selection of the main modeling targets - a *processor* and a *generator* - investigated and modeled in this thesis is guided by three key factors:

1. The musical device must exhibit time-variant and nonlinear characteristics.
2. Current neural modeling of such devices should still face significant challenges with respect to our objectives and selected requirements.
3. Adequate and appropriate data for study must be publicly available or obtainable directly from devices that are readily accessible.

Analog Dynamic Range Compressors (DRCs) and Acoustic Pianos meet all these criteria. Both exhibit time-variant and nonlinear characteristics, and current neural modeling techniques struggle to emulate these devices accurately. Moreover, a significant amount of ongoing research is focused on both. Regarding data availability, there are example of publicly accessible datasets for acoustic pianos, but only one for a well-known optical DRC. At the start of this research, predicting

whether such datasets would be suitable and sufficient for the project’s duration was challenging, as needs might evolve and new requirements emerge over time. Therefore, the selection of these devices was also influenced by the fact that the Department of Musicology at the University of Oslo owns a variety of analog compressors and acoustic pianos, including those that are electromechanically actuated. These instruments are maintained in good condition and are readily accessible, allowing for the collection of additional or different data as new needs appear.

Collecting data from the readily available devices is a delicate task. Since this task involves repeated and often time-consuming human operations, there is a non-zero probability of inaccuracies or even errors, which then risk propagating and biasing the evaluation of a model, whether positively or negatively. Therefore, systematic and robust routines were tested and established for data collection. Moreover, the equipment used to gather data requires calibration, along with eventual compensation for coloration and/or biases it may introduce, which are identified during preliminary trial runs. The selection of the type, combination, and duration of input stimuli—whose processed or generated sounds need to be recorded—is carefully chosen. This ensures the dataset, while of a manageable size, still represents the device as comprehensively as possible since data-driven techniques require a dataset that captures the device’s range of behaviors. Additionally, the collected data is examined for integrity and computationally refined to eliminate minor inaccuracies and temporal misalignments typically incurred during manual operation. Publicly available datasets are also scrutinized for integrity, compatibility, and consistency with data directly collected from devices. This is crucial to predict how differences might influence the outcome of model evaluations, allowing for more robust comparisons of results from models trained on distinct datasets. Such diligence is necessary because a standard for collecting and formatting datasets from musical devices for modeling applications has yet to be established.

The central and essential part of the methodology pertains to the investigation and design of suitable neural architectures for the modeling task. This typically begins with baseline models or those considered state-of-the-art, including architectural variations deemed appropriate based on preliminary experiments or findings from other works in non-sound and music application domains. Improvements to the architecture are introduced gradually and systematically evaluated, testing each variation in a representative set of cases. As previously discussed, the requirements for real-time processing, interactivity, and low latency are integral to the entire process. These criteria are used to screen out options that do not adequately meet these standards. However, designing the architectures is only part of the equation, as the extent to which they can accurately model a musical device also depends on the training strategy, selection of appropriate loss functions, and optimizing parameters such as learning rates, batch sizes, the division of datasets into training, validation, and test sets. These aspects, too, must be expressly designed and optimized for the task at hand.

Models and training frameworks must then be implemented and initially tested with relatively small datasets to assess correctness and stability. The

implementation framework must be efficient: training can span days or weeks when dealing with audio device modeling using raw audio for input and output, particularly with large datasets. Furthermore, owing to the large amounts of data involved, special strategies and schemes must be employed to ensure that the computational resources required, especially memory, remain within the limits of the available workstations. This often leads to bypassing utilities offered by deep-learning frameworks and re-implementing similar functions in a more memory-efficient manner, such as breaking down training algorithms to reduce memory use. Given the time-consuming nature of training, models are continuously saved during training. This allows for rolling back to earlier stages, resuming experiments, or avoiding the risk of overfitting that could degrade previously optimal models.

Training processes typically measure modeling accuracy exclusively against the loss function selected for training. However, a broader range of quantitative metrics is employed to assess the accuracy of the predictions on the test set more comprehensively. The test set should represent new data, not utilized for training, that can be used to evaluate the model’s generalization ability—a challenging task to conceptualize compared to traditional classification or regression in machine learning. In addition to these metrics, informal listening tests and visual inspection of various signal plots are carried out to discard modeling options that produce audible artifacts, glitches, or other short-duration inaccuracies, which loss functions and other average-based evaluation metrics may not capture effectively. The chosen metrics assess both the quality and limitations of the models and are carefully selected to reflect different aspects of the predicted sound. They include both time-based and frequency-based errors and are tailored for each specific modeling task. When possible, perceptually relevant evaluation metrics are also employed. Mature models that perform the best according to the quantitative metrics—following extensive development and deemed sufficiently accurate—are subsequently subjected to listening tests. These tests provide a final, comprehensive assessment of model fidelity.

Comparing with baselines or state-of-the-art models is an essential component of the evaluation method. However, figures published in existing literature are often unsuitable for direct comparison, as they may be based on slightly different modeling cases or datasets. Therefore, a number of state-of-the-art models and baselines have been re-implemented and trained in a manner that allows for a fair comparison of qualitative results. This also ensures the generation of comparable data for listening tests, especially when such models are not readily available off the shelf.

The overarching methodological approach to this research, which is applied to both *processors* and *generators*, is illustrated in Figure 3.1. This figure outlines the iterative nature of the process: the outcomes of each stage inform previous stages for subsequent design choices, which may lead to improvements, additional validation, or further refinement of the contributions of this work. Iterations may also include collecting more data from the same or similar devices to gauge how well the proposed neural modeling techniques generalize to the task of modeling musical devices. The scope and frequency of these iterations are inherently limited by the time required to generate results that influence the next phase of development,

which, in turn, is constrained by the computational resources available for this research, as detailed in Section 3.1.2.

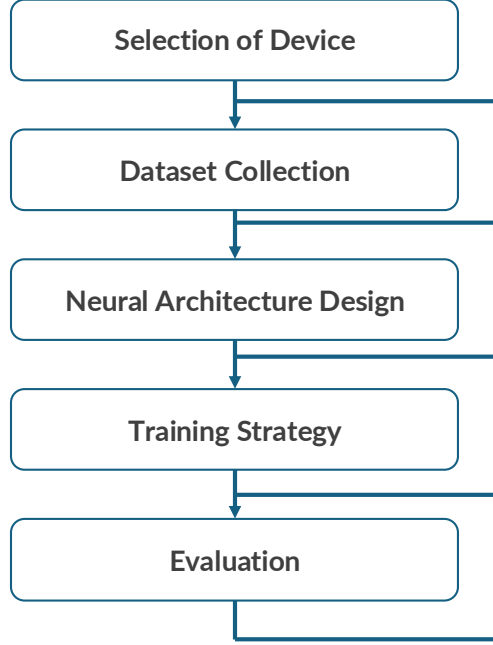


Figure 3.1: Experimental flow: selection of the musical device, the data collection, design of learning strategy, and evaluation. The investigation includes several trials that can lead to reconsidering previous steps.

3.1.1 Types of Neural Network Layers

The research presented in this thesis investigates neural modeling techniques, leading to the development of novel architectures and modeling approaches. We utilize various types of neural network layers as foundational building blocks. This section provides a formal review of the main types of neural network layers, which include convolutional, recurrent, attention, and state-space layers. These layers can be viewed as evolved from the simplest and most traditional type, the feed-forward Fully Connected (FC) layer, which is mathematically defined as follows:

$$\mathbf{y} = \phi(\mathbf{W}\mathbf{x} + \mathbf{b}), \quad (3.1)$$

where $\mathbf{y}$ is the output, $\phi(\cdot)$ is the activation function, $\mathbf{x}$ the input, $\mathbf{W}$ and $\mathbf{b}$ are the weights matrix and bias terms. This formulation is equivalent to Equation (2.1.2) but considers the whole layer of neurons rather than a single one. Therefore, the output and bias term are now vectors with a size equal to the number of neurons N , and the weight is a matrix of dimension $N \times I$, where I denotes the size or dimensionality of the input.

Convolutional Layers

As the name suggests, convolutional layers (LeCun et al., 1998) perform the convolution operation. The convolution is a linear operation that involves multiplying a set of weights with the input. The multiplication is performed between an array of input data and a two-dimensional array of learnable weights, called a filter or a kernel. The filter is smaller than the input data, and the type of multiplication applied between a filter-sized patch of the input and the filter is a dot product. A dot product is an element-wise multiplication between the filter-sized patch of the input and filter, which is then summed, always resulting in a single value. A filter smaller than the input allows the same filter to be multiplied by the input array multiple times at different points on the input. Specifically, the filter is applied systematically to each overlapping part or filter-sized patch of the input data, left to right, top to bottom. Considering a 3D matrix $\mathbf{x}$ with dimensionality $c_I \times r_I \times w$ as input, the layer applies a filter $\mathbf{F}$ with dimensionality $c_F \times r_F \times w$ dimension, where w is identical for both input and the filter. The filter moves upon the matrix's c_I , r_I dimension from left to right, multiplying that part of the input with the filter accumulating the results. The filter moves with a step size called stride. If considering a stride of one, the output has dimensionality $(c_I - c_F + 1) \times (r_I - r_F + 1)$. The output $\mathbf{y}$ of the convolutional layer represents specific features in the input and is called a feature map. Therefore, the feature map is computed as:

$$\mathbf{y} = \phi(\mathbf{F} * \mathbf{x} + \mathbf{b}), \quad (3.2)$$

where $\mathbf{b}$ the bias terms, and the $\phi(\cdot)$ is the activation function, which, though not commonly used in convolutional layers, can be included.

The convolutional layer described earlier corresponds to the most common case, which is 2D convolution, typically used for processing data with two independent variables, such as images. However, since the primary focus of this work is audio signals, which have time as the only independent variable, the convolution is applied in just one dimension.

Recurrent Layers

In recurrent layers, information flows bi-directionally (Rumelhart, Hinton and Williams, 1985), whereas in layers previously discussed, information is fed only in a forward direction - from input to output. Within recurrent layers, the neurons maintain a hidden state, which is updated at each time step based on the current input and the previous hidden state. This mechanism creates a feedback loop which helps to keep information from past inputs. In particular, at every step corresponding to the sampling instant of the sequential data that the network is processing, these layers process the input by computing both the output and an update to their internal states, which will influence the computation at the next step. This characteristic allows these layers to exhibit temporal dynamic behavior. The layer utilizes internal states, which act as memory, to model sequential information of arbitrary length. In theory, the layers should remember all past information, but empirical studies have demonstrated that their memory capacity is typically limited to a fairly short temporal span (Bai, Kolter and

Koltun, 2018). At each time step n , the output $\mathbf{y}_n$ and the states $\mathbf{h}_n$ are computed as follows:

$$\begin{aligned}\mathbf{h}_n &= \phi(\mathbf{W}\mathbf{h}_{n-1} + \mathbf{U}\mathbf{x}_n + \mathbf{b}) \\ \mathbf{y}_n &= \mathbf{V}\mathbf{h}_n + \mathbf{c},\end{aligned}\tag{3.3}$$

where $\mathbf{W}$, $\mathbf{U}$ and $\mathbf{V}$ are weights' matrices, and $\mathbf{b}$ and $\mathbf{c}$ the bias vectors, which are learned during the training process. The function ϕ is typically nonlinear, such as the hyperbolic tangent. The calculation of $\mathbf{h}_n$ is based on the current input and the hidden state from the previous time step. This helps the layer retain information from previous time steps, which is used when processing the current input. In the recurrent layer, there are three types of connections: input-to-hidden connections, hidden-to-hidden recurrent connections, and hidden-to-output connections. These connections are parameterized by the matrices $\mathbf{U}$, $\mathbf{W}$, and $\mathbf{V}$ respectively. Notably, these weight matrices are shared across all time steps.

The vanilla recurrent layer presents issues with vanishing or exploding gradients (Y. Bengio, Simard and Frasconi, 1994): gradients become either too small or too large during back-propagation due to recurrent connections. Indeed, the gradient of a parameter depends on several previous inputs and outputs. Consequently, the gradient can exponentially decay or increase as it propagates backward through time. This problem becomes more pronounced when multiple layers are involved, resulting in earlier layers ceasing to learn effectively. To address this issue, two variants called Long Short Term Memory (LSTM) (Hochreiter and Schmidhuber, 1997) and Gated Recurrent Unit (GRU) (Cho et al., 2014) have been proposed. These layers improve the memory capacity by incorporating internal mechanisms known as gates, which regulate the flow of information. The following equations regulate the functioning of LSTM layers:

$$\begin{aligned}\mathbf{f}_n &= \gamma(\mathbf{W}_f\mathbf{h}_{n-1} + \mathbf{U}_f\mathbf{x}_n + \mathbf{b}_f) \\ \mathbf{j}_n &= \gamma(\mathbf{W}_x\mathbf{h}_{n-1} + \mathbf{U}_x\mathbf{x}_n + \mathbf{b}_x) \\ \mathbf{c}'_n &= \theta(\mathbf{W}_c\mathbf{h}_{n-1} + \mathbf{U}_c\mathbf{x}_n + \mathbf{b}_c) \\ \mathbf{c}_n &= \mathbf{f}_n \circ \mathbf{c}_{n-1} + \mathbf{j}_n \circ \mathbf{c}'_n \\ \mathbf{h}_n &= \mathbf{y}_n \circ \theta(\mathbf{c}_n) \\ \mathbf{y}_n &= \gamma(\mathbf{W}_y\mathbf{h}_{n-1} + \mathbf{U}_y\mathbf{x}_n + \mathbf{b}_y),\end{aligned}\tag{3.4}$$

where $\mathbf{f}$ is the forget gate, $\mathbf{j}$ the input gate, $\mathbf{c}'_n$ is the cell input activation vector, $\mathbf{c}$ is the cell state vector, $\mathbf{h}$ is the hidden state vector, $\mathbf{y}$ the output gate, and $\circ$ is the Hadamard product. The forget gate determines what information from previous steps should be kept or discarded from the cell state vector. The input gate selects what information from the current step is relevant and, hence, should be added to the cell state vector. The output gate determines the value of the next hidden state. Additionally, the cell state is updated by multiplying the forget and input vectors with new values that the neural network deems relevant. The forget vector and input vector are obtained using the sigmoid function, denoted by γ . For example, 0 values in the input or forget vectors indicate that the associated information has to be discarded, while values of 1 indicate that all information has

to be kept. Furthermore, the θ represents the hyperbolic tangent function used to bound the values in the cell state between -1 and 1 . Finally, $\mathbf{W}$ and $\mathbf{U}$ are the weights' matrices, and $\mathbf{b}$ are the vector bias terms, which are learned during the training process.

GRU layers are similar to LSTMs but remove the cell state. Indeed, GRU has only two gates. The update gate functions similarly to the forget and input gates of LSTM, as it determines which information to discard and which new information to add to the state vector. The reset gate, on the other hand, determines how much past information should be forgotten. Together, these gates regulate the flow of information in the GRU network. The equations governing the GRU functioning are as follows:

$$\begin{aligned} \mathbf{z}_n &= \sigma(\mathbf{W}_z \mathbf{y}_{n-1} + \mathbf{U}_z \mathbf{x}_n + \mathbf{b}_z) \\ \mathbf{r}_n &= \sigma(\mathbf{W}_r \mathbf{y}_{n-1} + \mathbf{U}_r \mathbf{x}_n + \mathbf{b}_r) \\ \mathbf{y}_n &= \mathbf{z}_n \circ \mathbf{h}'_n + (1 - \mathbf{z}_n) \circ \mathbf{y}_{n-1} \\ \mathbf{h}'_n &= \theta(\mathbf{W}_h(\mathbf{r}_n \circ \mathbf{y}_n) + \mathbf{U}_h \mathbf{x}_n + \mathbf{b}_y). \end{aligned} \quad (3.5)$$

where $\mathbf{z}$ is the update gate, $\mathbf{r}$ the reset gate, $\mathbf{y}$ the output vector, and $\mathbf{h}'_n$ the candidate activation vector.

Other recent variants have emerged from empirical studies that have shown that the limitations of recurrent layers can be mitigated by removing the nonlinear activation function ϕ in Equation (3.3), which is used to compute the hidden state (Orvieto et al., 2023). When a linear activation function is used, the memory capacity of the layers improves significantly. Consequently, a new type of layer called the Linear Recurrent Unit (LRU) has been introduced. Additionally, to enhance stability, the LRU incorporates initialization with a complex-valued diagonal recurrent matrix and an exponential parameterization. At the same time, a normalization scheme for the hidden states is designed to increase efficiency and accuracy.

Attention Layers

Attention layers evoke the idea of attention and perform a weighted mean reduction of the input: each element is weighted before contributing to the mean (Luong, Pham and Manning, 2015). The basic idea of attention is to allow the model to focus on different parts of the input when producing each element of the output. Instead of treating all input tokens equally, the attention mechanism dynamically assigns different weights (attention scores) to different parts of the input.

The attention layer takes as input the entire sequence of feature vectors $\mathbf{X}$, which is a matrix with dimensionality $l \times d$, where l represents the length of the sequence and d represents the number of features. The layer performs three projections to the input using learned weight matrices W_Q , W_K , and W_V , respectively, and creates three vectors: the query, the values, and the keys. Afterward, the layer computes the dot product of the query and the key, scaled by the feature size, obtaining the so-called attention scores. The scaling helps stabilize the training by preventing excessively large values. Finally, the softmax

function is applied to the attention scores to obtain the attention weights and use them, and it computes a weighted sum of the value vectors.

$$\begin{aligned} \mathbf{a} &= \text{softmax}\left(\frac{\mathbf{X}\mathbf{W}_Q(\mathbf{X}\mathbf{W}_K)^T}{d}\right) \\ \mathbf{y} &= \mathbf{a} \cdot \mathbf{X}\mathbf{W}_V, \end{aligned} \quad (3.6)$$

where $\mathbf{a}$ is the attention weights vector, d is the dimensionality of the input (or number of features), and $\mathbf{y}$ is the output vector. The softmax function is defined as:

$$\text{softmax}(\boldsymbol{\chi}) = \frac{e^{\chi_j}}{\sum_{k=1}^K e^{\chi_k}}, \quad (3.7)$$

with $\boldsymbol{\chi}$ the input vector.

Practically, the keys and query determine how to weigh the values according to the attention mechanism. The attention mechanism outputs a tensor one rank less than the keys, giving a scalar for each key corresponding to the attention the query should have for the key.

State-Space layers

State-space layers are based in the State-space formulation, introduced in 2.1.1 and operate using a continuous-time representation:

$$\begin{aligned} \frac{dh(t)}{dt} &= \mathbf{A}h(t) + \mathbf{B}x(t) \\ y(t) &= \mathbf{C}h(t) + \mathbf{D}x(t), \end{aligned} \quad (3.8)$$

where h and y are the states and output at a time t , and $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$, and $\mathbf{D}$ are matrices expressing linear mappings between the input $x(t)$, and the states $h(t)$ of the system. The $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$, and $\mathbf{D}$ dimensionalities are $(N \times N)$, $(N \times 1)$, $(1 \times N)$, and (1×1) , respectively. Discretization with a step size Δ is performed for $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$, and $\mathbf{D}$ using the bilinear method or zero-order hold, to obtain the following time-discrete linear recurrent system:

$$\begin{aligned} \mathbf{h}_n &= \bar{\mathbf{A}}\mathbf{h}_{n-1} + \bar{\mathbf{B}}x_n \\ y_n &= \bar{\mathbf{C}}\mathbf{h}_n + \bar{\mathbf{D}}x_n, \end{aligned} \quad (3.9)$$

or a convolutional form:

$$\begin{aligned} y &= \bar{\mathbf{K}} * x + \bar{\mathbf{D}}x \\ \bar{\mathbf{K}} &= \mathbf{C}\bar{\mathbf{A}}^n\mathbf{B} \end{aligned} \quad (3.10)$$

where h_n has the similarity with the recurrent hidden state in Equation (3.9), and $\bar{\mathbf{A}}$, $\bar{\mathbf{B}}$, $\bar{\mathbf{C}}$, and $\bar{\mathbf{D}}$ are calculated as functions of $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$, $\mathbf{D}$, and Δ . $\bar{\mathbf{K}}$ is the convolution kernel, and Equation (3.10) is a single convolution that can be computed very efficiently via Fast Fourier transform (FFT) since the convolution in the time domain is equivalent to multiplication in the frequency domain. Thus,

the FFT is used to accelerate convolutional operations. $\bar{\mathbf{A}}$, $\bar{\mathbf{B}}$, $\bar{\mathbf{C}}$, $\bar{\mathbf{D}}$ and Δ are learnable.

A frequently used State-Space Model (SSM) is called the Structured State Space (S4) model, which owes its ability to capture long-term dependencies to the use of a special state matrix $\mathbf{A}$ known as the "HiPPO matrix" (Gu, Dao et al., 2020), and is defined as:

$$\mathbf{A}_{n,k} = - \begin{cases} (2n+1)^{1/2}(2k+1)^{1/2}, & \text{if } n > k \\ n+1, & \text{if } n = k \\ 0 & \text{if } n < k \end{cases} \quad (3.11)$$

This matrix is designed to encode all the past input history in $h(t)$, finding a map from $x(t)$ to a higher dimensional space $h(t)$ that represents the compression of the history. The matrix allows the model to be conceptualized as a convolutional model that decomposes an input signal onto an orthogonal system of smooth basis functions. In this way, the state $h(t)$ encodes the history of the signal.

A variant of S4 is S4D (Gu, Gupta et al., 2022), which enhances the efficiency of the matrix multiplication with $\bar{\mathbf{A}}$ in Equation (3.10) by parameterizing its state matrix as a diagonal matrix. This parametrization allows the basis kernels to be represented by closed-form formulas in the form of normalized Legendre polynomials $L_n(t)$. As a result, the SSM decomposes the input signal $x(t)$ onto an infinitely long set of basis functions that are orthogonal with respect to an exponentially decaying measure.

Notably, in the S4 and S4D layers, all matrices are independent of the input. To solve this problem, another variant called the Selective State Space S6 (Gu and Dao, 2023) has been proposed. The model is designed to selectively propagate or forget information along the time axis depending on the current input. In the newer S6 proposal, $\mathbf{A}$ and $\mathbf{D}$'s parameters are kept independent of the input and learned during training, as in S4 and S4D. Instead, Δ , $\mathbf{B}$, and $\mathbf{C}$ are dependent on the input and computed using a linear FC layer. Given these changes, the conventional convolution form described in Equation (3.10) cannot be utilized with S6. In this new formulation, Δ acts as a control parameter, balancing the emphasis on either the current input or the existing state. A large value for Δ effectively resets the state $\mathbf{x}_n$ to prioritize the current input, whereas a small value causes the model to deemphasize the current input in favor of the previous state. Accompanying the S6 variant is a gating mechanism known as 'mamba,' which doubles the dimensionality of the input. One part of this expanded input is processed through a convolutional layer, followed by a Swish activation function, before being passed to the S6 layer. The other part undergoes a separate Swish activation and is subsequently multiplied by the output of the S6 layer.

Other parametrization for $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$, $\mathbf{D}$ have also been proposed (Alonso, Sieber and Zeilinger, 2024).

Activation Functions

The activation functions in the neurons of Artificial Neural Networks (ANNs), as illustrated by the single neuron in Equation (2.1.2), as well as in the more

complex layers described in previous sections, are used to generate each neuron's output by introducing memoryless nonlinearity to the network. Indeed, when the activation function is nonlinear, it has been shown that a two-layer neural network can serve as a universal function approximator. This principle is known as the Universal Approximation Theorem (Hornik, Stinchcombe and White, 1989). Nonlinear activation functions enable the network to capture complex behaviors embedded in the input-output pairs used for training. Although not a universal rule, most of the time, it is beneficial to use an activation function that bounds the output within a limited range, as gradient-based training methods tend to be more stable under these conditions. Without bounded outputs, the weight-updating process can become unstable. Another desirable property of activation functions is differentiability, which is essential for gradient-based optimization; nondifferentiable functions are used but can pose challenges during training. To date, a variety of activation functions has been proposed. The most common ones, which are also employed in the neural architectures presented in this thesis, are detailed below:

- Sigmoid

$$\phi(x) = \frac{1}{(1 + \exp(-x))} \quad (3.12)$$

- Hyperbolic Tangent

$$\phi(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} \quad (3.13)$$

- Softsign

$$\phi(x) = \left(\frac{x}{|x| + 1} \right) \quad (3.14)$$

- Rectified Linear Units (ReLUs)

$$\phi(x) = \max(0, x) \quad (3.15)$$

- Gaussian Error Linear Unit (GELU)

$$\phi(x) = x \cdot \frac{1}{2} \left[1 + \operatorname{erf}(x/\sqrt{2}) \right], \quad (3.16)$$

where

$$\operatorname{erf} = \frac{2}{\sqrt{\pi}} \int_0^z e^{-t^2} dt. \quad (3.17)$$

- Swish

$$\phi(x) = x \cdot \operatorname{sigmoid}(\beta x) \quad (3.18)$$

where β is a trainable parameter.

Computational Cost of Layers

For the types of layers reviewed above and utilized in the neural architectures proposed in this thesis, we have estimated the computational cost in terms of FLOPs (Floating Point Operations). The FLOP count encompasses all types of operations, including multiplication, addition, division, and subtraction (although from a hardware perspective, subtraction is redundant with addition, and costly divisions are generally substituted with equivalent multiplications where possible). A rough estimate of the corresponding Multiply and Accumulate operations (MACs) can be obtained by halving the FLOPs for most layers. This estimate can be useful for comparing computational requirements against platforms that feature MAC operators.

A summary of the FLOPs required to compute the output of various layer types is presented in Table 3.1, where the counts are parameterized by the attributes that define each layer’s characteristics. Specifically, I denotes the input size, and U indicates the number of neurons, often referred to as units. To simplify the expressions, I includes the bias term, treated as the weight of an input permanently set to 1. The FLOPs for state-based layers, such as RNNs and SSMs, are provided for scenarios where the input is a single temporal observation of the sequence; here, I represents the input’s dimensionality (e.g., multivariate). However, if the input includes segments of the sequence, hence observations at different time steps T , the FLOPs should be multiplied by the number of time steps. For the 1D convolutional layer, K denotes the kernel’s length, and S indicates the stride.

Table 3.1: FLOPs and the number of trainable parameters for the neural layers types and activation functions detailed in Section 3.1.1. I , U , K , and S refer to input size, number of neurons, kernel length, and stride.

Layer	FLOPs	Trainable Parameters
FF	$2 \cdot (I \cdot U)$	$(I + 1) \cdot U$
CNN 1D	$(\frac{I-K}{S} + 1) \cdot K \cdot U$	$(K + 1) \cdot U$
RNN	$2 \cdot (I \cdot U + U^2 + U)$	$U \cdot (I + U + 1)$
LSTM	$8 \cdot (I \cdot U + U^2 + U)$	$4 \cdot U \cdot (I + U + 1)$
GRU	$6 \cdot (I \cdot U + U^2 + U)$	$3 \cdot U \cdot (I + U + 1)$
Attention	$T \cdot (2 \cdot I^2 + 2 \cdot T \cdot I + 5 \cdot T)$	$4 \cdot I^2 + 4 \cdot I$
S4	$I \cdot (2 \cdot U^2 + 3U - 1)$	$U^2 + 2 \cdot U + 2$
S6	$I \cdot (2 \cdot U^2 + 7U + 1)$	$U^2 + 2 \cdot U \cdot (I + 1) + I + 2$
Activation		
Sigmoid	14	-
Tanh	10	-
Softsign	3	-
ReLU	1	-
GELU	17	-
Swish	14	1

Since layers may employ various activation functions, the FLOP count for the layers does not include the cost of the activation function. The computational cost

for activation functions is listed separately in the table. The number of required operations can vary significantly depending on the specific implementation, particularly for exponential functions, which typically require a series expansion or other numerical methods. For the computation of exponentials, an estimation of 10 FLOPs is used. These figures included in the table represent an estimate averaging the cost in the most common implementations.

These figures allow a first estimate of the computational complexity of any designed architecture. However, as discussed in Section 2.3.1, such estimates can't guarantee real-time computation on a given hardware platform, but it can definitely provide an initial estimate and an additional metric to compare architectures, as often it is fair to assess accuracy between architectures performing a comparable number of operations.

3.1.2 Computational Tools

All models described in this thesis, as well as the associated training, experiments, and evaluations, were implemented using TensorFlow ¹ in Python ². TensorFlow is a free, open-source software library for machine learning and artificial intelligence, with a particular emphasis on the training and inference of deep neural networks. It also provides a suite of tools, utilities, and community resources that support the implementation and execution of machine learning experiments. Additionally, the TensorFlow C++ API ³ and CppFlow library ⁴ were used to implement real-time Virtual Studio Technology (VST) plugins of selected models.

As discussed earlier, designing novel and more effective neural architectures requires exploratory and validation experiments, which involve the training of networks with datasets that can be notably large when working with raw audio as input and/or output. As a result, training sessions can span multiple days, and the total number of feasible experiments is constrained by the computational power at one's disposal. The experiments conducted for this thesis utilized shared computational resources available at the University of Oslo, as well as a dedicated workstation in the Department of Musicology exclusively allocated for this project. Table 3.2 summarizes the specifications of the machines used and also provides an estimate of the total time these machines were engaged in uninterrupted training of Tensorflow models.

This information may give readers experienced with similar deep learning frameworks an indication of the scale of experiments undertaken, which significantly surpasses the number of models for which results and evaluations are presented in the papers included in this thesis.

¹<https://www.tensorflow.org/>

²<https://www.python.org/>

³https://www.tensorflow.org/api_docs/cc/

⁴<https://github.com/serizba/cppflow>

CPU	Memory	GPU	Usage
EPYC 7282 - 16 Cores	256 GB	8x RTX 2080Ti - 11 GB	sh 6 mths
Xeon Gold 5215 - 8 Cores	32 GB	Tesla V100 - 8 GB	ex 6 mths
Ryzen 9 5900X - 12 Cores	94 GB	RTX 3090 - 24 GB	ex 24 mths
i7-11700KF - 8 Cores	64 GB	RTX 3070Ti - 8 GB	ex 24 mths
i9-14900KF - 24 Cores	192 GB	RTX 4090 - 24 GB	ex 3 mths
i9-14900KF - 24 Cores	192 GB	RTX 4090 - 24 GB	ex 3 mths
i9-14900KF - 24 Cores	192 GB	RTX 4090 - 24 GB	ex 3 mths

Table 3.2: Details of computational resources used for the research work described in this thesis, including their technical specifications and the estimated period of uninterrupted usage. In the Usage column, the following abbreviations are utilized: 'ex' for exclusive, 'sh' for shared, and 'mths' for months.

3.2 Processors

The first part of the dissertation considers musical devices that can be categorized as *processors*. As discussed in Chapter 2, successful attempts have been made to model these types of musical devices with artificial neural networks, particularly those related to distortion-type audio effects. These audio effects exhibit only negligible time dependencies between input and output signals. Instead, Dynamic Range Compressors (DRCs) present a more complex, nonlinear, time-variant behavior. Due to their complexity, it is still unknown how to accurately model such a device with neural network techniques. This subsection briefly reviews the internal mechanisms of compressors and the various types of circuits used in their analog implementations. It describes the selected devices from which data was collected, outlines designs and experiments to explore the suitability of different neural architectures, and summarizes the approach utilized to evaluate the modeling accuracy.

3.2.1 The Dynamic Range Compressor

The Dynamic Range Compressor (DRC) reduces the amplitude of the input signal by a certain amount when it surpasses a specified level (Smith III, 2007). The use of DRC is widespread in music production and live audio. DRCs are typically used to tame the louder parts or peaks of an audio signal so that the relative loudness of the quieter parts is increased. In other words, they compress the dynamic range in the loudness between different parts of the signal. Some DRCs allow usage in 'sidechain' mode, meaning that the gain reduction amount is determined by a signal different from the one to which the gain reduction is applied. Sidechain mode is typically used to avoid masking when two sounds, which usually have significant energy in the same frequency bands, are played simultaneously, such as a kick drum and a bass, or in radio broadcasting, to reduce the level of music when someone speaks.

Although not all analog compressors feature exactly the same controls to adjust compression settings, the functioning of a DRC is generally characterized

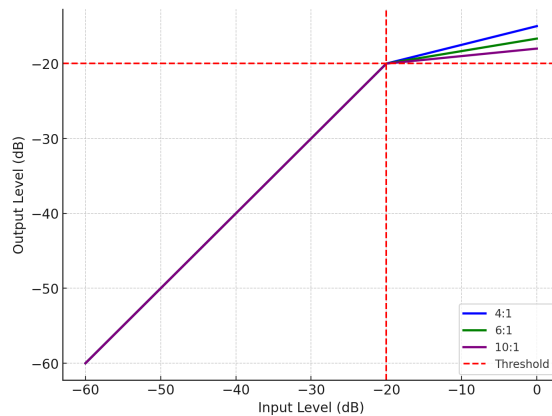


Figure 3.2: Examples of compression curves for three different ratio values.

by four main parameters: threshold level, compression ratio, attack, and release times. The threshold determines the input signal energy level that triggers the gain reduction process, while the ratio determines the amount of gain reduction applied, effectively compressing the signal. The amount of gain reduction is usually indicated as a ratio: for example, 4 : 1 ratio means that every 4 dB the input signal exceeds the threshold, it will be reduced to 1 dB above the threshold. The other parameters influence the timing behavior of the compressor. Indeed, the gain reduction is generally not applied or removed instantaneously but gradually. This allows some louder peaks to be only partially compressed and can cause the compression to carry over to subsequent parts of the signal that may fall below the threshold. Such behavior ensures a smoother and less evident transition in dynamic changes, providing a richer and more nuanced dynamic contour. Once the input signal exceeds the threshold, the attack parameter determines the duration of the interval in which the compressor gradually ramps up the gain reduction from zero to the target level determined by the ratio parameter. Conversely, the release time parameter controls how long it takes for the gain reduction to be gradually removed after the input signal level falls below the threshold. As a result, the relationship between input and output signal levels—and hence the response of the compressor—is nonlinear, and this nonlinearity is not static but time-variant, attributable to the nonzero attack and release times. The mapping between input and output signal levels in a DRC is typically known as a compression curve, which illustrates the compressor’s behavior after the attack transient has concluded. Figure 3.2 shows examples of compression curves with three different ratio values. Since the sound alteration process applied by the compressor results in an overall reduction of the signal’s loudness, most compressors also feature a make-up gain parameter. This gain follows the compression stage and is independent of it. Although the make-up gain is not part of the DRC’s core processing, it plays an essential role in achieving the common audio engineering objective of compression, which is to boost the level of quieter parts and reduce the level of louder parts. In some analog DRC units, the independent make-up gain circuit can also impart some sound coloring, contributing to the ‘warmth’ of the device’s response. However, from a modeling perspective, the make-up gain is

of less interest since it is an isolated and independent sub-component of DRCs. Furthermore, existing neural techniques can accurately model amplifiers, which can be used in cascade to a separate neural network modeling the compression mechanism. In DRCs, the compression curve can present different degrees of smoothness around the threshold level. The bend in the response curve between below threshold and above the threshold is called the 'knee,' and it can be abrupt or gradual. While analog compressors typically have a fixed knee that depends on the internal circuitry, digital compressors often allow the knee-width to be adjusted as a parameter.

As previously mentioned, the DRC is an audio effect of considerable complexity. It applies a nonlinear, input level-dependent gain $g(x(t))$ to the input signal $x(t)$, utilizing a variable attenuator to generate the output $y(t)$:

$$y(t) = g(x(t)) \cdot x(t). \quad (3.19)$$

Furthermore, the input-dependent gain, which is typically less than one, is not applied or removed instantaneously when the input signal crosses the threshold level. Instead, it changes gradually in response to the DRC attack and release parameters. These transient periods are often adjustable, can last for extended durations, and may differ from one another. Moreover, DRCs typically average input levels over a short time window with duration τ , estimating the level either as an absolute value, known as peak sensing, or as a Root-Mean-Square (RMS) value. Consequently, the gain applied at a given instant may be expressed as $g(x(t) \cdot v(t))$, where $v(t)$ is a rectangular window function non-zero only in $[t - \tau; t]$, indicating that current gain values depend on past input levels within this window.

For the sake of simplicity, in the following formulations, we will refer to the gain simply as $g(x(t))$. DRCs are fundamentally time-invariant systems, as described in Equation (2.2), yet they exhibit time-varying gain factors that can have complex profiles, particularly due to extended transient times. This complexity is particularly evident in the presence of audio signals featuring continuous and rapidly changing levels. These signals may cross the threshold before the transients have fully settled, resulting in an ever-changing gain factor over time. Consequently, the relationship between the gain factor and the input signal $x(t)$ can exhibit significant long-term dependencies.

The compression of the sum of two signals is generally not the same as the sum of the two signals compressed individually. Hence, the compression process is nonlinear, and the superposition condition of linearity fails:

$$g(x_1(t) + x_2(t)) \cdot (x_1(t) + x_2(t)) \neq g(x_1(t)) \cdot x_1(t) + g(x_2(t)) \cdot x_2(t). \quad (3.20)$$

Depending on the internal electronic components utilized, there are five different types of analog DRC: Variable Mu, VCA, FET, PWM, and optical. Variable Mu DRCs, also called delta-Mu, rely on vacuum tubes to reduce the signal level. The amount of gain reduction depends on the input signal level, which determines the reduction ratio. Higher input amplitudes lead to higher ratios. These compressors can have attack times ranging from 10 to 300 ms and offer a slower and often program-dependent release, between 300 ms and 2 s.

Program-dependent release means the release time dynamically adjusts based on the content and behavior of the input signal.

Voltage-controlled amplifiers (VCAs) can also be utilized as DRCs, allowing a wide range of attack and release times. A voltage-controlled amplifier applies gain depending on the control voltage level, which in this context is the compressed audio signal or the compressor's sidechain input. While many compressors could effectively be described as voltage-controlled amplifier circuits, VCAs traditionally utilize analog integrated circuits to determine the input voltage. This controls the resulting gain reduction, providing the cleanest and most controllable compression among all analog compressors. VCA DRCs offer the highest amount of gain reduction and the fastest response to the attack and release parameters. Attack times range between 0.1 and 30 ms, while release times from 5 ms to 2 s.

FET DRCs get their name from the field-effect transistor (FET), which is utilized to generate the gain reduction from the input signal passing. FETs are capable of both amplifying and attenuating signals, and they respond to an electrical field rather than voltage, as in VCA compressors. Therefore, FETs can respond extremely fast to input signals. Indeed, FET DRCs present fast attack times (typically 20 – 800 μ s). On the other hand, release times are slower and between 50 ms and 1 s.

PWM compressors utilize Pulse Width Modulation to measure and control the energy in an audio signal over time. The process involves sampling the audio signal at a very high rate, with the duration (or width) of each sample (or pulse) being adjusted (or modulated) to control the average energy. This modulation effectively provides signal attenuation. PWM compressors are capable of achieving fast attack and release times and tend to produce fewer audible artifacts, making them more robust in this regard. Attack times are from 0.01 ms to 10 ms, and release time 2 ms to 500 ms.

In optical DRCs, the amount of gain reduction is determined by a light-sensitive resistor, also known as a photocell, illuminated by a light-emitting element driven by the input audio signal. The light emission begins upon exceeding the threshold level. The input signal's amplitude determines the element's brightness, which, in turn, changes the resistance in the gain attenuation circuit. Optical compressors present a smoother knee, hence a more gradual compression. Due to the physical properties of the photoresistor, these compressors feature frequency-dependent compression and program-dependent release. These complexities make them particularly challenging to model. Additionally, they exhibit significant long-time-variant characteristics. Attack times range from 1 ms to 300 ms, and release time from 50 ms to several seconds. These aspects make them a challenging and noteworthy case study for this thesis.

As explained, DRCs can be based on various technologies, and understanding their functionality and mathematically deriving the computation for the gain $g(x(t))$ in Equation (3.19) is a complex task. Therefore, data-driven black box modeling can provide significant advantages for achieving accurate emulations.



Figure 3.3: TubeTech CL 1B (top) and Teletronix LA-2A (bottom) optical range dynamic compressors in the Mixing Suite ⁷ of the Department of Musicology at the University of Oslo.

3.2.2 Datasets

The TubeTech CL 1B⁵ and the Teletronix LA-2A⁶, shown in Figure 3.3, are two well-known analog optical compressors commonly used in audio engineering. These devices have been primarily targeted for modeling in the *processor* part of this thesis. Since the compressors are physically available, collecting data directly from these devices was possible. The compressor’s output was recorded for different settings of the control parameters that influence the compression process. For the CL 1B, these parameters include threshold, ratio, and attack and release times. For the LA-2A, the parameters are peak reduction and the mode switch. The selection of the compressor’s input signal, for which the output was recorded, has been carefully considered. The signal contains a fixed sequence of a large variety of sounds that cover different frequency ranges and dynamics, such as frequency sweeps from 20 Hz to 20 kHz, white noises with increasing amplitude (both linear and logarithmic), recordings of instruments like guitar, bass, drums (individual notes and riffs), vocals, piano, synthetic pads, and segments of electronic and rock songs. DRC affects the dynamic and, in particular, the transients of the input signals. The input signal has a duration of 3 minutes and 15 seconds. Extending the duration and, consequently, the resulting dataset size is not likely to lead to improved accuracy (Steinmetz and Reiss, 2022; Damskågg, Juvela and Välimäki, 2019).

The data was collected using a MOTU M4 ⁸ audio interface. The left output and input channels of the audio interface were connected to the input and output

⁵<https://www.tube-tech.com/cl-1b-opto-compressor/>

⁶<https://www.uaudio.com/hardware/la-2a.html>

⁸<https://motu.com/en-us/products/m-series/m4/>

of the compressor, respectively. The right output channel of the interface was looped back into its right input channel. Both left and right output channels reproduced the sequence of selected signals while recording took place on both the left and right input channels. Gains and recording levels were carefully adjusted to ensure matching levels on both channels when no compression was applied. The looped-back and re-recorded selection of input signals was then used to form the input-output pairs required for black-box modeling. This method allows for sample-level alignment between input and output signals, as latencies introduced by drivers or converters are not a concern. Additionally, any minor coloration introduced by the soundcard affects both input and output data equally. Should this not be the case, it might be mistakenly attributed to coloration by the analog compressor. The audio data was recorded at a sampling rate of 48 kHz, which is also the rate at which the models operate. A rate of 48 kHz offers a good balance between audio quality and the computational demands for training and inference in the considered models. These compressor input-output pairs were recorded for each combination of selected parameter values, influencing the resulting dataset size.

This research utilizes additional DRC datasets, including an existing public dataset of the Teletronix LA-2A (Colburn and Hawley, 2020) and other datasets collected using commercial software plugins. The Teletronix LA-2A online dataset was useful for comparing our approach to other existing architectures. While modeling digital datasets may have limited utility or relevance from an application perspective, this exercise provides valuable insights into the capability of the investigated neural architectures to model various types of DRCs. Furthermore, it allows us to assess the extent to which dataset features impact the model's accuracy. Indeed, collecting data from software plugins is significantly faster and more precise with respect to parameter values, besides enabling easier access to a broader collection of DRC plugins compared to their analog counterparts. Furthermore, since the dataset generation is fully automated using a software tool we developed (Fasciani, Simionato and Tidemann, 2024; Paper XI), it is feasible to efficiently collect large datasets that include a vast range of parameter combinations. Similarly, additional experiments were carried out to assert the generalization capability of the proposed neural architectures using other types of analog effects, such as overdrives and filters. All datasets were recorded using the same sequence of input signals selected for the TubeTech CL 1B and Teletronix LA-2A.

All collected datasets are publicly available online and detailed in papers I, II, III, V, VI:

- CL 1B and LA-2A analog datasets (Simionato and Fasciani, 2024e; Simionato and Fasciani, 2024d)
- Digital DRC datasets (Simionato and Fasciani, 2023d; Simionato and Fasciani, 2023b; Simionato and Fasciani, 2023c)
- Various digital and analog effects (overdrives, filters, equalizers, and saturators (Simionato and Fasciani, 2024b; Simionato and Fasciani, 2024a).

3.2.3 Architectures

The exploration of neural architectures begins from those proposed in recent literature for analog audio effects. The selected networks must be capable of learning time dependencies and nonlinearities while also predicting output signals using short input data segments. These segments should be small enough to ensure an input-output latency that complies with the requirements discussed in the previous chapter and is suitable for the typical audio processing workflow, which operates on data in blocks. Consequently, architectures featuring layers with internal states, such as Recurrent Neural Networks (RNNs) and the more recent State-Space Models (SSMs), are chosen. Stateful architectures are advantageous because they can memorize past information and utilize it for subsequent predictions, leveraging internal states rather than relying solely on the input received at each iteration. On the other hand, neural networks with only memoryless layers, such as Convolutional Neural Networks (CNNs) and feed-forward Fully Connected (FC), require longer input sequences to model accurately and, therefore, are not included in this study. However, an appropriate integration of these layers with RNNs and SSMs, which serve as the backbone of the structure, can significantly improve modeling accuracy.

The investigation drew inspiration from recent advances in Natural Language Processing (NLP) for sequence-to-sequence modeling and from the latest developments in SSM networks. The design of ad-hoc neural architectures for modeling DRCs includes improvements that utilize two Long Short-Term Memory (LSTM) layers acting as an encoder and a decoder (Simionato and Fasciani, 2022; Paper I), as well as variations incorporating a convolutional layer and FC layers in the encoder (Simionato and Stefano, 2023; Paper II). The latter architecture hinders the full exploitation of the internal states, and an improved approach to overcome this aspect adopts a gate mechanism to update the internal states (Simionato and Fasciani, 2024a; Paper V).

Neural architectures incorporating SSM are also explored, beginning with a comprehensive comparative evaluation (Simionato and Fasciani, 2024a; Paper V). The outcomes of this evaluation informed the final architecture proposed for modeling analog DRCs (Simionato and Fasciani, 2024c; Paper VI), which is designed with a data flow in the network that aims to support the emulation of the compression effect.

Additional studies are designed to determine whether these neural architectures could learn parametric audio processing influenced by the effect's parameters. A conditioning component is included in all proposed architectures. A range of techniques is explored, from the basic approach of incorporating parameter values directly as extra inputs (Simionato and Fasciani, 2022; Paper I) to more elaborate methods, including gated activation (GA), as implemented in the original WaveNet network (Oord et al., 2016), and Feature-wise Linear Modulation (FiLM) (Perez et al., 2018). The conditioning process is further enhanced by integrating conditioning information directly into the internal states of the LSTM (Simionato and Stefano, 2023; Paper II) and by combining the FiLM method with the Gated Linear Unit (GLU) (Dauphin et al., 2017). However, the first method precludes a stateful mechanism, while the second one leads to more flexibility. Finally,

the neural architectures developed are designed to present low computational complexity for inference and low latency to allow interactive control of the parameter, satisfying the modeling requirements detailed in Section 2.3.1.

3.2.4 Learning Strategy and Experiments Setup

Despite the need for specific adjustments to achieve accurate results for each dataset and modeling task, the learning strategy across all experiments involving different architectures and tasks shares several commonalities. All models are trained using the Adam (Kingma and Ba, 2014) optimizer, which is a standard choice in the literature. Since training RNNs, a gradient norm scaling of 1 (Pascanu, Mikolov and Y. Bengio, 2013) is added, which helps to avoid instability during the training.

Particular considerations have been made for the learning rate and batch size. The learning rate is the hyper-parameter that defines the adjustment in the weights of our network concerning the loss gradient descent. It determines how fast or slow the learning algorithm moves toward the optimal weights. The batch size is the number of samples to process before updating the weights and impacts the generalization accuracy (Keskar et al., 2017): larger batch sizes may result in faster convergence and training times but poorer generalization. In comparison, smaller sizes can aid generalization but may require additional epochs. Although there are studies on optimal learning rate and batch size in a non-convex optimization problem, their choice is not trivial. For this reason, they are often selected based on specific empirical evidence.

The initial learning rate is selected between 10^{-4} and $3 \cdot 10^{-4}$, which is a common choice in related works, and its suitability is also confirmed during the experiment. Adjustments are made to the learning rate throughout the research, adopting a rate that decays exponentially over time. On the other hand, the batch size is adjusted specifically for each experiment. Epochs and early stop conditions are chosen to ensure convergence, if possible.

The Mean Squared Error (MSE) is commonly used as a loss function in neural networks that generate raw audio because it generally results in the highest overall accuracy. The MSE measures the difference on a sample-by-sample basis between the audio output recorded from the target device and the one generated by the model being optimized. This choice is further supported by empirical evidence with the specific task and dataset of this thesis (Simionato and Stefano, 2023; Paper II), where results indicate better transient emulation than frequency domain loss functions based on spectrograms comparison at different resolutions. In addition, models trained with MSE as the loss function provided higher accuracy than those trained using other time domain loss functions, such as Mean Absolute Error (MAE) and Error-to-Signal Ratio (ESR) error. The normalization term helps the loss not to become negligible in segments where the target signal level is particularly low. However, this may affect the accuracy of low input levels but at the cost of transient accuracy.

3.2.5 Evaluations

The evaluation is pursued using a variety of quantitative metrics and inspection of the generated audio at first. In the last stage, the perceptual listening tests are carried out, specifically for the most accurate neural architectures. Among the selected metrics, we employed the MSE and MAE, which are commonly utilized in related works. While MAE estimates the average error of the audio signal, MSE weights larger mismatches. The magnitude of errors must be interpreted with respect to the dynamic of the signal. A normalized metric such as the ESR helps to cope with this, providing a metric that is easier to interpret. The ability to emulate the transients in sounds is an important aspect to evaluate, especially in DRCs. For this reason, the root-mean-squared error (RMSE) and spectral flux error (SF) have also been utilized. The first focuses on the energy envelope mismatch, and the second one measures the spectral change between two successive frames, thus how quickly the power spectrum of a signal is changing. It is computed as the squared difference between the normalized magnitudes of the spectra of the two successive frames. This error indicates how much the target and emulated onsets are matching. The compression effect often leaves the beginning of the sound unaltered but affects an immediate later portion of the sound, creating a fast change in dynamic. SF error gives insight into how well these changes are properly emulated. In addition to time-domain metrics, frequency-related metrics are also employed. Specifically, the multi-resolution Short-Time Fourier Transform (STFT) error quantifies the discrepancies between spectrograms at various frequency resolutions. Furthermore, two metrics that reflect the human auditory perception of sound have been considered: metrics based on Mel-Frequency Cepstral Coefficients (MFCCs) and the Perceptual Similarity Measure (PSM). Test losses and evaluation metrics are computed using the model’s weights that minimize the validation loss throughout the training epochs. To ensure fairness, we compare models with similar numbers of layers and trainable parameters that utilize input audio segments of the same length.

Informal listening of the generated audio, combined with visual inspection of the associated waveforms and spectrograms, is also employed to evaluate models. Indeed, the quantitative metrics described above are computed over the entire test set, or large portions of it, and hence represent only the average accuracy of the model. Short-duration or low-energy glitches and artifacts—perceptible when not masked, particularly at certain frequencies—typically go undetected by quantitative metrics but can be easily and immediately identified by listening to the audio or looking at its visual representations. Ultimately, only a selection of models, which includes minor variations of the most accurate neural architectures as well as state-of-the-art models, are subject to systematic listening tests carried out using the Multiple Stimuli with Hidden Reference and Anchor (MUSHRA) method.

3.3 Generators

The second part of the dissertation considers Acoustic Pianos, musical devices that can be categorized as *generators*. Acoustic piano modeling is a topic of great interest and presents significant challenges to achieving an accurate but efficient solution that includes all its sonic nuances. Indeed, despite a long history of research and development, a significant market and user base, and state-of-the-art commercial or prototype emulations, they still lack certain features of acoustic pianos. These acoustic instruments, despite being time-invariant systems, as in Equation (2.2), exhibit particularly complex characteristics, responding to inputs in a nonlinear and time-varying manner. This subsection briefly reviews the sound generation mechanisms of acoustic pianos, describes the selected pianos and associated datasets, outlines the designs and experiments conducted to explore the suitability of different neural architectures, and summarizes the evaluation approach.

3.3.1 Acoustic Piano

The acoustic piano is a musical instrument whose sound is produced by strings struck by hammers. Specifically, the piano produces sound when its keys are depressed; the energy is transferred to the hammer via a mechanical device called the *action*. The hammer, covered in felt, travels upwards and hits the piano string(s), vibrating the string(s). The vibrations are transferred to the soundboard via the bridge, which also vibrates. Finally, the vibration of the soundboard compresses and pushes the air around it, which forms sound waves transmitted to a listener's ear. The piano is a 300 years old musical instrument, and it has gone through countless design iterations and refinements until today (Giordano and Jiang, 2004). In the 1700s, the piano was invented by Bartolomeo Cristofori. The piano was designed to enhance the clavichord and harpsichord, offering a more expressive playing method by allowing musicians to control the loudness of each keystroke, thereby providing richer dynamics. The name 'pianoforte' is derived from this aspect: the instrument allows the playing of pianissimo (soft sound) and fortissimo (loud sound). A piano can be Upright or Grand, as visible in Figure 3.4. Their difference, besides the size, is in their action mechanisms. The grand piano presents a horizontal action: the hammers return to their resting position under their own weight. The upright piano action is vertical. Hammers rely on springs to return to the at-rest position. In addition, grand pianos typically have a larger soundboard than an upright. This means grand pianos sound louder and richer.

Scientific piano studies were pioneered by Helmholtz (2007) in the late 19th century, although active research only began in the 1920s (Suzuki and Nakamura, 1990). The piano consists of different components: the action, the hammer, and its interaction with string, soundboard vibrations, and acoustic radiation.

Hammer-string interaction

The interaction between hammer and string is an important subject as it is where vibrations start. The hammering force, its amplitude, contact point, and



Figure 3.4: Grand (left) and upright (right) Pianoforte. The grand piano is a Steinway & Sons concert grand piano in high polish polyester finish, model D-274 ⁹. The upright piano is a Steinway & Sons upright piano in mahogany finish, model K-52 ¹⁰.

the contact duration influence the excitation of the string and, subsequently, the soundboard dynamics. The hammer is covered by felt, compressing upon impact. This results in hysteretic behavior and a nonlinear relationship between the hammer force and displacement. Early models that have been developed (Suzuki and Nakamura, 1990; Boutillon, 1988) found a nonlinear finite-difference hammer scheme modeling the felt with a hysteretic spring. In more sophisticated models (Chaigne and Askenfelt, 1994; Stulov, 2005; Vyasarayani, Birkett and McPhee, 2009) parameters were extracted from experiments. The relationship between the hammer force and displacement can be described as follows:

$$F(u(t)) = H_0 \left[u^p(t) - \frac{\epsilon}{\tau_0} \int_0^t u^p(\xi) \exp\left(\frac{\xi - t}{\tau_0}\right) d\xi \right] \quad (3.21)$$

where $F(u(t))$ is the force exerted by the hammer, $u(t)$ is the compression, H_0 is the hammer stiffness, and p is the exponent that together describes the elastic parameters of the hammer. The hereditary parameters are ϵ , the hereditary amplitude constant, and τ_0 , the relaxation time. Air humidity also considerably influences hammers (H_0 and τ_0 are particularly sensitive), and their parameters change across the piano's compass. The complexity arises if including the flexibility of the hammer shank, which also influences the interaction between the hammer and string (Izadbakhsh, McPhee and Birkett, 2008; Chabassier and Duruflé, 2014).

In recent years, an experimental framework has been established to more accurately and consistently study the hammer-string interaction (Birkett, 2013), and a numerical method has also been proposed to re-construct the hammer forces from string velocity measurements (Chaigne, 2016).

String

The piano strings determine the pitch and tonal color of the piano sound, characterized by features such as double decay, inharmonicity, precursor signals, and phantom partials. The piano sound has two decay phases at different rates (Weinreich, 1977). When considering the strike of one string, the double decay derives from the string vibrating vertically and horizontally. The two vibrations have different decay rates. As the string is coupled to the bridge, its energy is dissipated quickly, making up the initial fast decay. As the vertical displacement reduces, horizontal displacement becomes more dominant and exhibits the second slower decay. The phenomenon with vertical and horizontal vibrations is known as *double polarisation*. On the other hand, when the hammer strikes duplets or triplets of strings, the initial vibrations of the strings are in phase with each other. The forces exerted on the bridge are also in phase, and more energy is transmitted to the bridge. However, the vibrations slowly get out of phase due to detuning among the strings. The out-of-phase forces cancel each other, reducing the net bridge forces and slowing the energy transfer. The different energy transfer rates thus result in different decay rates for the strings. As the vibrations go in and out of phase, beatings can also be observed (Nakamura, 1989; Thomas et al., 1998).

The simplest model for describing the transverse vibrations of a string assumes that the point of the string moves only transversely in a vertical plane along a line that is orthogonal to the string at rest, and the d'Alembert's equation governs the displacement field:

$$\rho A \partial_t^2 u_s - T_0 \partial_x^2 u_s = 0 \quad \forall x \in (0, L), \quad t > 0 \quad (3.22)$$

where u_s is transverse displacement, and, assuming that the string is homogeneous and of constant cross-section, ρ denotes its volumetric mass, A is the area of its section, L this string length, and T_0 is its tension at rest. The corresponding propagation velocity is

$$c = \sqrt{T_0 / \rho A}. \quad (3.23)$$

The real string presents stiffness, which causes inharmonicity. Inharmonicity occurs when the string's partials are not exact integer multiples of the fundamental frequency. This phenomenon can be described by various models, such as the Euler-Bernoulli beam theory (Chaigne and Askenfelt, 1994), Timoshenko beam theory (Chabassier, Chaigne and Joly, 2013), or, more recently, shear beam theory (Ducceschi and Bilbao, 2016), which is a slightly reduced version of the Timoshenko model.

Starting from the Euler-Bernoulli beam equation, the motion of the stiff piano string perpendicular to the soundboard can be described by the following:

$$\partial_t^2 u_s - c^2 \partial_x^2 u_s + \epsilon c^2 L \partial_x^4 u_s = 0, \quad (3.24)$$

where ϵ is the string's stiffness parameter, c is the transverse wave velocity of the string. The stiffness parameter ϵ is equal to $\frac{K^2 EA}{L^2 T_0}$, and K is the radius of gyration. The natural frequencies can be derived from the lossless dispersion relation, which results from substituting a harmonic wave, $y(x, t) = y_0 \exp(\omega t - kx)$ into Equation

(3.24), yielding to $\omega^2 = c^2 k^2 (1 + \epsilon L k^2)$, where k is the wave number and ω the angular frequency (Podlesak and Lee, 1988). The wave velocity of the string can now be found from the dispersion relation via $c(\omega) = \omega/k$. Finally, for pinned strings $y(0, t) = y(L, t) = 0$, the shape of mode j at position x is $y(x) = \sin(jx\pi/L)$, and the natural angular frequencies are

$$w_j = \pi c(\omega)/L = 2\pi j F0(\sqrt{1 + B j^2}). \quad (3.25)$$

The frequency of the first *partial* of the string without stiffness is $F0 = c/2L$, and the inharmonicity factor is $B = \frac{\pi^2 E S K^2}{T L^2}$.

On the other hand, if taking into account the transverse displacement of the stiff string and the fact that each cross-section of the string can rotate with an angle of ϕ_s to the normal plane to the so-called neutral fiber of the string, we can derive the following system using Timoshenko's beam model:

$$\begin{cases} \rho A \partial_t^2 u_s - T_0 \partial_x^2 u_s + A G \kappa \partial_x (\phi_s - \partial_x u_s) = 0 & x \in (0, L), \quad t > 0 \\ \rho A \partial_t^2 \phi_s - E I \partial_x^2 \phi_s + A G \kappa (\phi_s - \partial_x u_s) = 0 & x \in (0, L), \quad t > 0 \end{cases} \quad (3.26)$$

where I is the inertia momentum of the string's cross-section, and G is the shear module of the material composing the string. The parameter κ is called "Timoshenko's parameter", or "shear correction factor" and is a dimensionless parameter between 0 and 1. From there, we can find a similar expression (the full derivation can be found in (Chabassier, Chaigne and Joly, 2013)):

$$w_j = \pi c(\omega)/L = 2\pi j F0(1 + B j^2). \quad (3.27)$$

The closer model to reality is not clear, but in practice, for real piano strings, since T_0/EA is significantly smaller than one, there are not many differences. The choice of model is usually based on practical reasons.

The piano model can be more accurate when introducing the phenomenon called tension modulation. It is one of the most important phenomena in stricken string vibrations. This modulation arises from the interaction between the string's mechanical stretching and its viscoelastic response. Unlike a purely elastic spring, the tension modulation in a piano string allows for the generation of shear deformation, resulting in changes in its length during oscillation. Therefore, the tension is not constant during the oscillation and can be approximated according to Hooke's law (Bank and Sujbert, 2005). The tension T is now a function of (x, t) :

$$T = T_0 + EA((\partial_x u_s)^2 + (1 + \partial_x u_l)^2)^{1/2} - 1, \quad (3.28)$$

where u_l is the longitudinal displacement. Indeed, longitudinal vibrations are also created by the stretching of the string during the collision with the hammer. When the hammer is in contact with the string, it is elongated slightly compared to its initial length. This stretching is just a displacement along the longitudinal direction and creates free-response longitudinal waves. Longitudinal waves and transverse waves are coupled due to the influence of transverse vibrations on string tension (Kurmyshev, 2003) and contribute to generating the so-called phantom partials in strings vibrating at high amplitudes. The amplitude of these waves is

small compared to the transverse ones. The exact generation process of phantom partials is yet unclear and could also be affected by non-string components of the piano (Neldner, 2020). However, experiments show the relation between the transverse and longitudinal displacements (Legge and Fletcher, 1984). In particular, experiments indicate that the amplitude of longitudinal displacements is proportional to the square of transverse ones. Even if small, the effect of these displacements can be observed on the soundboard motion and the sound spectrum. Lastly, the speed of longitudinal waves is approximately ten times greater than that of transverse waves, which is close to 300 m/s:

$$\frac{c_l}{c_t} \geq 10, \quad (3.29)$$

where $c_l = \sqrt{E/\rho}$. This implies that the initial motion of the bridge and soundboard, as well as the initial portion of the sound (known as the 'attack'), results solely from the longitudinal motion of the string. These small initial signals that precede the transverse displacement are called precursors (Askenfelt, 1993; Giordano and Korty, 1996; Fontana et al., 2024).

If w_n and w_m are two transversal modal frequencies of a string, additional partials will be generated at $2w_n$, $2w_m$, and $w_n \pm w_m$. Partial at $2w_n$ and $2w_m$ exhibit decay times that are halved compared to those at w_n and w_m (Legge and Fletcher, 1984; Bank, 2009), and they are called even phantom partials. The odd phantom partials are instead given by the sum and difference of consecutive transverse partials and the multiplication of the related coefficients (Bank and Sujbert, 2005):

$$w_k^l = w_n \pm w_m \quad (3.30)$$

where $|n - m| = 1$, and k is the longitudinal mode index.

Another important effect of tension modulation is the creation of transverse motion of the bridge, which has different consequences. First, it can excite the mode of the string initially absent from the vibration (typically, this happens over a time of order 1 s); secondly, the nonlinear coupling with the two possible transverse polarisations of the string vibration induces the creation of forced-response longitudinal waves (Etchenique, Collin and Moore, 2015).

Soundboard and radiation

The soundboard represents the resonator, which amplifies the string vibration coming from the bridge so that it is audible to the human ear. The soundboard is a large, thin wooden plate made of many smaller planks glued together. Ribs are glued to the soundboard to improve its stiffness and longevity, while treble and bass bridges are glued to it to couple the strings and the soundboard. The soundboard is also tapered and curved slightly upwards, forming a frown shape. The soundboard can be characterized by its features, such as vibrational patterns in different frequency ranges, mechanical mobility, and radiation patterns. The first few resonant frequencies are influenced by the presence of rim, bridge, and downbearing forces from the string (Ege, 2009b). The average modal frequency spacing is between 20 and 30Hz regardless of the dimensions of the piano, which

range from 500Hz. Soundboard's modal loss and shapes are also experimentally studied (Berthaut, Ichchou and Jézéquel, 2003; Suzuki, 1986; Ege, Boutillon and Rébillat, 2013; Moore and Zietlow, 2006).

Soundboard mechanical mobility defines the loudness and sustain of a piano tone. Strings transmit an initial bulk of energy to the soundboard to make up a piano's loud 'attack' tone and follow up with a slower energy transmission for the slow-decaying 'sustain' tone. Nakamura (1983), Conklin Jr (1996) and Giordano (1998) studied the admittance of the string forces at the bridge of a piano. In a first soundboard radiation study (Suzuki, 1986), it is observed that below 80Hz the radiation efficiency is particularly low, and the efficiency fluctuates up to 1kHz. More recent radiation studies (Boutillon and Ege, 2013; Trévisan, Ege and Laulagnet, 2017) demonstrated the effect of the ribs or string coupling points on the soundboard's radiation.

When modeling, the soundboard damping is considered a linear phenomenon of modal nature, with the attenuation increasing with the frequency of the modes. Therefore, the damping function is supposed to be increasing and have a sub-linear growth at infinity (Ege, 2009a):

$$S_{loss}(f) \leq Af + B \quad \text{with } A, B > 0. \quad (3.31)$$

Lastly, the following equation has been found to match better the experimental data (Chabassier, Chaigne and Joly, 2014):

$$S_{loss}(f) \leq Af + B\sqrt{f} + C \quad \text{with } A, B, C \geq 0. \quad (3.32)$$

Piano Action

The piano action is a mechanism translating the depression of the keys into the motion of a hammer, which, in turn, creates sound by striking the string. Although the piano action does not directly affect the piano sound, it holds an important role as it is the main input for pianists to control the sound generation. The piano action mechanism is intrinsically nonlinear (Suzuki and Nakamura, 1990). Lieber (1985) investigated the relationship between the energy input and the key and the key travel time, while Askenfelt and Jansson (1987) studied the contact timing and motions of various components in the piano action. Several grand piano action models have been proposed (Gillespie, 1994; Gillespie, 1996; Hirschhorn, McPhee and Birkett, 2005; Izadbakhsh, McPhee and Birkett, 2008), presenting different numbers of components and complexity. The flexible shank is also included in the latter. The action models of upright piano have also been derived (Masoudi, Birkett and McPhee, 2009; Masoudi and Birkett, 2015). Lastly, Lozada et al. (2017) proposed a grand piano model based on non-smooth dynamics, in which forces of the components are simulated instead of displacements.

As presented, acoustic piano modeling is another challenging task; the instrument's functioning principles are not completely known and several aspects are difficult to model accurately when aiming for real-time applications. Therefore, data-driven approaches can provide significant advantages for modeling complex piano features without the need to describe them using physical knowledge.

3.3.2 Dataset

The datasets utilized for the piano experiments were collected from Yamaha Disklavier MX100A. The MX100A is controlled via MIDI messages, allowing accurate actuation of the piano. The sound was captured using an Earthworks M50 microphone placed inside the piano case a few centimeters away from the instrument strings to minimize the adverse effect of the room acoustic on the recorded data. The keyboard range was limited to two octaves in the mid-range of the piano, from C3 to B4, and therefore, the mic was placed in the center of this range. A MOTU M4 provided interfacing for both audio and MIDI, as also detailed in (Simionato, Fasciani and Holm, 2024; Paper VI).

The choice of stimuli for which the piano sound was recorded was considered carefully. As detailed in Chapter 2, previous attempts to model acoustic instruments with the Differentiable Digital Signal Processing (DDSP) framework utilized recordings of music performances as datasets. In contrast, the proposed approach adopts a different perspective, starting with recordings of individual notes played at varying velocities.

The dataset also includes recordings of a selection of trichords, a different type of stimulus involving the simultaneous triggering of three keys. This allowed for the capture and analysis of the coupling among different keys, which could then be compared against the sum of the individual keys recorded separately. Trichords were also recorded at various velocities, which were kept identical for each set of three keys, including all minor chords playable within the C3 to B4 range. All collected datasets are publicly available online (Simionato and Fasciani, 2023a) and detailed in papers VII, VIII, and IX.

An additional public dataset (Papetti, Avanzini and Fontana, 2018) has been used for additional experiments. This dataset is recorded from Yamaha Disklavier grand and upright piano recordings. The grand piano model is DC3 M4, and the upright model is the DU1A with control unit DKC-850. A subset of this dataset is used for the experiments, comprising the same individual notes at different velocities as those recorded from the Disklavier MX100A. However, recordings of chords are not available in this case.

3.3.3 Architectures

The input for *generators* significantly differs in both type and quantity from what we provide to *processors*. Indeed, *processors* are designed to receive an input that mirrors the type and amount of information they emit at the output, which is typically an audio signal. This is with the exception of a few additional input values that represent the settings of control parameters. The inputs for *generators* lie in the domain of symbolic music representation, such as MIDI, which may also include a few additional expressive parameters for altering the instrument’s timbre. This type of information is generally processed at rates significantly lower than audio since it reflects user inputs with limited bandwidth or rates of change. Often, these inputs remain constant for extended periods, such as the duration of an entire note. In contrast, neural networks typically require large amounts of information-rich data to approximate a function accurately, especially complex

ones. For this reason, the DDSF framework is particularly well-suited for this modeling task. A pure black-box approach might struggle due to the vast 'distance' between input and output data in terms of domain, dimensionality, and rates.

The approach taken here to model the piano integrates physical knowledge of the sound generation mechanism with machine learning, utilizing the DDSF (Simionato, Fasciani and Holm, 2024; Paper VII). Initially, the goal is to accurately emulate piano notes, taking only velocity as the variable parameter. Later developments will incorporate additional aspects of the piano's sound generation mechanism, such as key coupling, re-striking of played keys, and various scenarios that can occur during an actual piano performance.

The designed models prioritize the most significant perceptual features: inharmonicity, sound decay rates, and beatings. The inharmonic nature of sound, the internal and air losses of the string, the double polarization, the generation of phantom partials, and the loss of soundboards are included in the model functioning. All these aspects are described with physics formulas, theoretically or experimentally derived (Tan, 2017; Chabassier, Chaigne and Joly, 2013; Bank and Sujbert, 2005; Ege, 2009b).

A separate model is trained for each key, which avoids the challenges associated with learning a wide range of pitches using a single model that would require a significantly larger network. This strategy yields smaller models that are more computationally and memory efficient. Having one model per key is theoretically consonant with fully emulating a piano, as the polyphony can be as extensive as the number of keys. To some extent, this approach is comparable to sample-based sound synthesis, which is commonly used for pianos; however, only a few parameters, rather than the complete audio samples, need to be stored for each key.

Modeling methods have been developed to generate both the initial impulsive part of the key press and the background noise. This leads to a set of three sub-models that emulate the percussive, noisy, and harmonic components of the sound, capable of predicting the correct signals even at velocities not present in the training set. Moreover, these models are designed to operate incrementally, predicting only a few audio samples at a time, which facilitates low-latency, interactive implementations. Key coupling and re-striking, aspects that sample-based methods struggle to accommodate due to their high memory demands, can be emulated with the addition of small auxiliary networks without notably increasing the system's complexity—unlike with physical modeling methods (Simionato and Fasciani, 2025; Paper VIII). The generalization ability of this approach is assessed by applying it to various pianos, confirming its versatility and efficacy.

A second parallel modeling direction considers the '*generators*' as '*processors*'. Building on the advancements made in *processor*-based methods, similar approaches are applied to piano modeling. Specifically, starting with a recorded note at a particular velocity, the network acts as an 'effect' that alters the sound to emulate stronger or weaker stimuli corresponding to different velocities. This approach utilizes neural architectures designed for dynamic range compression and other nonlinear, time-variant audio effects, adapting them to the task of piano

modeling. Consequently, the proposed methodology maintains low computational complexity, cost, and storage requirements while meeting the demands for low-latency and interactivity.

3.3.4 Learning Strategy and Experiments Setup

For the DDSP models, an accurate sample-level accuracy between the model outputs and the dataset target outputs is not guaranteed. As a result, frequency domain loss is preferred because they do not perform sample-by-sample comparisons. Additionally, the loss function is normalized to ensure the model is accurate even when the signal’s amplitude is low, such as during the last stage of the note decay. Different resolutions are investigated to find the best combination.

A combination of other losses is investigated to improve the tuning of aspects that can not be learned directly from spectrograms. Specifically, information extractable from the dataset, such as a subset of partial distributions and the RMS energy envelope, is utilized. The partial distributions are matched using a specially conceived loss function called Cent loss, based on the Cent unit. The RMS energy envelope is compared using the time domain loss function, which aids in learning the signal onset—a perceptually significant aspect.

Achieving stability in the DDSP framework has proven challenging and requires an attentive training process. Therefore, specific learning strategies are designed, and the models are trained in multiple stages, starting from the component responsible for computing the partial distribution. Once the frequency of the partials is correctly learned, the network is fine-tuned to match their partial amplitudes. The learning rate is set similarly to the *processor* case, and the batch size is adjusted specifically for each experiment.

3.3.5 Evaluations

Similar to the *processor* case, the evaluation was conducted utilizing objective metrics, visual analyses, informal listening, and ultimately, systematic listening tests using the Multiple Stimuli with Hidden Reference and Anchor (MUSHRA) method.

The monitored metrics included the root-mean-squared (RMS) error and frequency-related metrics such as the multi-resolution short-time Fourier transform (STFT). The RMS error must be interpreted concerning the signal’s dynamics, while the multi-resolution STFT error evaluates the differences in spectrograms at varying frequency resolutions. This assessment is crucial for accurately determining how well the model emulates partials’ amplitude evolution. Additionally, the investigation incorporated metrics that reflect human auditory perception, such as Mel-Frequency Cepstral Coefficients (MFCCs) and the Perceptual Similarity Measure (PSM). Similarly to the *processor*, the test losses and evaluation metrics are computed using the model’s weights that minimize the validation loss throughout the training epochs.

In the initial stage, akin to the compressor case, the evaluation also involved informal listening to the generated predictions and a visual inspection of the

corresponding waveforms and spectrograms. Quantitative metrics alone are insufficient for evaluating perceptually sensitive content, such as the attack phase or the decay rate of a note. Consequently, systematic listening tests were conducted using the MUSHRA method only for the final models.

3.4 Summary

The chapter began by presenting the general approach taken during the research and detailing the neural network typologies used in this thesis. This includes descriptions of their computational complexities. Additionally, the chapter provided an overview of the tools utilized to implement the neural models and conduct computationally demanding experiments. The second and third parts of the chapter introduce the modeling tasks chosen for the dissertation: the optical Dynamic Range Compressor (DRC) within the *processors* category and the Acoustic Piano within the *generators* category. For each task, the chapter offers a detailed description of the working principles of the selected devices, discusses the dataset collection process, provides a high-level overview of the designed neural architectures and learning strategies, and outlines the experimental setup and evaluation methods.

Chapter 4

Summary of Papers

This chapter presents and discusses the 11 papers included in this thesis, providing an overview of how each paper corresponds to the research questions introduced in Section 1.3. The papers are arranged as follows: those related to the neural modeling of *processors* are presented first, in chronological order. This is followed by papers pertaining to neural modeling of *generators*, also in chronological order. The final two papers focus on tools that aid in modeling tasks. Figure 4.1 illustrates the relationship between each research question and the papers included in this thesis. The three research questions are primarily addressed in a linear fashion, treating *processors* and *generators* separately for the most part. However, the concept of researching modeling techniques for the emulation of musical devices—with the aim of achieving the same playability as the original instruments—is, to some extent, applicable to all of the papers. The chapter concludes with a synthesis of the main findings and contributions from this collection of papers.

4.1 Papers

This section provides the motivation and discussion for each of the papers. Each paper is attached in full at the end of this thesis.

4.1.1 Paper I

Reference: Simionato, Riccardo, & Fasciani, Stefano (2022). Deep Learning Conditioned Modeling of Optical Compression.

In: *Proceedings of the 25th International Conference on Digital Audio Effects (DAFx20in22)*, Vienna, Austria.

Motivation

This paper presents the earliest proposal of a novel architecture for modeling audio effects that exhibit nonlinear and time-variant characteristics. The work employs an Encoder-Decoder LSTM architecture, drawing inspiration from sequence-to-sequence modeling techniques, to learn the behavior of an optical Dynamic Range

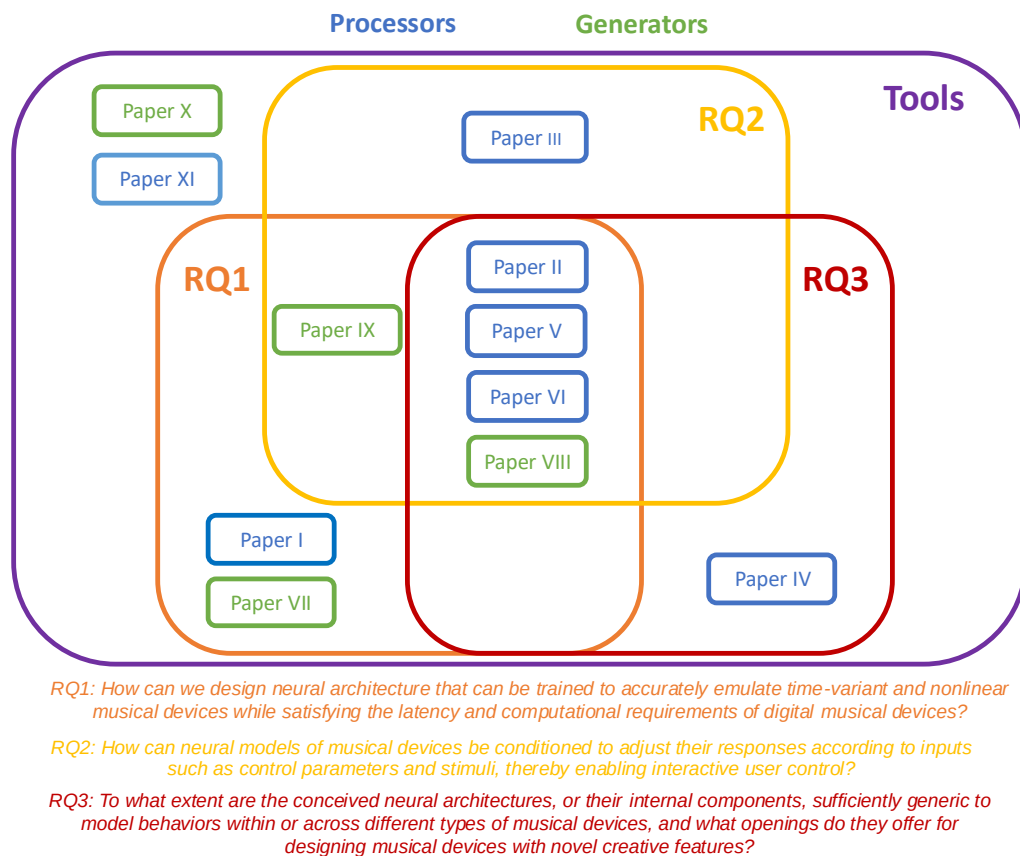


Figure 4.1: The three research questions addressed in this thesis and the corresponding papers.

Compressor (DRC). The paper aims to assert whether the novel architecture offers improvements over the LSTM-based networks, which have been used in recent years for audio effect modeling. In this case, the compressor’s timing behavior is fixed and not subject to parametric variation.

Discussion

In the initial experiments, the goal is to design neural architectures capable of increasing modeling precision for audio effects that exhibit significant long-term dependencies between the input and the processed output signals. The dataset for this study is acquired directly from the optical DRC, the TubeTech CL 1B. It includes recordings at various settings for the threshold and ratio parameters, while the attack and release times are held constant at their minimum values, 0.5 ms and 50 ms, respectively. Comparative experiments consider Fully-connected (FC), Long-Short Term Memory (LSTM), and the proposed Encoder-Decoder LSTM (ED-LSTM) architecture. The architecture detailed in the paper enables the sharing of internal states between two LSTM layers; the first encodes the input samples, which are then processed with improved accuracy by the second LSTM layer. Networks are designed to process 2 to 16 input samples and generate one output sample at a time. All models are trained using the Mean Square Error

(MSE) and evaluated with a test set that considers threshold and ratio values, included in or excluded from the training set, which are used to condition the network. Once normalized, the conditioning approach considers the parameters as additional inputs to the network, similar to the audio samples.

Results indicate that the ED-LSTM architecture outperforms others, having lower MSE and Error-to-Signal Ratio (ESR). However, all architectures produce aliasing-like artifacts, which are less prominent in the ED-LSTM. Also, results indicate a positive correlation between the size of the input-output and accuracy. From these experiments, it became apparent that all architectures struggled to learn the attack phase of compression. Specifically, they failed to accurately model the initial 0.5 ms of the signal post-threshold exceedance, typically after a sharp attack, which should remain unaltered. Although the ED-LSTM model exhibited the least inaccuracies during the attack phase, all models tended to behave as though compression was applied almost instantaneously. Lastly, distortion is noticeable under heavy compression settings—characterized by low thresholds and high ratios—particularly when processing sounds with significant low-frequency energy, such as kicks and basses.

4.1.2 Paper II

Reference: Simionato, Riccardo, & Fasciani, Stefano (2023). Fully Conditioned and Low-latency Black-box Modeling of Analog Compression.

In: *Proceedings of the 26th International Conference on Digital Audio Effects (DAFx20in23)*, Copenhagen, Denmark.

Motivation

This work builds upon the previous investigation into modeling nonlinear and time-variant audio effects. The focus remains on dynamic range compression; however, in this study, a fully parametric emulation is pursued to include variable attack and release times. The motivation behind this work is to demonstrate that the novel architecture can learn variable temporal behaviors without significantly increasing the sizes of the network and the length of the input signal used at each inference step. The variable attack and release times introduce significantly longer temporal dependencies between the input and altered output signals than those explored in prior related research. Another aspect evaluated in the paper is the suitability of the proposed architecture for dynamic range compression in general rather than for a specific compressor device.

Discussion

This work extends the dataset from Paper I for the optical DRC TubeTech CL 1B, and includes also recordings at various attack and release time settings. To assess modeling accuracy, additional datasets from other compressors are utilized, including a public dataset for the Teletronix LA-2A (Steinmetz and Reiss, 2022), and other datasets from a selection of software plugins, automatically are generated utilizing a prototype version of the tool described in Paper XI.

A variant of the Encoder-Decoder LSTM architecture is designed, replacing the encoder with a convolutional layer and a FC layer. The convolutional layer processes a short segment of the input signal, recent-past samples, while the FC layer processes the conditioning information. These are combined to output the internal states for the decoder, with parameter values included in the internal states-sharing process. The model takes as input a block of 32, 64, or 128 input audio samples, generating as output a block with half of the input size. Experiments are conducted with different loss functions. The convolutional layer improved accuracy while simultaneously reducing the model’s computational cost. Additionally, the parameter information is processed and more efficiently incorporated into the model than in the previously proposed architecture. From experiments, it emerges that the most accurate model is the one with a 64 sample as the input segment and requiring 3,280 FLOPs per sample. The same configuration is the most accurate among the experiments with datasets collected from different compressors, showing that it can accurately learn the nonlinear time-variant characteristics of various DRCs. The proposed architecture delivers accuracy comparable to state-of-the-art models based on Temporal Convolutional Networks, with the added benefit of significantly reduced input-to-output latency. However, perceptual accuracy under heavy compression settings remains limited, and audio artifacts are still present due to misalignment between the last and first samples in consecutive blocks of predicted samples. Finally, although the model successfully learned long attack and release time settings, the current formulation restricts the full exploitation of the LSTM’s internal states.

4.1.3 Paper III

Reference: Simionato, Riccardo, & Fasciani, Stefano (2024). Conditioning Methods for Neural Audio Effects.

In: *Proceedings of the 19th Sound and Music Computing Conference (SMC24)*, Porto, Portugal.

Motivation

All musical devices, particularly audio effects, are parametric systems. Each effect offers a range of sound alterations based on the settings of the control parameters. Consequently, neural models must learn multiple representations of the device from data. The effect parameters should be used to condition the neural network’s prediction process. This work investigates different conditioning methods in the context of modeling dynamic range compression and distortion in order to identify the best strategy.

Discussion

The paper evaluates different conditioning approaches and determines how they should be integrated into the modeling architectures. The conditioning methods are based on Feature-wise Linear modulation (FiLM) (Perez et al., 2018), Gated Linear Unit (GLU) (Dauphin et al., 2017), and Gated Activation (GA) (Oord et

al., 2016), and are placed at different stages of the neural architecture, based on Structured State Space models (Gu, Gupta et al., 2022). Four datasets are collected, comprising two distortion units and two DRCs, each including two parameters. Distortion and dynamic range compression effects are already established as validated modeling tasks. Additionally, having a diverse set of datasets to experiment with suggests further general validity of the findings. All placement combinations for the three conditioning methods are evaluated, ensuring a consistent number of parameters across comparisons. Subsequently, the nonlinear transformation is introduced using the Feature-wise modulation method.

The results indicate that conditioning methods perform better when placed at the end of the network, with Feature-wise modulation outperforming Gated Activation (GA). Gated Linear Units (GLU) showed improvements for compressors, while Gated Convolutional Units (GCU) are more effective for distortion effects. The study also finds that placing the conditioning layer at multiple stages does not lead to improvements. A noteworthy finding is that the models are more accurate when applying the processing due to conditioning parameters to the encoded representation of the input signal rather than processing the raw input signal according to the conditioning information and subsequently adding the nonlinear processing given by the State Space layer, contrary to intuitive expectations. This means the model is more efficient in modulating the signal after processing it than processing a modulated signal. Additionally, nonlinear modulation proved beneficial for distortion effects, where parameters may exhibit a stronger nonlinear relationship.

4.1.4 Paper IV

Reference: Simionato, Riccardo, & Fasciani, Stefano (2024). Hybrid Neural Audio Effects.

In: *Proceedings of the 19th Sound and Music Computing Conference (SMC24)*, Porto, Portugal.

Motivation

This work exploits neural networks' ability to act as universal interpolators in a creative modeling application. Neural networks can interpolate between observed data, which is also shown when conditioning the networks using the device's parameters. The paper further investigates the interpolation capabilities of neural networks when trained with recordings collected from different audio effects, introducing additional conditioning inputs to achieve audio processing that blends various effects.

Discussion

To conduct this work, a specialized dataset is created that includes recordings from analog devices such as a preamplifier circuit, a Dynamic Range Compressor (DRC), and a tape recorder. A set of artificial control parameters is introduced to guide the network in producing specific effects. By adjusting these control

parameters, hybrid audio effects can be achieved, creating a control space for a sound alteration process that does not exist in the physical devices.

This work demonstrates that a single neural model is capable of learning diverse nonlinear sound alteration characteristics, such as those produced by distortion, compression, and delay. In particular, the architecture was able to learn significantly different processes affecting dynamics, harmonic generation, saturation, and time delay. Notably, the networks successfully learned and reproduced the time lag delay between input and output waveforms. Furthermore, the conditioning approach, developed in Paper III, effectively processes the internal representation of the input signal to represent various audio processes. This context adds complexity since the conditioning navigates different audio processing and not the same parametric audio processing as in previous papers. This demonstrates the robustness and versatility of the neural architecture in modeling a range of audio effects in a single network.

4.1.5 Paper V

Reference: Simionato, Riccardo, & Fasciani, Stefano (2024). Comparative Study of State-based Neural Networks for Virtual Analog Audio Effects Modeling. *Submitted to EURASIP Journal on Audio, Speech, and Music Processing.*

Motivation

Neural networks, particularly Recurrent Neural Networks (RNNs), have proven adept at accurately learning analog audio effects with diverse characteristics. The resulting models can be conditioned with designs that achieve low latency and computational complexity. This paper explores state-based architecture for modeling a variety of analog audio effects, including the recent Linear Recurrent Unit (LRU) and State Space Models (SSM). The paper performs a comparative study for different modeling tasks that consider only low-latency and computationally efficient architectures to enhance the knowledge of these different technologies' different qualities and defects. The outcome serves to determine the best approach to be applied to optical compressors and, at the same time, explore the extent to which the different models can learn various effects.

Discussion

This work experiments on analog and software datasets of distortions, saturation, compression, equalizer, and low-pass filter. Four different architectures, utilizing different types of state-based layers and with a similar number of trainable parameters, are evaluated using different time- and frequency-based metrics.

From this work, it emerged that SSM-based models are better candidates when modeling devices with long-time dependencies, such as compressors. LSTM-based methods performed better when modeling distortion, but they also introduced more artifacts than other models considered in this study. Encoder-decoder models, LRU and SSM, are generally preferable for saturation and compression audio effects. These results guided the search for the neural architecture to

model the compression effect toward the SSMs. Finally, the work highlighted that although a different effect is modeled with different accuracy using the same architecture, it is not true for all audio effects. In particular, all models considered in this study showed poor accuracy in learning the low-pass filtering sound alteration process.

4.1.6 Paper VI

Reference: Simionato, Riccardo, & Fasciani, Stefano (2025). Emulating Optical Compressors with Selective State Space Models.

In: *Journal of the Audio Engineering Society*, 2025.

Motivation

In Paper V, State Space Models, both Structured (S4D) (Gu, Gupta et al., 2022) and Selective (S6) (Gu and Dao, 2023) variants, have demonstrated higher accuracy in musical devices featuring time-variant characteristics, such as optical Dynamic Range Compressors (DRCs). In optical DRCs, the input signal influences the time-variant characteristics. The S4D model does not directly depend on the input segment signal, while the S6 links the model variables to the input segment and, for this reason, could be advantageous. Previous work did not show significant superiority of the S6 over S4D. The work described in this paper further explores how to employ the S6 variant to improve the modeling of optical DRCs, in particular, by combining it with the Mamba gating mechanism. This paper is informed by all insight gained in previously described papers and proposes the most accurate state-of-the-art neural model of an optical DRC usable in live audio applications.

Discussion

The paper presents the neural architecture for modeling DRCs, shown in Figure 4.2, which provides the highest accuracy. The design is informed by findings from Papers I, II, III, and V. The model is based on State Space Models (SSMs) and incorporates conditioning methods that combine Feature-wise Linear Modulation (FiLM) and Gated Linear Unit (GLU) layers, as can be seen in Figure 4.3. The model is conceived for low-latency and real-time applications, requiring approximately 1290 FLOPs per sample and producing one output sample at each inference iteration from the 64 most recent input samples.

Empirical results validate the proposed model using datasets from two analog optical DRCs and from their software plugins, which allows capturing a wider range of compression parameter settings. The model’s performance is compared with variants of the proposed networks and a baseline model from the literature. Evaluation is based on various time- and frequency-domain metrics, perceptual-based metrics, and perceptual listening tests. Additionally, the paper investigates the correlation between model accuracy and the sampling density of control parameters in the dataset. Insights gained from this aspect can inform data collection designs, which can be a time-consuming task. Sampling more values

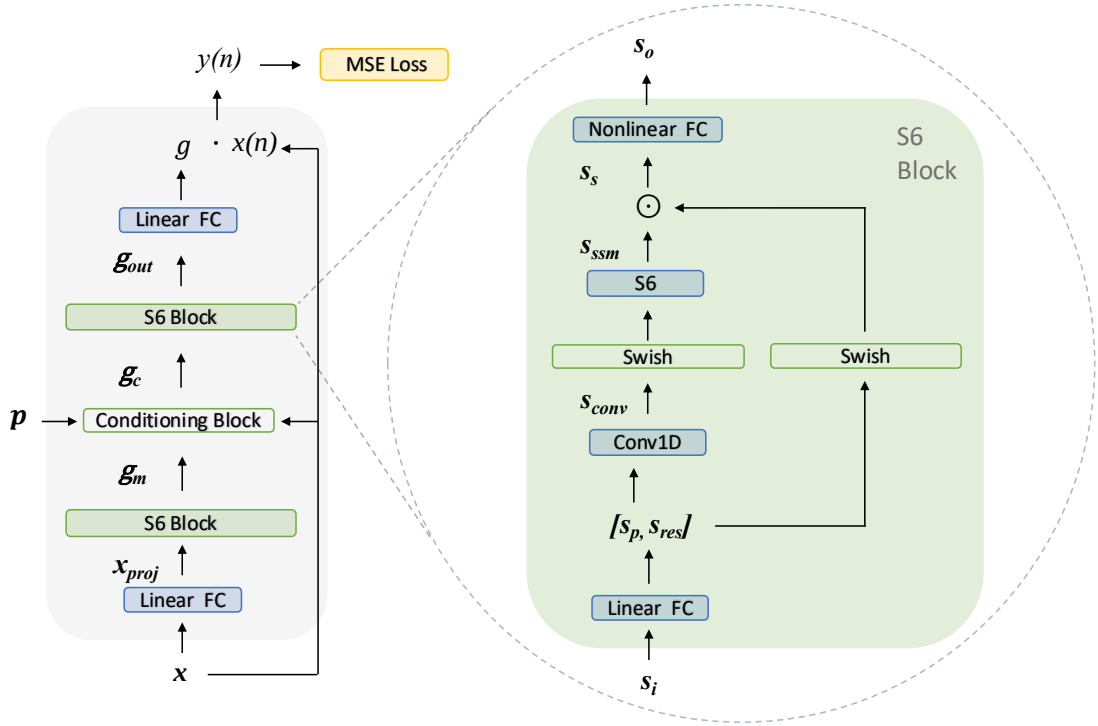


Figure 4.2: Neural architecture proposed in Paper VI for modeling analog optical dynamic range compressors. The 64 most recent input samples $\mathbf{x} = [x(n), \dots, x(n - 63)]$ are fed to a linear Fully connected (FC) layer and subsequently to an S6 block (detailed on the right). The resulting vector passes through the conditioning layer (detailed in Figure 4.3) and another identical S6 block before reaching the output layer, which is a linear FC layer with one unit generating the coefficient g . The output is computed by multiplying g by the current input sample $x(n)$ and yields the current output sample $y(n)$. The model is trained using the Mean Squared Error (MSE) as the loss function. A detailed description of the architecture is included in Paper VI.

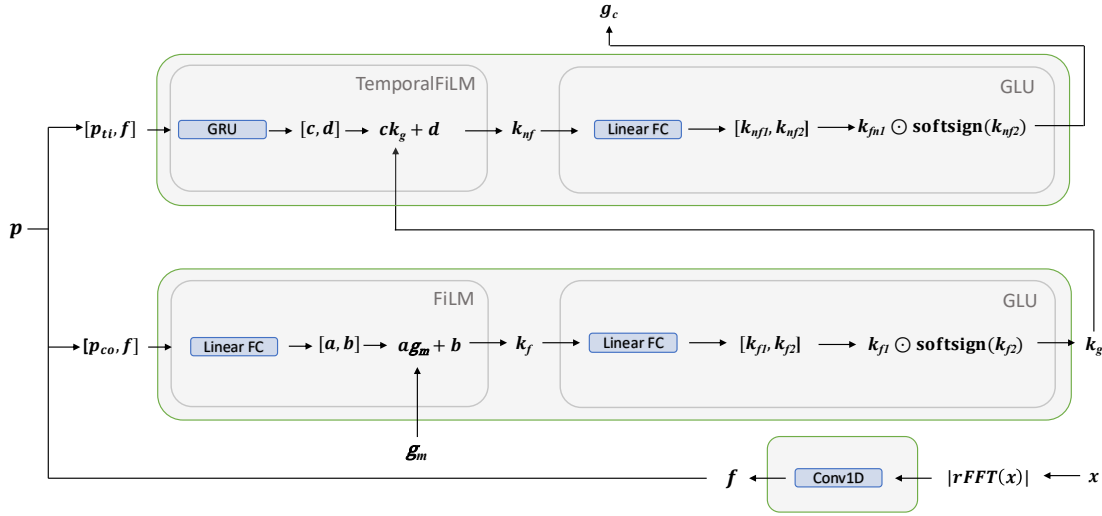


Figure 4.3: Conditioning block featured in the architecture shown in Figure 3.3. The conditioning is based on Feature-wise Linear Methods (FiLM) and Gate Linear Unit (GLU). $\mathbf{x}$ vector include the 64 most recent input samples, while $\mathbf{p}$ the device's parameters values. $\mathbf{g}_m$ is the output of the first S6-block, and $\mathbf{g}_c$ is the input of the second S6-block, as can be seen in Figure 4.2. A detailed description of the architecture is included in Paper VI.

for each compression control parameter, hence including more recordings in the dataset, does not necessarily lead to increased accuracy and generalization ability of the trained model; instead, there appears to be an optimal balance between too many and too few control parameter sampling points.

The model's ability to capture the nonlinear time-variant response of compressors and emulate the range time behaviors determined by the devices' control parameters was assessed. Experiments indicated that the combination of fast attack and slow release presented the most challenging compression scenario to emulate, yielding the highest errors across all proposed models trained on individual compression settings. In conclusion, the S6 model outperformed other proposed variants in all experiments. This was confirmed by quantitative metrics and by listening tests, where the model was perceived as the closest to the target, surpassing also the digital emulation counterparts.

4.1.7 Paper VII

Reference: Simionato, Riccardo & Fasciani, Stefano & Holm, Sverre (2024). Physics-informed Differentiable Method for Piano Modeling. In: *Frontiers in Signal Processing, volume 3, 2024*.

Motivation

The paper presents the initial work for the *generators*. In particular, it details the proposal to model the nonlinear and time-variant characteristics of the piano instrument. Although black box models generating raw audio have been proposed,

they are not compliant with the requirements for the emulations of musical devices. Instead, a Differentiable Digital Signal Processing (DDSP) framework (Engel, Hantrakul et al., 2020) that combines traditional digital signal processing and machine learning can map MIDI to sound. Earlier work in literature confirmed the potential of this approach by synthesizing piano sound at 16kHz starting from datasets, including full piano performances. On the other hand, the resulting network presents a high number of trainable parameters, resulting in computationally complex and inappropriate for real-time interactive use of such emulations. This paper aims to validate an alternative approach that is still built upon the DDSP but utilizes a different type of piano recording and incorporates architectural improvements to enhance computational efficiency.

Discussion

The paper presents the initial experiment aimed at validating a novel methodology as a suitable approach for modeling piano instruments. The network design incorporates physical knowledge of the piano’s inharmonicity and partial temporal unfolding. Aspects such as partial decay and inharmonicity, which significantly influence the perception of piano sounds, are included, and their physical descriptions derived from the literature to be integrated into the model using differentiable operations. The method primarily focused on the harmonic content of the sounds, omitting the noisy and percussive components.

The models are trained using a multi-resolution STFT and RMS loss to tune the partials’ temporal unfolding and a loss measuring the difference between the first 6 partials frequency, called Cent loss, to tune the partial distribution. The paper compares models producing 240 or 24 audio samples at each inference iteration, finding that accuracy is higher when producing small blocks of audio samples. Despite not utilizing a comprehensive physical model, the results show that the audio generated follows the partial temporal envelope and accurately matches the inharmonicity of the target piano. The model demonstrated the ability to learn from a relatively small dataset. Also, this model was designed to present low-latency and involves fewer trainable parameters than state-of-the-art approaches. The experiments utilize a dataset collected directly from a Disklavier MX100, which includes single notes played at different MIDI velocities. The recorded data was downsampled to 24 kHz, and only two octaves in the middle register are considered. The model synthesizes audio at 24kHz sampling rate, deemed sufficient to cover partials of notes in the considered range. Starting with a single note and embedding physical formulas within the model results in an approach that requires minimal data and parameters.

4.1.8 Paper VIII

Reference: Simionato, Riccardo & Fasciani, Stefano (2025). Sines, Transient, Noise Neural Modeling of Piano Notes.

In: *Frontiers in Signal Processing, volume 4, 2025.*

Motivation

The previous paper demonstrated that the selected approach for handling piano modeling is suitable for the intended purpose. This work further refines the method, aiming to enhance the sound generation description and perceptual fidelity. Paper VII presents several perceptual limitations due to the need to incorporate more aspects of sound generation to achieve satisfactory results. The paper explores how to include all the aspects happening during the piano sound generation. More accurate physical formulas for generating piano notes alongside considerations for the percussive and noise aspects of the sounds are included. Finally, most modeling challenges come from the coupling effects between different keys or mechanical components of the piano. The paper aims to demonstrate the feasibility of including this aspect in the proposed modeling framework.

Discussion

Building upon the previous model, enhancements were made in harmonic generation, and two additional modules are included to account for the sound's harmonic, percussive, and noise components. The HPPS method (Driedger, Müller and Disch, 2014) is employed to target these components separately. More sophisticated physics formulations are integrated into the model. The harmonic module features an improved description of partial temporal unfolding, incorporating two transverse polarizations and longitudinal displacements. The percussive part is generated using a deep neural oscillator that produces the DCT (Discrete Cosine Transform) form of the percussive component of the piano sound. Given the complexity of physically describing the percussive component, efforts are made to identify a form that could be easily synthesized. The DCT form consists of a sine-based waveform and can be described with a limited number of points. Conversely, the noise component is filtered as white noise using a learnable time-varying filter, a technique previously validated in similar works. Figure 4.4 shows the modules comprising the resulting neutral architecture.

Trichord emulation is achieved using a convolutional neural network that processes the sum of the single notes and replicates the sound of a real trichord, as shown in Figure 4.5. This approach treats trichord emulation as a *processor* by processing the sum of the single notes and adding the coupling effect as an audio effect. The conditioning method designed for DRC has also proven effective for this task. The architecture training is still based on multi-resolution STFT, RMS, and Cent loss, but following the insights from Paper VII, one output sample is produced at each inference iteration.

Results indicate a significant improvement in partial distributions and temporal unfolding matching compared to Paper VII. The model presents overall good accuracy for the percussive and noise components. Perceptual tests reveal that the model effectively captures the characteristics of the target piano, producing perceptually realistic sounds. Additionally, the chord emulation produces more accurate results compared to the simpler sum of individual notes.

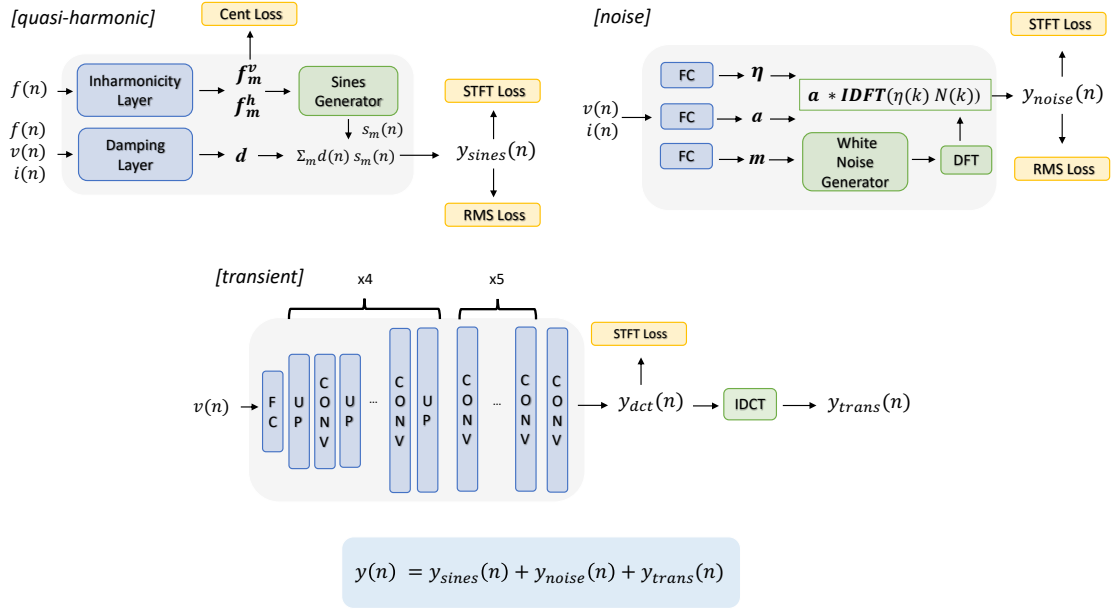


Figure 4.4: Neural architecture designed for the piano instruments and based on Differentiable Digital Signal Processing framework. It consists of three sub-modules: (quasi-)harmonic, noise, and transient modules. The (quasi-)harmonic module computes the harmonic partials distribution, and the partials' damped amplitude coefficients and generates the related sines signals composing $y_{harm}(n)$. The noise module generates the noise $y_{noise}(n)$ by predicting filter magnitudes to be convolved in the frequency domain with a generated white noise. The transient module uses a stack of upsampling and convolutional layers to generate the waveform $y_{dct}(n)$ representing the Discrete Cosine Transform form of the transient component $y_{trans}(n)$. The inputs for the modules are the key frequency $f(n)$, velocity $v(n)$, and time index $i(n)$ or a subset of them. The utilized loss functions include the STFT at different resolutions (STFT loss), the RMS energy (RMS loss), and the first 6 partials frequency distribution (Cent loss). Different sub-modules are trained with different loss functions. A detailed description of the architecture is included in Paper VIII.

4.1.9 Paper IX

Reference: Simionato, Riccardo & Fasciani, Stefano (2024). Neural Sampled-based Piano Synthesis.

Submitted to 2025 IEEE International Conference on Acoustics, Speech, and Signal Processing.

Motivation

From the previous paper emerged the feasibility of approaching piano synthesis from the perspective of *processors* rather than *generators*. In particular, the model can process the recording of a piano note, generating all variations based on MIDI information utilized as network conditioning. This work further investigates this concept by reformulating the problem and by exploiting the findings from the *processor* works.

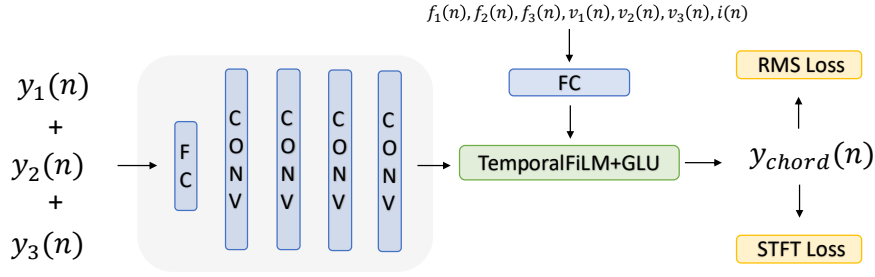


Figure 4.5: Neural architecture to model the coupling among different keys. It consists of convolutional layers, and the temporal FiLM method is combined with GLU to condition the network based on the keys playing. The input vector is the sum of the three separate note sounds, while their frequencies $[f^1(n), f^2(n), f^3(n)]$, velocities $[v^1(n), v^2(n), v^3(n)]$, and time index $i(n)$ compose the conditioning vector. The training utilizes a loss considering the STFT at different resolutions (STFT loss) and a loss considering the RMS energy (RMS loss). A detailed description of the architecture is in Paper VIII.

Discussion

The paper aims to develop a modeling approach for generating piano sounds, which is an alternative to those previously proposed. Specifically, a single recording of a piano note is used as input of the network, trained to learn how to modify the signal to generate the sound corresponding to other velocities. While utilizing this approach to model multiple keys with different pitches is unlikely to provide the necessary accuracy and/or lightweight complexity, modeling velocities of the same key is accurate and efficient. This approach enabled the creation of a sample-to-sample model with a minimal number of parameters, achieving perceptually accurate results. To a certain extent, this approach can be viewed as sample-based synthesis, where neural modeling is employed to minimize the memory requirements for sample storage while improving the accuracy of responses at different velocities, where the relationship between the stimulus and the generated sound is nonlinear.

Experiments are conducted to compare SSMs with other network types and to determine whether the networks perform better when tasked with adjusting the sound for stronger or weaker stimuli. The results indicated that the models were more effective in the latter case, which involves processing the sound recorded at the highest velocity. This approach also allows the utilization of neural architectures designed for Dynamic Range Compression and other nonlinear, time-variant audio effects, adapting them for Acoustic Piano modeling. The proposed methodology maintains low computational complexity, cost, and storage requirements while presenting low-latency and allowing interactive control.

4.1.10 Paper X

Reference: Simionato, Riccardo & Fasciani, Stefano (2023). A Comparative Computational Approach to Piano Modeling Analysis.

In: *Proceedings of the Sound and Music Computing Conference (SMC23), Stockholm, Sweden.*

Motivation

When developing models of musical instruments, such as the acoustic piano case this thesis is considering, it is critical to identify aspects that lead to relevant audible improvements. Listening tests are usually exploited to achieve this goal. On the other hand, from listening tests, it is not trivial to identify the differences between different signals, and they require a significant amount of time. Computational analysis techniques can help and support the modeling process. The idea is explored in the context of piano modeling. Numerical approaches are used to identify and emphasize the differences between digital models and acoustic pianos. These approaches can be instrumental in establishing more objective evaluation metrics to complement perceptual tests and validate the quality of a specific digital model.

Discussion

This study explored the use of a numerical approach as a supplementary method to assess differences among various models or the quality of a specific emulation. It emphasizes how computational methods can provide novel insights into piano analysis and modeling, complementing perceptual analyses. The proposed method identifies audio descriptors that show discriminative differences between types of piano models when these instruments are subjected to specific stimuli. Recordings were collected from upright acoustic and synthetic pianos, excited by single-played notes, triads, and repeated notes. Linear Discriminant Analysis (LDA) projection was then utilized to maximize the separability between the types of pianos (acoustic, sample-based models, and physics-based models) while minimizing within-type variance. LDA projects the feature vectors into a lower-dimensional space with dimensionality, or a number of components, equal to the number of classes minus one. Finally, a correlation analysis is utilized to determine which features are the most discriminative in distinguishing the various types.

The results revealed significant differences in the sound generated by the considered types of pianos, particularly in terms of spectral descriptors and constant-Q transform coefficients. The tool offers valuable insights into the modeling process while simultaneously serving as an additional measure for evaluating digital models.

4.1.11 Paper XI

Reference: Fasciani, Stefano & Simionato, Riccardo & Tidemann, Aleksander (2024). A Universal Tool for Generating Datasets from Audio Effects. In: *Proceedings of the 19th Sound and Music Computing Conference (SMC24)*, Porto, Portugal.

Motivation

Datasets are pivotal components in machine learning research. For audio effects, data collected manually tend to be error-prone and inaccurate, in addition to

consuming a significant amount of time. This work was initiated to address these challenges for the specific needs of this thesis research, facilitating and expediting the collection of accurate and reliable datasets.

Discussion

This paper describes a software tool developed to facilitate data collection from any type of audio effect, ranging from software plugins to hardware devices, accommodating interfaces including MIDI, Gate/CV, or manual control for the parameters. The tool fully automates the generation of datasets, allowing an arbitrary number of parameters with arbitrary ranges and resolutions across four different modalities. The recordings offer sample-level accuracy in terms of alignment, which is crucial for machine-learning models that work with raw audio input and output. Furthermore, the tools provide a feature-rich, easy-to-use graphical user interface. The resulting datasets are organized as a collection of audio files and parameters stored in a comma-separated values (CSV) file.

4.2 Summary of Paper Contributions

The first six publications of this thesis fall into the *processor* category. The contributions of Paper I, and in particular Paper II, demonstrate that state-based neural networks in dedicated configurations are capable—to a certain extent—of learning parametric nonlinear and time-variant characteristics of audio effects. These configurations present input-output dependencies that can span up to 10 seconds. They process audio in small blocks, introducing almost negligible latency and requiring a limited number of operations per audio sample. These works highlight that better structures to encode past information can be formulated. Indeed, the Encoder-Decoder architecture has a different mechanism for updating the internal states of the neural networks and presents better accuracy for compressor modeling tasks than LSTM architectures. Paper II also proposes a new modeling strategy, using a sliding window on the input signal to provide auxiliary information for the network. This mechanism aims to lower the vanishing of the information in the network’s states when encoding the past information from the signal. By doing so, the states are updated according to a ‘summary’ from a block of samples, carrying more information instead of one sample per iteration. Moreover, the window slides with a hop smaller than the block size; hence, a sample at a given time index is presented at the input of the network multiple times. Paper II demonstrates how 0.33 ms past signal, together with the proposed internal state mechanism of the encoder-decoder architecture, is sufficient to track the time behavior of the Dynamic Range Compressor (DRC). An encoder can be used to translate past information and conditioning parameters into a form that the decoder can use to perform the output signal prediction more accurately. The states-sharing strategy can efficiently encode the signal information and conditioning information. Additionally, internal states-sharing is efficient when infusing the parameters information into the network. On the other hand, the mechanism can preclude the full exploitation of the internal states

of RNNs since they are overwritten at every inference iteration that generates a block of output samples. An improved solution that overcomes this issue is provided in Paper V.

Papers II, IV, V, and VI show that the proposed architectures can model different Dynamic Range Compressors (DRCs) and different audio effects as well. However, they have limitations and different accuracy. In papers II and VI, the same model learns different DRCs. Paper IV uses the same model to learn three significantly different audio effects. Finally, Paper IV demonstrates that the proposed architectures, in principle, can also accurately model other types of audio effects, such as saturation, distortion, compressors, and equalizers. However, they were shown to have inadequate accuracy when learning low-pass filters. This suggests that the proposal is less effective with effects that remove a substantial portion of an audio signal’s frequency content. Also in this case accurate learning was also achieved with networks presenting low-latency and low computational complexity.

The research contribution of Paper III provides valuable insights into how to use effect parameters to effectively condition a neural architecture. In particular, it contributes by identifying the optimal stage at which the conditioning process should be implemented within state-based architectures and determining which conditioning techniques lead to more accurate modeling of parametric audio effects. The investigation led to the conclusion that using a dedicated layer to encode the input signal before conditioning the data flowing within the network is preferable. Additionally, it was found that the combination of FiLM and GLU constitutes a flexible and efficient technique for conditioning networks. In addition, findings indicate that FiLM with nonlinear transformation can benefit some modeling of audio effects, in which the relationship between parameters and sound alteration process is strongly nonlinear. These results mark a significant advancement in neural model design, and the insights from Paper III have been utilized in all subsequent works, including modeling of *processor* and *generators*.

Papers V and VI contribute by validating State Space Models (SSMs), recently integrated into machine learning frameworks, and Linear Recurrent Units, a recent RNN variant, for audio effect modeling. These can provide advantages thanks to their ability to track past signal information. The study concludes that State space models, although having slightly higher accuracy for cases including time-variant behaviors, do not outperform more traditional types of networks, such as RNNs, in distortion modeling tasks.

Paper VI integrates all the previous contributions in an architecture and learning strategy, which provides the best accuracy compared to other state-of-the-art models and other proposed variants. From this work emerges a model featuring the FiLM+GLU conditioning method, S6-based layers as a recurrent structure to encode the past from the signal, and a sliding window infusing into the network frequency-domain information later processed using a convolutional layer. The paper offers additional insights, such as the effect of parameter sampling density in the dataset. It concludes that sampling control parameters at a high resolution—thereby working with large datasets—does not always yield a better approximation of parameter combinations not included in training data.

Additionally, it reveals that a fast attack coupled with a slow release presents the most challenging scenario to emulate in DRC.

The neural models featured in Papers II, IV, V, and VI have been developed into prototype Virtual Studio Technology (VST) plugins for practical demonstration of their real-time computation capabilities. However, from a software development standpoint, these plugins are not yet ready for public release and distribution, as they require intricate configuration of low-level dependencies.

Paper VII investigates the neural modeling of *generators*, taking Piano instruments as a case study. The paper’s contribution is an enhanced, Differentiable Digital Signal Processing-based model that integrates knowledge of physics, resulting in a more flexible and efficient piano emulation. The proposed model learns to generate the sound of single notes, obviating the need for large, deep neural networks and vast amounts of data to achieve high accuracy. Paper VIII further contributes in the same vein by integrating models for the transient and noise generation components of piano sound, and it incorporates more sophisticated physics equations to enhance the quasi-harmonic content generation.

Papers VII and VIII provide an approach for designing gray box models of musical instruments, including those with strong nonlinear responses, by utilizing neural-based, data-driven techniques to yield computationally efficient emulations. The proposed model presents fewer trainable parameters and lower intrinsic latency compared to other state-of-the-art models documented in the literature. In addition, the model learns and achieves good accuracy even with small datasets. Training the model to generate sound associated with individual notes facilitates the generation, leveraging knowledge that can be easily extracted from recordings. This initial achievement holds its own against physical modeling and sample-based synthesizers, offering lower computational costs and memory usage, respectively. Moreover, this approach does not compromise on flexibility; it enables the emulation of any specific piano and provides the potential to synthesize timbres based on the learned datasets.

Given its complex sound generation mechanism, the piano is a challenging acoustic instrument to model. For example, phenomena such as the coupling between different keys when playing chords are not trivial to model. Paper VIII also contributes to this aspect: starting with the sounds of notes synthesized individually, a subsequent network—trained as a black-box model—can process the sum of the generated notes to introduce the coupling effect between the keys. The sum of individual notes and the actual chord is close enough that large networks are not required, and the conditioning information, which includes the keys and velocities involved, can be incorporated using the conditioning strategy that emerged from previous papers on audio effects. Listening tests have highlighted that a neural network can accurately emulate key coupling when playing chords. By utilizing this approach, we envision that modeling other aspects, such as the sound nuances generated by the re-striking of already vibrating strings, could be achieved.

The contribution of Paper IX is a novel neural modeling philosophy for *generators*, once again considering the piano as a case study, and conceptualizes instrument emulation as a hybrid between sample-based synthesis and a neural

audio effect. Particularly, the modeling problem can be linked to an audio effect modeling task, where the model is trained to alter a specific sound signal recorded from the piano. For instance, a recorded note at a given velocity is processed to render the same note at different velocities. In doing this, the model approximates the nonlinear relationship between the input energy and the resulting sound. This method meets the predefined requirements outlined in this thesis. The architecture is designed to ensure zero-sample latency and a reduced network size while still maintaining high accuracy.

Finally, this thesis also contributes a computational comparative method and software tools that can support research in musical device modeling. The former, detailed in Paper X, facilitates the evaluation of digital models of musical instruments by identifying aspects that are still not accurately modeled. The latter, described in Paper XI, provides an automated and accurate means to collect datasets from any audio effect.

Chapter 5

Conclusions

In this final chapter, the findings and contributions from the papers included in this thesis are discussed in the wider context of the research questions presented in Chapter 1. The research questions are explored from multiple perspectives throughout the publications submitted as part of this thesis. This chapter begins by revisiting the research questions and answering them. It then provides a general discussion before describing the potential impact in research and development of emulation of musical devices. The chapter concludes by summarizing avenues for future research.

RQ1: How can we design neural architecture that can be trained to accurately emulate time-variant and nonlinear musical devices while satisfying the latency and computational requirements of digital musical devices?

Results described in this dissertation show that neural networks can emulate time-variant and nonlinear characteristics, but in particular, neural networks featuring state mechanisms, when serving as a backbone in neural architectures, offer low latency and efficient designs. The papers presented in this thesis demonstrate that models based on recurrent and state space layers can accurately learn to emulate parametric nonlinear and time-variant devices while limiting computational costs and processing small blocks of samples. Conversely, other network types, such as fully connected and convolutional networks, cannot meet these criteria because they require longer blocks of samples as input, which is detrimental to the model’s intrinsic latency.

Papers II and VI present an optical Dynamic Range Compressor (DRC) neural model that features a small number of parameters and input-output latency. Paper V demonstrates these types of networks, with a maximum of 800 internal parameters, can emulate different *processors* and still satisfy the musical device requirements described in Section 2.3, and paper IV demonstrates that this type of architecture can also learn different complex effects with different complexity while keeping the number of parameters less than 900.

Similarly, this question is addressed in Papers VII and VIII for the case of *generators*. The case study for the *generators* considers Acoustic Pianos. The work carried out for this category suggests that nonlinear and time characteristics

of piano instruments can be emulated by hybrid models based on spectral modeling and Differentiable Digital Signal Processing. Specifically, the sound generated by striking a single piano key requires nearly 10 seconds to fully decay, a duration that is comparable to the longer release time found in optical DRCs. Results from these works suggest that strategies such as dividing the modeling problem into separate sub-modules and incorporating physics knowledge into them lead to compact network models. These networks can be configured to synthesize audio samples in small blocks, potentially down to one sample at a time, while capturing the specific timbre of the piano from which the training dataset was recorded. This gray-box approach eliminates the need for the neural model to comprehend complex sound generation features, such as the inharmonicity of the frequency distribution, the generation of different strings' polarizations, and frequency-dependent decay. As a result, it enables the use of smaller network models for the above aspects. Finally, the thesis presents an alternative approach to modeling acoustic instruments using neural techniques. The findings in Paper IX suggest that treating *generators* as *processors* is a viable hybrid alternative, blending sample-based synthesis with neural modeling, offering accurate emulation, low latency, as well as computational and memory efficiency, as demonstrated in the case study of the acoustic piano.

RQ2: How can neural models of musical devices be conditioned to adjust their responses according to inputs such as control parameters and stimuli, thereby enabling interactive user control?

The research described in this thesis demonstrated that parametric musical devices can be effectively modeled by feeding the networks with additional information, which enables them to learn multiple representations of the sound alteration or generation process. Results from Paper III suggest that the best approach is for the network first to find a suitable representation for the input signal and then apply the conditioning operations. This adds valuable insight into how to design the architecture for parametric devices, resulting in conditioned neural networks. Besides showing different possibilities, Paper III also indicates other design insights, such as the use of convolution and nonlinearity for the conditioning of distortion audio effects. Paper IV demonstrates that the conditioning technique is also effective when applied to models trained to emulate more than one audio effect controlled by artificially created parameters. Paper VIII suggests a trade-off regarding the sampling density of control parameter values in the dataset used for training models. Specifically, a larger dataset with more sampling points within the parameter range does not necessarily guarantee higher accuracy when predicting the audio signal associated with control parameter combinations not included in the training data.

The case of *generators* presents a different challenge. In general, acoustic instruments do not have parameters like those found in analog synthesizers that can significantly alter the generated timbre. However, the stimulus provided by the player, representing the energy that triggers acoustic sound generation, as well as the manipulation of other mechanical elements, may cause noticeable variations in the generated sound. Hence, in this case, the parametric information is provided at the input of the networks, such as feeding the MIDI-like information directly

into the sound-generating layers. This thesis has demonstrated that the models are capable of embedding various representations of the note-generation mechanism, reflecting changes in velocity and pitch. Consequently, the architecture produces partial envelopes and frequency distributions that differ significantly from key to key. The proposed conditioning techniques remain valuable for addressing additional challenges in the synthesis of *generators*' sound, such as the interaction among multiple keys. In scenarios where keys are struck simultaneously, they can influence each other's sound. The associated sound-altering simulation can be informed by contextual information, such as which keys are being played and their respective velocities. This approach has been found to be effective in providing sufficient perceptual accuracy.

RQ3: To what extent are the conceived neural architectures, or their internal components, sufficiently generic to model behaviors within or across different types of musical devices, and what openings do they offer for designing musical devices with novel creative features?

Papers II and VI demonstrate that neural architectures designed to model specific analog optical DRCs maintain accuracy when applied to modeling compressors with different characteristics, considering also the various types of control parameters featured to condition the network's emulation of the compression process.

Paper V suggests that the effectiveness of the proposed neural architectures in modeling audio effects can be extended to other types, characterizing audio alteration processes. The proposed architectures can accurately model effects such as distortion, compression, saturation, and equalization. However, this is not universally applicable to all *processors*; for instance, all models evaluated in Paper V failed in the case of low-pass filtering. This indicates that while Recurrent Neural Networks and State-Space Models have similar generalization capabilities, their key differences lie in performance. Specifically, their ability to utilize the pursued methodology is confined to effects primarily involving time-domain alterations. Other types of effects that were out of the scope of this thesis, such as time-based effects (Martínez, Benetos and Reiss, 2020), like chorus, flanger, and phaser. These effects pose different challenges, although nonlinearities and time-variant behavior are less pronounced and may require a different modeling approach. Preliminary experiments, which are not detailed in this thesis, have revealed the proposed architecture's unsuitability in modeling this particular group of audio effects.

For the case of *generators*, the generalization capability is explored in Paper VIII. Here, the classes of instruments that can be modeled with the same architecture are restricted by the physical formulas infused into them. In paper VIII, we found the model learned to emulate different types and specific acoustic piano models, such as upright and grand piano types. The same methodology can be applied to other strings-instruments such as guitars and basses, sharing the main physical phenomena, although minor instrument-specific adjustments are necessary.

Lastly, the findings of Paper IV demonstrated that a single neural model can learn to process multiple audio effects with varying degrees of nonlinearity and

timing behaviors. These results pave the way for the design of hybrid musical devices that do not have corresponding physical counterparts, thereby significantly expanding the range of creative applications of such technologies.

5.1 Discussion

The primary goal of this thesis has been to investigate the application of deep learning techniques for modeling nonlinear and time-variant musical devices that modify or create sound signals. This exploration takes into account key aspects of musical device emulation, including computational complexity, latency, and interactivity, all of which are essential for realizing a playable emulation.

The contributions of this work aim to benefit the modeling of musical devices exhibiting complex physical behaviors and inaccessible knowledge of their functioning to apply white box modeling. Indeed, the frameworks explored in this thesis offer a simplified parameters-tuning process based on the device's recordings. These frameworks are differentiated among two categories that were primarily identified. Musical devices can present very different behaviors from each other, but in particular, there is a fundamental distinction when considering their modeling: some musical devices take audio and produce audio, while others take stimuli and produce audio. This aspect forced the exploration to branch into the two categories of *processors* and *generators*. This identification led to different design considerations and the need to define the methodology accordingly. Since there are a variety of musical devices, the research questions are answered considering two cases of study chosen to be complex and representative of the open challenges in the respective categories: the analog optical Dynamic Range Compressor and the Acoustic Piano.

The papers included in the thesis demonstrated that thoughtful design of deep learning architectures, exploiting a variety of layers, can model musical devices, solving limitations in the accuracy or usability of the earlier proposals. This research led to musical device emulations that satisfy all the requirements for being playable as the target musical device. These results emerged for both categories. To a different extent, machine learning techniques enhanced the modeling methodology for nonlinear and time-variant musical devices, giving good accuracy and interactive capability. A more accurate neural model of Dynamic Range Compressor than state-of-the-art proposals emerged from Paper VI. The research shows how to adjust the model output according to the control parameters. In the *processors* category, all the models are conditioned with different numbers of parameters, which govern different aspects of the input alteration. In the *generators* category, the models are effectively guided by the velocity values for the single note generation and by the velocity and pitches of the played notes for the emulation of trichords.

Another quality emerging from this thesis is the flexibility of some modeling techniques, which can be generically utilized in several contexts. Although it cannot be extended to all, the same architecture can be used for different types of other *processors*, including the case of *generators* when formulated as a *processor* problem, as done in Paper VIII and Paper IX. The learning strategy, although

needing minor adjustments, is shown to be not specific but effective for different modeling cases. Similarly, the designed conditioning block proved to be effective in audio-to-audio learning. The methodology exploited for the *generators* considers the infusion of knowledge that is partly shared among different instruments. In particular, with the example of the piano, the formulas used to build the models are valid for other string-excited instruments. The material parameters, which are actually different among the instruments, are learned by the networks. However, additional consideration may be needed to adapt the model to different types of stimuli; for example, plucked strings in the guitar can reach higher amplitude displacements and create a frequency glide effect.

This dissertation also provides insight into the hyperparameters and loss functions. Mean squared error loss is preferable for the dynamic range compressor task and for other audio effects considered when evaluating the universality of the proposed architectures. Mean Squared Error (MSE) loss helps to emulate the bigger change in amplitude than Mean Absolute Error (MAE) and Error to Signal Ratio (ESR) error. The multi-resolution Short-Time Fourier Transform (STFT) loss has resulted in more efficiency when combined with the Root-Mean-Squared (RMS) loss in the piano modeling case. This is because the multi-resolution STFT loss generally causes more instability during training. Additional constraints are beneficial when STFT loss is utilized. For this reason, STFT loss has been avoided when tuning the partial distributions since it has led to unstable learning. While the batch size on the *processors*' category is close to the standard choices, for the *generators*, the batch size was selected as large as possible when generating the single note using the Differentiable Digital Signal Processing framework. Small batch size leads to more difficulty in tuning the weight to simulate the high decay rate presented in the initial part of the note and the lower decay rate and amplitudes of the release part. The updates of the weights after the network have been seen in both portions, benefiting the learning of this aspect. The learning rates presented more changes throughout the papers and slightly diverged from standard choices. In particular, a time-based schedule that is exponentially decaying has generally been preferred to a fixed learning rate since it has led to better performance.

Finally, providing insight and knowledge on the literature on current methodologies is part of the continuation of the dissertation. Papers V, VI, and VII provide a systematic overview of machine-learning approaches for modeling analog audio effects and acoustic instruments.

To conclude, the main contributions of the thesis can be summarised as follows:

Novel neural architecture for *processor* and *generator* musical devices.

Novel neural-based techniques have been designed, including specialized architectures, to model nonlinear and time-variant musical devices that present low computational complexity and minimal latency.

Novel condition approaches. Approaches to condition the models have been conceived. The conditioning techniques allow the model to learn and replicate the different behaviors a musical device exhibits when changing input parameters and/or stimuli.

Insight into musical device modeling learning strategy. Knowledge of the use of hyper-parameters and loss functions for training parametric audio-to-audio and parametric symbolic music-to-audio neural networks has been developed during the experiments carried out for this thesis.

Open-access data set. All data collected and produced in this dissertation is available online. It includes open-source code for all proposed neural architectures and training approaches, public datasets, and associated trained models for a variety of musical devices collected from hardware and software musical devices.

5.2 Impact

The research undertaken for this thesis provides new insights and methodologies for modeling nonlinear and time-varying behaviors. These can be applied to other analog *processors* and acoustic *generators*, potentially influencing related fields of research and development.

From a research perspective, the contributions of this thesis facilitate further experimentation with various audio effects. The proposed architecture can serve as a foundational model for implementing targeted adjustments to accommodate effects that, until now, have been challenging to model, such as filters, effects with internal parameters modulated by low-frequency oscillator, and reverbs. Notably, these effects could benefit from the stateful blocks used in the architecture designed for compressors. With minor modifications, the approach to modeling the acoustic piano may be extended to other string-excited instruments, and the underlying principles could also be applied to non-string instruments such as woodwinds. Additionally, *generators* such as analog synthesizers present another avenue for exploration; although they exhibit a much simpler excitation principle compared to acoustic instruments, the architectural framework from this thesis, supplemented with circuit-level knowledge, may be adapted to model them effectively.

The contributions of this thesis also have the potential to make a significant short- to medium-term impact on the audio plugin industry, which is partly focused on developing playable software emulations of musical devices. This potential can be harnessed further with the support of appropriate frameworks that facilitate the collection and training of models using end-user data. This aligns with the creative possibilities that machine learning presents, allowing users to combine data from different devices to craft their unique hybrid emulations.

The methodologies developed in this work extend beyond the musical domain and can be advantageous for tasks in physical system identification and time series forecasting. In these contexts, the stringent requirements for low latency and low computational complexity can be relaxed, allowing a greater emphasis on prediction accuracy.

The innovative conditioning methods devised can facilitate the integration of external information into the network within a multi-modal context, enabling the processing of diverse data types beyond parameter values. Moreover, the dataset curated for this research and made publicly accessible has the potential for a wide

range of uses, including analysis, classification, and automatic mixing, beyond its initial purpose for modeling.

5.3 Future Work

The presented architecture for modeling the Dynamic Range Compressor effect has been shown to accurately capture compression characteristics. However, the current loss function averages errors over extended temporal segments of the output signal. A loss function that places greater emphasis on errors during sharp input transients could prove beneficial for enhancing model precision, specifically during the attack phase of compression when faced with fast attack times. Reducing the batch size might seem like a viable solution, but it comes with the drawback of significantly increasing training time.

In the context of acoustic piano modeling, a perceptual modeling inaccuracy is noticeable in the emulation of the onset of individual notes' sounds, potentially attributable to the underestimation of the damping coefficients. More specifically, it has been noted that higher partials exhibit an excessively slow energy decay, indicative of inaccuracies in the learned parameters governing the damping envelope's mathematical exponent. Providing an initial estimate of expected values could facilitate the training process, especially when the learnable parameters could have large values, which may be challenging to tune from random initialization. Additionally, a more accurate physical formulation for the damping coefficient could prove advantageous as well. Improvements in model training could also be achieved by adjusting the loss functions employed. While the multi-resolution Short-Time Fourier Transform loss has aided in learning the sound generation process, it may contribute to instability during training. This instability could stem from weight updates of significant magnitude.

Other temporal and mechanical interdependencies happen in an acoustic piano, other than trichords, such as more keys being played at the same time or keys struck with associated strings already vibrating. These are challenging to model using sample-based synthesis or physical modeling. The proposed modeling approach can offer a viable solution, given that the required datasets are developed, which represents another challenge.

Although the networks presented are relatively small, further size reduction could be explored through techniques like knowledge distillation. This method involves using the outputs of a trained, larger network or the latent space from its intermediate layers to train more compact networks. Larger models often yield higher accuracy due to their increased parameter count. Knowledge distillation can facilitate the retention of this higher accuracy while shrinking the network size to levels comparable to those discussed in this thesis.

Lastly, an aspect highlighted throughout this thesis that needs further investigation is the use of nonlinear transformation when using the proposed conditioning method. Nonlinear transformation can describe a more complex influence of the parameters on the sound alteration or generation process. Further experiments on the design of polynomial equations performing the transformation may be considered.

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